

CLINICAL STUDY PROTOCOL

A multicentric, randomized, double-blind, parallel, placebo-controlled phase III study to assess the safety and efficacy of sovateltide in patients with acute cerebral ischemic stroke.

Protocol Number	Sovateltide/ACIS/IND2022
Version/Date	Version 3.1/25 May 2025
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Clinical Trial Phase	Phase III
Sponsor	Pharmazz EU Limited (A Wholly Owned Subsidiary of Pharmazz, Inc.) 50 W 75th Street, Suite 105 Willowbrook, IL 60527, USA Tel. No. +1 (630) 780-6087
Investigational Product	Sovateltide for Injection

CONFIDENTIALITY STATEMENT

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1 GENERAL INFORMATION

1.1 Protocol Details

Protocol Title A multicentric, randomized, double-blind, parallel, placebo-controlled phase III study to assess the safety and efficacy of sovateltide in patients with acute cerebral ischemic stroke.

Protocol Number Sovateltide/ACIS/IND2022

Version No./Date 3.1/25 May 2025

1.2 Sponsor Details

Sponsor Pharmazz EU Limited
(A Wholly Owned Subsidiary of Pharmazz, Inc.)
50 W 75th Street, Suite 105
Willowbrook, IL 60527, USA
Tel. No. +1 (630) 780-6087

1.3 Person Authorised to Sign the Protocol for the Sponsor

Sponsor Signatory Name: Prof. Anil Gulati
Pharmazz, Inc.
50 W 75th Street, Suite 105
Willowbrook, IL 60527, USA
Phone: +1 (630) 780-6087

1.4 Investigator(s) Responsible for Conducting the Study and Study Site(s)

This information is documented separately in the dossier to be submitted to the regulatory authority.

1.5 Clinical Laboratory and other Medical and Technical Department(s) and Institutions involved in the Study

This information is documented separately in the dossier to be submitted to the regulatory authority.

PREPARATION AND APPROVAL OF THE PROTOCOL

Protocol prepared by:	
Medical writing responsible person	Dr. Amaresh Ranjan, PhD Pharmazz, Inc. 50 W 75th Street, Suite 105 Willowbrook, IL 60527, USA Phone: +1 (630) 780-6087 E-mail: amaresh.ranjan@pharmazz.com <i>AKRanjan</i> 05/25/2025 Signature and date: Ms. Shruti Gulati, MS Pharmazz, Inc. 50 W 75th Street, Suite 105 Willowbrook, IL 60527, USA Phone: +1 (630) 780-6087 Email: shruti.gulati@pharmazz.com <i>Shruti Gulati</i> 25-May-2025 Signature and date:
Protocol reviewed by:	
Reviewer	Dr. Dharmesh Shah, MD, DNB, DM Address: Pharmazz India Pvt. Ltd. H-6, Site-C, Surajpur Industrial Area, Greater Noida-201307, India Phone: +91 8800015065 E-mail: dharmesh.shah@pharmazz.com <i>D. H. Shah</i> Signature and date: 25- MAY-2025
Protocol approved by:	
Sponsor	Prof. Anil Gulati, MD, PhD Pharmazz, Inc. 50 W 75th Street, Suite 105 Willowbrook, IL 60527, USA Phone: +1 (630) 780-6087 Email: anil.gulati@pharmazz.com <i>Anil Gulati</i> 05/25/2025 Signature and date:

AGREEMENT ON THE FINAL PROTOCOL

The investigator agrees to conduct the study as outlined in this protocol and in accordance with local regulations and current International Council for Harmonisation-Good Clinical Practice guidelines. Any modification to the protocol must be approved in writing by the sponsor, the institutional review board/institutional ethics committee as may be required by applicable regulations.

The investigator agrees, by written consent to this protocol, to fully co-operate with monitoring and audit checks by allowing direct access to patient's records, including source data, by authorized individuals representing the sponsor.

Approved consent in writing:

Name

Signature

Date

PROTOCOL SYNOPSIS

Title	A multicentric, randomized, double-blind, parallel, placebo-controlled phase III study to assess the safety and efficacy of sovateltide in patients with acute cerebral ischemic stroke
Protocol Number	Sovateltide/ACIS/IND2022
Study Objectives	<u>Primary objective</u> The primary objective of the study is: - To determine whether sovateltide causes a good functional outcome in patients with acute cerebral ischemic stroke assessed by modified Rankin Scale (mRS) score of 0-2 at day 90 post-randomization. <u>Secondary objectives</u> The secondary objectives of the study are to determine whether sovateltide causes: - A good functional outcome in patients with acute cerebral ischemic stroke assessed by the National Institute of Health Stroke Scale (NIHSS) score of <6 at day 90 post-randomization. - A good functional outcome in patients with acute cerebral ischemic stroke assessed by the Barthel Index (BI) score of ≥90 at day 90 post-randomization. - An excellent functional outcome in patients with acute cerebral ischemic stroke assessed by a modified Rankin Scale score of 0-1 at day 90 post-randomization. - Change in Quality-of-life (QoL) as assessed by EuroQol-EQ-5D-5L and Stroke-Specific Quality of Life (SS-QOL) at days 30, 60, and 90 post-randomization. - Incidence of recurrent cerebral ischemic stroke within 90 days post-randomization - Incidence of mortality within 90 days post-randomization. - Incidence of symptomatic Intracerebral Hemorrhage (ICH) within 24 (± 6) hours of randomization. - Incidence of radiographic Intracerebral Hemorrhage (ICH) within 24 (± 6) hours of randomization. - Alteration in cognition at days 30 and 90 measured by Montreal Cognitive Assessment (MoCA) Test. - Any adverse events (AE) or serious adverse events (SAEs) are associated with sovateltide.

	<u>Exploratory objectives</u> The exploratory objectives of the study are to determine the following: • A good functional outcome in patients with acute cerebral ischemic stroke assessed by mRS score of 0-2 at day 30 post-randomization. • A good functional outcome in patients with acute cerebral ischemic stroke assessed by the NIHSS score of <6 at day 30 post-randomization. • A good functional outcome in patients with acute cerebral ischemic stroke assessed by the BI score of ≥ 90 at day 30 post-randomization. • Alberta Stroke Program Early CT (ASPECT) Score for stroke severity at baseline and identify sites of ischemic lesions. • The subtype of the index acute cerebral ischemic stroke according to TOAST classification by day 30 post-randomization.
Study Endpoints	<u>Efficacy endpoints</u> <u>Primary endpoint</u> The primary efficacy endpoint of the study is: • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale score of 0-2 on day 90 post-randomization. <u>Secondary endpoints</u> The secondary efficacy endpoints of the study are: • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with NIHSS score of <6 on day 90 post-randomization. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with BI score of ≥ 90 on day 90 post-randomization. • The proportion of acute cerebral ischemic stroke patients having an excellent functional outcome with a modified Rankin Scale score of 0-1 on day 90 post-randomization. • Change in QoL as assessed by EuroQol EQ-5D-5L and by SS-QOL from baseline to days 30, 60, and 90 post-randomization. • The proportion of patients with recurrent ischemic stroke within 90 days post-randomization.

	• Number of deaths within day 90 post-randomization. • The proportion of patients with symptomatic ICH within 24 ($\pm$ 6) hours of randomization. • The proportion of patients with radiographic ICH within 24 ($\pm$ 6) hours of randomization. • Change in MoCA score at days 30 and 90 post-randomization. • The proportion of patients with adverse events (AEs) and serious adverse events (SAEs). <u>Exploratory endpoints</u> The exploratory endpoints of the study are to: • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a mRS score of 0-2 on day 30 post-randomization. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with NIHSS score of <6 on day 30 post-randomization. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with BI score of ≥ 90 on day 30 post-randomization. • Compare baseline Alberta Stroke Program Early CT (ASPECT) Score for stroke severity and identify sites of ischemic lesions. • Compare the subtype of the index acute cerebral ischemic stroke by TOAST classification by day 30 post-randomization.
Study Design	This is a multicentric, randomized, double-blind, parallel, placebo-controlled phase III clinical study to assess the safety and efficacy of sovateltide along with the standard of care in patients with acute cerebral ischemic stroke. A total of 514 patients (257 in each treatment arm) with acute cerebral ischemic stroke will be enrolled in this study. The enrolment period of the study will be approximately 15 months, and the total duration of the study will be approximately 18 months. For an individual patient, the duration of the study will be 3 months (90 days), including 4 study visits: visit 1/Day 1 (screening/baseline/treatment visit), visit 2/Day 30 $\pm$ 5, visit 3/Day 60 $\pm$ 5 (telephonic visit), and visit 4/Day 90 $\pm$ 5 (End of Study). A total of 514 patients will be randomized 1:1 into 2 treatment groups after meeting the eligibility criteria: • Group 1 (Active Group): sovateltide + standard of care

	Group 2 (Control Group): matching placebo for sovateltide + standard of care Central randomization will be stratified by early vs. late enrollment (<12 hours vs. ≥ 12 hours) as well as usage or non-usage of IV thrombolytic agents at enrollment. Although the usage of IV thrombolytics is recommended within 4.5 hours after symptom onset (varying by regional approvals), operationally, there will be some patients who receive IV thrombolytics on time but may enroll late. Therefore, the stratification of IV thrombolytics will be done among the patients that will be enrolled either early (< 12 hours) or late (≥ 12 hours). To allow adequate opportunity for enrollment into each of the strata, a minimum of 200 patients per stratum in early vs. late enrollment and a minimum of 70 patients will be included in each sub-stratum of IV thrombolytic usage and non-usage in the early (< 12 hours) but not in the late (≥ 12 hours), enrollment group. Sovateltide or matching placebo will be administered as an intravenous (IV) bolus over 1 to 2 minutes within the window of 24 hours after the onset of the stroke. In the sovateltide group, 3 doses of sovateltide, at a dose of 0.3 $\mu\text{g/kg}$, will be administered as an IV bolus over 1 to 2 minutes every 3 hours ± 1 hour on days 1, 3, and 6 (total dose/day: 0.9 $\mu\text{g/kg}$). In the control group, 3 doses of the placebo will be administered as an IV bolus over 1 to 2 minutes every 3 hours ± 1 hour on days 1, 3, and 6 post-randomization. Patients will be provided the best standard of care in both groups. Every effort will be made to administer the study drug at the same time on days 1, 3, and 6. The day 6 dosing has to be performed within a time window of ± 1 day. Those patients discharged prior to day 6 will be called back to the healthcare facility for the remaining doses. The standard of care to be provided to the patients shall be as per the particular hospital setup for acute cerebral ischemic stroke. Each patient will be monitored closely throughout his/her hospitalization for the qualifying stroke and will be followed for 3 months from randomization. Each patient will be assessed for efficacy and safety parameters over 3 months from randomization at a clinic visit. Each patient will be contacted by telephone for a brief (<30 minutes) clinical and QoL assessments at visit 3/Day 60 from randomization.
Study Center (s)	Approximately 65 study centers in the United States, Canada, United Kingdom, and Europe (the demographics and standard of treatment being similar in these countries) will participate in the study.

Safety Assessments	• AEs will be collected and evaluated for relatedness, severity, seriousness, and outcome • C-SSRS assessment for suicidal ideation • Physical examination • Vital signs (respiratory rate, heart rate, systolic and diastolic blood pressure (BP), body temperature) • ECG assessment for off-target QT effects • Clinical laboratory parameters • Assessment for immunogenicity potential of sovateltide
Efficacy Assessments	The efficacy assessments in the study are as follows: • mRS • NIHSS • BI • QoL by EuroQol-5D-5L and SS-QOL • Clinical evaluation for signs or symptoms of recurrent stroke • Mortality and time-to-death • Radiographic and symptomatic ICH within 24 ($\pm$ 6) hours post-randomization • MoCA Test • ASPECT Score • TOAST classification of index acute ischemic stroke
Diagnosis and Main Inclusion Criteria/Exclusion Criteria	A patient will be eligible for inclusion in the study if they fulfill the following criteria: 1. Adult males or females aged 18 – 80 years of age 2. Consent obtained per national laws and regulations, and in accordance with the applicable ethics committee requirements prior to study procedures 3. A stroke is ischemic in origin that is diagnosed clinically and/or radiologically confirmed by Computed Tomography (CT) scan or diagnostic magnetic resonance imaging (MRI) prior to enrolment. No hemorrhage as proved by cerebral CT/MRI scan. 4. Cerebral ischemic stroke patients presenting within 24 hours after the onset of symptoms with NIHSS score of ≥ 8 and < 20, NIHSS Level of Consciousness (1A) score < 2 at the time of screening. This includes cerebral ischemic stroke patients who completely recovered from earlier episodes before having a new or fresh stroke having a pre-stroke historical measure of mRS score of 0-2.

	5. The patient is <24 hours from the time of stroke onset when the first dose of sovateltide is administered. Time of onset is when symptoms began; for stroke that occurred during sleep, time of onset is when the patient was last seen or was self-reported to be normal 6. Reasonable expectation of availability to receive the full sovateltide/placebo course of therapy and to be available for subsequent follow-up visits A patient will not be eligible for inclusion in this study if they meet any of the following exclusion criteria: 1. Patients receiving endovascular therapy or is a candidate for any surgical intervention for the treatment of stroke, which may include but not limited to endovascular techniques. 2. Patients classified as comatose are defined as a patient who requires repeated stimulation to attend or is obtunded and requires strong or painful stimulation to make movements (NIHSS Level of Consciousness (1A) score ≥ 2) 3. Evidence of intracranial hemorrhage (intracerebral hematoma, intraventricular hemorrhage, subarachnoid hemorrhage (SAH), epidural hemorrhage, acute or chronic subdural hematoma (SDH) on the baseline CT or MRI scan 4. Known pregnancy and lactating women 5. Known medical history of neurological (other than current acute ischemic stroke) or psychiatric condition that, in the investigator's opinion, would confound the neurological and functional evaluations, lead to further deterioration of neurological status, or interfere with participation in this study 6. Concurrent participation in any other therapeutic clinical trial 7. Evidence of any other major life-threatening or serious medical condition that would prevent completion of the study protocol impair the assessment of outcome, or in which sovateltide therapy would be contraindicated or might cause harm to the patient
Test Product, Dose, Route, Regimen	The test product is sovateltide. It is available as a lyophilized injection containing 30 µg of sovateltide in a 5.0 mL vial, to be reconstituted in 5 mL normal saline (0.9% Sodium Chloride Injection). Three doses of 0.3 µg/kg will be administered as an IV bolus over one to two minutes every 3 hours $\pm$ 1 hour on day 1. The dose will be repeated on days 3 and 6 post-randomization. The study drug will be administered as an IV bolus dose over 1 to 2 minutes within 24 hours of the stroke

	onset. The day 6 dosing has to be performed within a time window of ± 1 day. Those patients discharged prior to day 6 will be called back to the healthcare facility for the remaining doses.
Reference Product (Placebo), Dose, Route, Regimen	The reference product is a matching placebo for sovateltide. It is a sterile lyophilized injection consisting of excipients formulated without the active ingredient in a 5.0 mL vial, to be reconstituted in 5 mL of normal saline (0.9% Sodium Chloride Injection). Three doses will be administered as IV boluses over one to two minutes every 3 hours ± 1 hour on day 1. The dose will be repeated on days 3 and 6 post-randomization. The day 6 dosing has to be performed within a time window of ± 1 day. Those patients discharged prior to day 6 will be called back to the healthcare facility for the remaining doses.
Statistical Methodology	<u>Sample size calculation</u> The sample size for this study is calculated based on the results of our recently completed phase III clinical study in patients of cerebral ischemic stroke in India (CTRI/2019/09/021373; NCT04047563). An enrollment ratio of 1:1 is considered for the present phase III study. The primary endpoint of the study is the modified Rankin Scale score at day 90 post-randomization. The key secondary endpoints are (NIHSS) score of ≤ 6 and Barthel Index (BI) score of ≥ 90 at day 90 post-randomization. Treatment with placebo is expected to obtain 55.71% of patients having an mRS score ≤ 2 at day 90 post-randomization, and treatment with sovateltide is hypothesized to obtain 80.60% of patients to have an mRS score ≤ 2 at day 90 post-randomization. In the referenced phase III study, an odds ratio of 3.302 was obtained; however, to account for variation in different countries, a conservative approach was used and set the odds ratio to 2.45. Therefore, a sample size of 274 is required to achieve 90% power for the primary analysis to show significance in log-scale relative risk, i.e., $\log(\text{OR}) > 0$ in the proportion of patients with favorable outcomes between the Sovateltide and placebo arms. The above sample size calculation assumes the above-mentioned proportions for placebo and sovateltide with a Type I error probability of 0.025 (one-sided for assessing the patients who completed the 90-day follow-up assessments). For the National Institute of Health Stroke Scale (NIHSS) score of ≤ 6 at day 90 post-randomization a sample size of 166 is required to achieve 90% power for the analysis to show significance in log-scale relative risk i.e., $\log(\text{OR}) > 0$ in the proportion of patients with favorable outcomes between the Sovateltide and placebo arms. The above sample size calculation assumes a placebo proportion of 0.7, Sovateltide proportion

of 0.9104 and Type I error probability of 0.025 (one-sided for assessing the patients who completed the 90-day follow-up assessments.

For the Barthel Index (BI) score of ≥ 90 at day 90 post-randomization a sample size of 436 is required in order to achieve 90% power for the analysis to show significance in log-scale relative risk i.e., $\log(\text{OR}) > 0$ in the proportion of patients with favorable outcomes between the Sovateltide and placebo arms. The above sample size calculation assumes a placebo proportion of 0.4714, Sovateltide proportion of 0.6269 and Type I error probability of 0.025 (one-sided for assessing the patients who completed the 90-day follow-up assessments. The ITT principle will be applied to the analysis and, therefore, to safeguard against dilution of the treatment effect associated with an approximate 15% non-adherence rate (due to lost to follow-up, consent withdrawal, treatment crossovers, and stroke mimics), the above sample needs to be inflated by a factor of $1/0.85$ or 1.1765. Therefore, a total sample size of 512.94 rounding it to 514, is required for this study.

Analysis Sets

Randomized (RAN): Randomized set is defined as all patients who are randomized in the study.

Intent-to-treat (ITT): The ITT set is defined as all randomized patients that receive at least one application of the study medication.

Per-protocol (PP): The PP set is defined as those patients in the ITT population who completed day 90 of treatment and for whom there were no serious protocol deviations.

Safety (SAF): The SAF set is defined as all randomized patients who receive at least one application of study medication.

Safety analyses

All safety parameters will be analyzed based on the SAF set. In addition, the number and percentage of patients with treatment-emergent adverse events will be presented. The treatment-emergent adverse events (events started on or after the first dose of study medication or events present prior to the first dose of study medication but increased in severity based on preferred term) will be summarized by MedDRA primary System Organ Class (SOC) and Preferred Term (PT).

Adverse events will be summarized by presenting the number and percentage of patients having at least one AE, having an AE in each primary system organ class, and having each individual AE (preferred

	term). Summaries will also be presented for AEs by severity (CTC Toxicity grade) and for study treatment-related AEs. The most common adverse events reported ($\geq z$ % in any group for each preferred term in the table by SOC and PT) will be presented in the clinical study report by descending frequency according to their incidence in the overall group, starting from the most common event. Here, the threshold value z is set to 5 (%), but it may be updated following a review of the dry run outputs. Separate summaries will be provided for death, serious adverse events, other significant adverse events leading to treatment discontinuation, and adverse events leading to dose adjustment. For the legal requirements of ClinicalTrials.gov, two required tables on treatment emergent adverse events which are not serious adverse events with an incidence greater than 5% and on treatment-emergent serious adverse events and SAE suspected to be related to study treatment will be provided by system organ class and preferred term on the safety set. If, for the same patient, several consecutive AEs (irrespective of study treatment causality, seriousness, and severity) occurred with the same SOC and PT: • A single occurrence will be counted if there is ≤ 1-day gap between the end date of the preceding AE and the start date of the consecutive AE • More than one occurrence will be counted if there is > 1-day gap between the end date of the preceding AE and the start date of the consecutive AE For occurrence, the presence of at least one SAE/SAE suspected to be related to study treatment/non-SAE has to be checked in a block, e.g., among AEs in a ≤ 1-day gap block, if at least one SAE is occurring, then one occurrence is calculated for that SAE. The number of deaths resulting from SAEs suspected to be related to study treatment and the number of deaths resulting from SAEs irrespective of causality will be provided. The adverse events will be summarized for the following subgroup. • Sex (male vs. female) • Age group (< 65, ≥ 65 yrs)
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	• IV thrombolytics vs. no IV thrombolytics • Race <u>Efficacy analysis</u> Primary Endpoint: The scientific objective guiding the primary analysis is to estimate the treatment effect of sovateltide compared to placebo for the target population on the modified Rankin Scale (mRS) score of 0-2 on day 90 post-randomization. The treatment effect of interest shall be defined as: • not missing the dose on day 6 • and regardless of missing the dose on day 6 The primary efficacy endpoint, mRS score of 0-2 on day 90 post-randomization, will be analyzed based on the data from the ITT according to the treatment group and the stratification factors to which the participants were randomized. In line with the treatment effect of interest, the primary observation period will include all the available data for patients who missed the dose on day 6 as well as those who discontinued. Since the treatment effect of interest is on treatment, participants who were randomized but not treated will be excluded from this analysis. The primary observation period will be defined as the time from the date of the randomization to the last study day. The last study day will be EOS/90 days, and in case of discontinuation/lost to follow-up, the last study day will be the date of discontinuation/lost to follow-up. A logistic regression model with stratification and treatment group as factors and baseline value as a covariate will be used for analysis. The estimates of the odds ratio for the mRS score of 0-2 on day 90 post-randomization between treatment groups and their 95% confidence intervals will be provided. The following hypothesis will be tested: The primary statistical null hypothesis is: • H_1: The log odds ratio for mRS score of 0-2 on day 90 post-randomization in the sovateltide group vs. placebo group is ≤ 0 • H_{1A}: The log odds ratio for mRS score of 0-2 on day 90 post-randomization in the sovateltide group vs. placebo group is > 0
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	This hypothesis will be tested using a one-sided test, with a z-score corresponding to the log of the odds ratio for the mRS score of 0-2 on day 90 post-randomization between the two groups with 0.025 α-level. The mRS score of 0-2 on day 90 post-randomization will be summarized by the treatment group. The choice of the Hypothetical strategy is based on the following: • The fact that the patient is exposed to the drug only till day 6 and the discontinuation after day 6 is independent of the treatment effect. For this purpose, assuming the missing is “Missing at Random”, multiple imputations will be used to impute missing responses. The missing day 6 doses will still have the drug effect but may not be as extensive as if the patient completed the course on day 6. For this purpose, the available response on day 90 itself will be used. No new analysis will be performed, and the analysis mentioned above will suffice. The sensitivity analysis of the primary efficacy endpoint will include all ITT patients, with patients grouped according to the treatment assigned at randomization. Patients with missing day 90 assessment will not be included to check the sensitivity of the imputation done for the primary endpoint. A logistic regression model with stratification and treatment group as factors and baseline value as a covariate will be used for analysis. The estimates of the odds ratio for the mRS score of 0-2 on day 90 post-randomization between treatment groups and their 95% confidence intervals will be provided. This hypothesis will be tested using a one-sided test, with a z-score corresponding to the log of the odds ratio for the mRS score of 0-2 on day 90 post-randomization between the two groups with 0.025 α-level. Secondary Endpoints: For all secondary endpoints, estimates and confidence intervals will be presented, but no formal statistical hypothesis testing will be conducted except for the key secondary endpoints. Randomized but not treated participants will be excluded from the analyses below. The statistical tests of all secondary endpoints will be carried out at a two-sided 5 % significance level.
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	The following hypotheses will be tested sequentially and are included in the hierarchical testing strategy and type-I-error will be set such that a family-wise type-I-error of $\alpha = 2.5\%$ (one-sided) is kept. The graphical approach of (Bretz et al. 2009) for sequentially rejective testing procedures is used to illustrate the testing strategy: <pre> graph TD H1((H1 1)) -- 1 --> H2((H2 0)) H2 -- 1 --> H3((H3 0)) H3 -- 0 --> H4((H4 0)) </pre> The null hypotheses for key secondary objectives are as follows: • H₂: The log odds ratio for NIHSS score <6 on day 90 post-randomization in the sovateltide group vs. placebo group is ≤ 0 • H₃: The log odds ratio for Barthel Index (BI) score ≥ 90 on day 90 post-randomization in the sovateltide group vs. placebo group is ≤ 0 • H₄: The log odds ratio for mRS score of 0-1 on day 90 post-randomization in the sovateltide group vs. placebo group is ≤ 0 The secondary efficacy variables involved in the above testing strategy will be analyzed analogously to the primary endpoints. i.e., a logistic regression model with stratification and treatment group as factors and baseline value as a covariate. The estimates of the odds ratio on day 90 post-randomization between treatment groups and their 95% confidence intervals will be provided. Missing data will be imputed as specified in section 2.5.3 in the statistical analysis plan. The Multiple Imputation (MI) method with 50 iterations creating 100 copies will be used to impute the missing assessments of the respective endpoint at day 90 if the day 90 assessment is missing. The testing sequence will continue to H₂ at familywise Type I error rate at a two-sided 0.05 level only if H₁ has been rejected at 0.025 (one-sided). Similarly, the testing sequence will continue to H₃ at familywise Type I error rate at a two-sided 0.05 level only if H₂ testing has been rejected and the testing sequence will continue to H₄ at familywise Type I error rate at a two-sided 0.05 level only if H₃ testing has been rejected.
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	Logistic regression will be repeated for: • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with National Institute of Health Stroke Scale (NIHSS) score <6 on Day 90. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with Barthel Index (BI) score ≥ 90 on Day 90. • The proportion of acute cerebral ischemic stroke patients having an excellent functional outcome with a modified Rankin Scale score of 0-1 on day 90 post-randomization. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale score of 0-2 on day 30 post-randomization. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with National Institute of Health Stroke Scale (NIHSS) score <6 on day 30 post randomization. • The proportion of acute cerebral ischemic stroke patients having a good functional outcome with Barthel Index (BI) score ≥ 90 on day 30 post randomization. Mixed model repeated measures analyses (MMRM) will be used for analyzing the below parameters with treatment, stratification factor, visit, treatment visit interaction as factors, and baseline score as a covariate. • Change in QoL as assessed by interviewer-administered EuroQol EQ-5D-5L from baseline to days 30, 60 and 90 post-randomization • Change in QoL as assessed by interviewer-administered SS-QOL from baseline to days 30, 60 and 90 post-randomization • Change in MoCA score from day 6 to days 30 and 90 post-randomization The Cochran–Mantel–Haenszel test (CMH) for stratified categorical data of proportion will be used for analyzing the below parameters for comparison between treatment groups if the number allows. • The proportion of patients with recurrent ischemic stroke within 90 days post-randomization • Number of deaths within day 90 post-randomization.
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	• The proportion of patients with symptomatic ICH within 24 ($\pm$ 6) hours of randomization. • The proportion of patients with radiographic ICH within 24 ($\pm$ 6) hours of randomization. • The proportion of patients with adverse events (AEs) and serious adverse events (SAEs). Descriptive statistics will be generated for all clinical laboratory tests performed (actual values and changes from baseline, where applicable) for three groups of laboratory tests (Hematology, Biochemistry, and Liver function test) by laboratory test and treatment group. Change from baseline will only be summarized for patients with both baseline and post-baseline values and will be calculated as follows: Change from baseline = post-baseline value – baseline value Analysis in vital sign measurement using descriptive summary statistics for the change from baseline for each post-baseline visit will be performed by vital sign and treatment group. Change from baseline will only be summarized for patients with both baseline and post-baseline values and will be calculated as follows: Change from baseline = post-baseline value – baseline value Exploratory Analyses: The comparison of the ASPECT scores will only be made at baseline using a simple t-test. The subtypes of TOAST classification, which are binary in nature, will be compared only at day 30 post-randomization. These subtypes will be compared using the chi-square test.
Ethical Aspects	The study will be conducted in compliance with the consolidated International Council for Harmonisation-Good Clinical Practice (ICH-GCP), the Declaration of Helsinki, the European Union (EU) Clinical Trials Regulation No. 536/2014, and other national law(s) in the countries where the trial takes place. Consent will be obtained per national laws and regulations, and in accordance with the applicable ethics committee requirements prior to study procedures. This protocol and any amendments will be submitted to a properly constituted Institutional Ethics Committee, in agreement with local regulations, for formal approval of the study conduct.

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LIST OF ABBREVIATIONS

AD	Alzheimer's disease
AE	Adverse Event
ALP	Alkaline Phosphatase
ALT	Alanine Aminotransferase
ASPECT	Alberta Stroke Program Early CT
AST	Aspartate Aminotransferase
BI	Barthel Index
BMI	Body Mass Index
BP	Blood Pressure
CI	Confidence Interval
CSR	Clinical Study Report
CT	Computed Tomography
CTC	Common Toxicity Criteria
DOA	Date of Admission
DOD	Date of Discharge
DSMB	Data Safety Monitoring Board
ECG	Electrocardiogram
eCRF	Electronic Case Report Form
EMA	European Medicines Agency
EOS	End of Study
ET	Endothelin
FDA	Food and Drug Administration
GSH	Glutathione
Hb	Hemoglobin
ICF	Informed Consent Form
ICH	Intra Cerebral Hemorrhage
ICH-GCP	International Council for Harmonisation-Good Clinical Practice
IEC	Institutional Ethics Committee
IP	Investigational Product

IRB	Institutional Review Board
ITT	Intent-to-Treat
IV	Intravenous
IWRS	Interactive Web Response System
LAR	Legally Acceptable Representative
LPLV	Last Patient, Last Visit
MAP	Mean Arterial Pressure
MCAO	Middle Cerebral Artery Occluded
MCH	Mean Corpuscular Hemoglobin
MCHC	Mean Corpuscular Hemoglobin Concentration
MCV	Mean Corpuscular Volume
MDA	Malondialdehyde
MedDRA	Medical Dictionary for Drug Regulatory Affairs
MoCA	Montreal Cognitive Assessment
MRI	Magnetic Resonance Imaging
mRS	Modified Rankin Scale
MTD	Maximum Tolerated Dose
NIHSS	National Institute of Health Stroke Scale
NO	Nitric Oxide
PI	Principal Investigator
PP	Per Protocol
QoL	Quality-of-Life
RBC	Red Blood Cells
RTSM	Randomization and Trial Supply Management
SAE	Serious Adverse Event
SAF	Safety Population
SAH	Sub Arachnoid Hemorrhage
SAP	Statistical Analysis Plan
SAS [®]	Statistical Analysis System [®]
SD	Standard Deviation

SDH	Sub Dural Hematoma
SOC	System Organ Class
SOD	Superoxide Dismutase
SS-QOL	Stroke-Specific Quality of Life
SUSAR	Suspected Unexpected Serious Adverse Reaction
TK	Toxicokinetics
WBC	White Blood Cells

2 INTRODUCTION

2.1 Background

Stroke, also known as a cerebrovascular accident, is the second leading cause of mortality and a third leading cause of serious long-term disability and morbidity worldwide [1]. Every year, it affects nearly 13.7 million people, and about 5.5 million people do not survive [2]. In the United States, the overall stroke prevalence is estimated to be around 2.7%. A stroke occurs every 40 seconds, and death due to a stroke occurs every 4 minutes. One out of every twenty deaths in America is due to stroke, making it the 5th leading cause of death in this country. Annually, there are more than 795,000 people who have a stroke nationwide, out of which 137,000 result in death. About 610,000 people have a first or new stroke, and 185,000 who survive a stroke will have another within 5 years. Projections for 2030 estimate an additional 3.4 million U.S adults to have a stroke, representing 3.9% of the adult population [3-5].

A stroke is a syndrome defined as an abrupt neurological outburst due to impaired blood flow to part of the brain [2]. There are two types of stroke: hemorrhagic stroke and ischemic stroke. A hemorrhagic stroke follows the rupture of a weakened blood vessel in the brain, causing accumulation of blood and compression of the surrounding brain tissue. An ischemic stroke follows a blocked blood vessel by a thrombus (blood clot) or embolism. In both types of strokes, the specific region of the brain supplied by the affected blood vessel is deprived of oxygenated blood, causing local hypoxia that damages the brain tissue and cells [6, 7]. Both types of stroke are very serious; however, ischemic stroke is more common.

The global burden of ischemic stroke is nearly 4-fold greater than hemorrhagic stroke, with close to 87% of the total incidence of stroke attributed to acute cerebral ischemic stroke (ACIS) [8, 9]. ACIS is a critical care emergency caused by a significant reduction in blood flow to the brain by a blood clot or embolism. This nearly halts cerebral blood flow to the affected region leading to neuronal death [10]. Neuronal cell death is followed by plasma membrane disruption, organelles swelling, cell contents leaking into extracellular space, and loss of neuronal function. Other events include inflammation, excitotoxicity, free radical-mediated toxicity, cytokine-mediated cytotoxicity, impaired blood-brain barrier, and oxidative stress [2].

Clinically, ACIS presents as neurologic deficits of sudden onset with symptoms dependent upon the brain region involved. Nearly 90% of ACIS cases are supratentorial in origin. If the left hemisphere is involved, symptoms seen include aphasia, right-sided hemiparesis, and right-sided hemianopia. If the right hemisphere is involved, symptoms seen include left hemispatial neglect, left hemiparesis, and left hemianopia [11]. The specific clinical features of an acute ischemic stroke include sudden onset of the following: numbness or weakness in the face, arm, or leg (especially on one side of the body), confusion, difficulty in speech, abnormal vision and gait, dizziness and loss of balance, lack of coordination, and/or severe headache of unknown cause [12].

ACIS can be further classified based on etiology as large-artery atherosclerosis, cardioembolism, small-vessel disease, other determined causes, and cryptogenic causes. To preserve brain tissue viability, compensatory mechanisms, including cerebral vasodilation, are initiated upon cerebrovascular compromise to maintain or increase cerebral blood flow to the affected brain tissue. However, in time, cerebral blood flow starts to decrease, leading to hypoperfusion and the formation of an irreversibly damaged ischemic core surrounded by an ischemic penumbra- viable and reversibly injured brain tissue [13]. Since the penumbra is at elevated risk for infarction, it is crucial to begin stroke therapy as soon as possible to salvage as much of the penumbra as early as possible [10, 14].

Therapeutic management of ACIS is a multidisciplinary approach with a primary goal of revascularization and limiting neuronal injury. A stroke team consists of emergency medicine physicians, neurologists/neurosurgeons, radiologists, nurses and advanced care providers, clinical pharmacists, therapists, technicians, and laboratory personnel [15]. Currently, the only FDA-approved pharmacological agent for ischemic stroke is a tissue plasminogen activator (t-PA). It is a thrombolytic agent which restores blood flow by breaking down a clot. However, timing is crucial in the administration of t-PA as the therapeutic time window is very narrow, and the patient must receive it within 4.5 hours of the onset of stroke symptoms. Therapeutic administration after this timeframe can result in hemorrhagic transformation, leading to additional brain damage [3]. Nevertheless, even upon timely administration of t-PA in ACIS, only about 30% of patients obtain stroke resolution having minimal or no disability at the 90-day mark [15, 16]. Over the past few years, endovascular therapy has been developed as an alternative treatment option considering the narrow therapeutic time window of t-PA. Endovascular therapy includes intra-arterial thrombolysis, intravascular stent, and mechanical thrombectomy [17]. Stroke management has significantly reduced mortality in the past decade; however, the number of disabled survivors has increased, with stroke remaining a leading cause of disability in the United States and globally. Functional deficits of a stroke may include speech, visual, cognitive, sensory, and motor impairments, which have a negative impact on the patient's quality of life [18]. While the current therapeutic options treat stroke patients by removing the clot, there is no therapeutic available in the market which alleviates neurological deficit and repairs cerebral damage. The pathophysiology of cerebral ischemia is complex, involving hypoxia, vascular damage, inflammation, apoptosis, and other events triggering neuronal cell damage and death. Many neuroscientists are engaged in thoroughly comprehending the pathophysiology of the stroke to help develop novel therapeutic agents that target anticoagulation, neuroprotection, and neuroregeneration through various mechanisms. It has been demonstrated that endothelin signaling regulates these pathophysiological events in the mammalian neurovascular system; therefore, the endothelin system is regarded as a potent target for drug development in stroke [19].

The discovery of endothelin (ET) and ET receptors in various tissues has paved the way for developing therapeutics using novel targets [20]. Endothelin is an endogenous vasoregulatory peptide having 3 isoforms: ET-1, ET-2, and ET-3 [21]. ET-1 is the main member of the endothelin

family synthesized by almost all cell types, with significant amounts produced by neurons, astrocytes, and glial cells in the central nervous system [22, 23]. Endothelin peptides interact with ET_A and ET_B receptors to bring about various effects throughout the body. ET_A receptors are found on smooth muscle cells mediating a contractile effect, while ET_B receptors are confined to vascular endothelial cells as well as neurons, astrocytes, and glial cells in the brain [24, 25]. ET_A and ET_B receptors are present throughout the brain, with ET_B receptors being more prominent. The expression of ET_B receptors is further enhanced among astrocytes following brain injury, along with increased production of ET in damaged neurons. The search for novel neuroprotective therapeutics is ongoing, with the neural ET system as a potential target given its possible roles in the pathophysiological responses of brain damage [20]. Studies have shown that ET_B receptors have a significant role in neuronal survival and proliferation under pathological conditions by releasing angiogenic and neurotrophic factors [23, 26]. Additionally, it was demonstrated that ET levels were elevated in rats following cerebral ischemia. The increased ET levels coinciding with a decreased regional blood flow in ischemic areas of the brain following experimental stroke initiated considerable discussion regarding its possible involvement in the pathophysiology of ischemic stroke. This led to the investigation of several ET antagonists for the treatment of focal ischemic stroke [27]. Furthermore, it was discovered that ET_B receptor blockade leads to exacerbated damage of the CNS following cerebral ischemia, and ET_B receptor deficiency leads to a poorer outcome following cerebral ischemia, suggesting that ET_B receptors play a significant role in the protection of brain tissue following an ischemic stroke [19, 28, 29]. This evidence steered us to investigate the role of ET_B receptors in a focal stroke model by selectively activating these receptors with sovateltide, a highly selective ET_B receptor agonist.

Sovateltide (PMZ-1620, IRL-1620) is a highly selective ET_B receptor agonist and a synthetic analog of ET-1. We and others have conducted studies to determine the effects of sovateltide upon its interaction with neural ET_B receptors and have found that it enhances angiogenesis and neurogenesis and promotes neural repair and regeneration [30-41]. In a rat model of ischemic stroke, sovateltide was found to be neuroprotective as well as enhance angiogenic and neurogenic remodeling. In addition, Sovateltide significantly improved survival and reduced neurological and motor function deficits while effectively decreasing infarct volume, edema, and oxidative stress [34-36].

Sovateltide was also found to be safe and well tolerated in healthy human volunteers in phase I clinical trial (CTRI/2016/11/007509). A phase II study was also conducted in patients with acute ischemic stroke, where sovateltide demonstrated significant improvement when compared to the standard of care (CTRI/2017/11/010654, NCT04046484). Our recent phase III study conducted in patients of acute ischemic stroke demonstrated improved favorable functional and neurological outcomes at 3 months compared to standard of care (CTRI/2019/09/021373, NCT04047563). Additionally, phase II clinical trials are ongoing for Alzheimer's disease (CTRI/2017/12/016394) and spinal cord injury (CTRI/2018/12/016667). Toxicological studies have also been completed

in mice, rats, and dogs, with results indicating that sovateltide is well tolerated and has a good safety margin.

A prospective, multi-centric, randomized, double-blind, parallel, saline-controlled phase II clinical study (CTRI/2017/11/010654; NCT04046484) was conducted in forty patients with cerebral ischemic stroke, of which 36 completed 90-day follow-up. All patients received standard treatment and were randomly assigned to either control (saline; n=18; 11 male and 7 female) or PMZ-1620 group (n=18; 15 male and 3 female). Clinical outcome parameters such as the National Institute of Health Stroke Scale (NIHSS), Modified Rankin Scale (mRS), Barthel Index (BI), EuroQol Index, and Stroke-Specific Quality-of-life (SS-QoL) Index for cerebral stroke were determined. Standard treatment and care were provided to all the patients. All the patients received saline or PMZ-1620 within 24 hours of the onset of the stroke. The number of patients that received saline within 20 hours of the onset of stroke was 14 out of 18, whereas those that received PMZ-1620 were 10 out of 18. NIHSS on day 6, when compared to day 1 showed a significance value of $P=0.0495$ in saline and $P<0.0001$ in PMZ-1620 cohorts. Similarly, mRS on day 6 when compared to that observed on day 1 showed a significance value of $P=0.0859$ in the saline cohort, while in the PMZ-1620 cohort, it was $P<0.0001$. BI on day 6, when compared to day 1 was similar ($P=0.3948$) in the saline cohort, whereas it significantly ($P<0.0001$) improved in the PMZ-1620 cohort. PMZ-1620 produced a significantly quicker recovery in patients with acute cerebral ischemic stroke. In saline 12.50% while in PMZ-1620 cohort 87.50% (odds ratio 10.18; 95% CI 10.18 to 120.50; $P=0.0201$) patients showed an improvement of ≥ 6 in NIHSS from baseline (enrollment time) at day 6 of treatment, while at 90 day 43.48% in saline and 56.52% patients in PMZ-1620 cohort (odds ratio 2.275; 95% CI 0.5785 to 8.310; $P=0.2714$) showed an improvement of ≥ 6 in NIHSS. An improvement of ≥ 2 in mRS was observed in 39.13% and 60.87% of patients in saline and PMZ-1620 cohorts, respectively, at 30 day of treatment. Similar improvement in mRS was observed at 90 day of treatment (odds ratio 5.250; 95% CI 1.00 to 27.63; $P=0.0519$). An improvement of ≥ 40 in BI in the saline cohort was 30.00%, while it was 70.00% of patients in the PMZ-1620 cohort at 30 days of treatment (odds ratio 5.833; 95% CI 1.217 to 21.26; $P=0.0172$). At 90 days of treatment, 36.00% and 64.00% of patients in control and PMZ-1620 cohorts showed an improvement of ≥ 40 in BI (odds ratio 12.44; 95% CI 1.502 to 147.10; $P=0.0112$). A change of 80 or more in the EuroQol scale was 0% of patients in saline and 100% of patients in the PMZ-1620 cohort at 60 days (odds ratio ∞ ; 95% CI 0.9391 to ∞ ; $P=0.0389$) and at 90 days (odds ratio ∞ ; 95% CI 1.00 to ∞ ; $P=0.0332$). Similarly, a change of 150 or more in the SS-QOL scale was 22.22% of patients in saline and 77.78% of patients in the PMZ-1620 cohort at 60 days (odds ratio 4.773; 95% CI 0.9278 to 25.06; $P=0.0665$) and at 90 days (odds ratio 5.250; 95% CI 1.00 to 27.63; $P=0.0519$). The number of patients with 100% recovery achieving NIHSS score of 0 ($P=0.04791$), mRS of 0 ($P=0.1193$), and BI of 100 ($P=0.02795$) was more in the PMZ-1620 cohort compared to saline. PMZ-1620 treatment did not have any effect on hemodynamic, biochemical, or hematological parameters. No incidence of drug-related adverse events was reported. No mortality and no incidence of recurrent ischemic stroke occurred in any of the enrolled patients.

A prospective, multicentric, randomized phase III study was conducted in acute cerebral ischemic stroke patients aged 18 years through 78 years. Clinical Trials Registry (CTRI/2019/09/021373; NCT04047563). Patients with radiologic confirmation of ischemic stroke could be enrolled if presenting up to 24 hours after the onset of symptoms and with a modified Rankin Scale (mRS) score of 3 to 4 and NIHSS score of greater than 5. Patients with intracranial hemorrhage and those receiving endovascular therapy were excluded. Patients were randomized in a 1:1 ratio, 80 patients to the sovateltide group and 78 patients to the control (saline) group. Every patient received the standard of care (SOC) and was followed for 90 days. The study drug (normal saline or sovateltide (0.3 µg/kg) was administered in three doses administered as an intravenous bolus over 1 minute at an interval of 3 hours $\pm$ 1 hour on day 1, day 3, and day 6 (total sovateltide dose of 0.9 µg/kg/day). The primary objectives were to determine the neurological outcome based on the National Institute of Health Stroke Scale (NIHSS), modified Rankin Scale (mRS), and Barthel Index (BI) scales from day 1 through day 90. In addition, quality-of-Life was measured by EuroQol EQ-5D (EQ-5D) and stroke-specific quality-of-life (SS-QoL) at 60 days and 90 days of treatment. A total of 158 patients with acute cerebral ischemic stroke were enrolled in this study, of which 137 completed 90-day follow-up. Patients received saline (n=78; 48 male and 30 female) or sovateltide (n=79; 53 male and 26 female) within 24 hours of the onset of stroke. Patients received the investigational drug at 17.31 ± 0.74 and 17.48 ± 0.65 hours (mean $\pm$ SEM, $p=0.8609$) of stroke onset in control and sovateltide cohorts, respectively. The baseline characteristics and SOC in both cohorts were similar. Alberta Stroke Program Early Computed Tomographic Score (ASPECTS) mean value was similar in the control (7.44) and sovateltide (7.61) groups, indicating that the extent of infarction was similar in both the groups. Sovateltide treatment resulted in significantly improved neurological outcomes compared to the control cohort. The mRS score is the most widely used primary outcome measure in trials for acute stroke interventions. The median mRS score at 90 days was significantly ($p=0.008$) lower in the sovateltide group (1.00, 95% CI 1.179 to 1.776) than in the control group (2.00, 95% CI 1.753 to 2.447). The distribution of scores on the mRS at 90 days in the intention-to-treat population shows an ordinal shift across the range that highly favors sovateltide therapy. An improvement of ≥ 2 points on the mRS was observed in 52.86% and 76.12% of patients in the saline and sovateltide groups, respectively ($p=0.0045$). A mean improvement in the NIHSS was 0.9610 (95% CI -0.1244 to 2.046, $p=0.1217$) and 1.267 (95% CI 0.3176 to 2.216, $p=0.0017$) in the control and sovateltide groups, respectively on day 2 from the day of randomization. Similarly, on day 3, an improvement in NIHSS from the day of randomization was 1.981 (95% CI 0.8843 to 3.077) in the control group and 3.050 (95% CI 2.086 to 4.014) in the sovateltide group. At day 90, an improvement in the NIHSS of ≥ 6 points was seen in 64.29% and 82.09% of the saline and sovateltide groups, respectively ($p=0.0190$). At day 90, BI score improvement of ≥ 40 points was seen in 76.12% of patients in the sovateltide group vs. 61.43% in the saline group ($p=0.0639$). At 90 days mRS (difference between means -0.6224 ± 0.2303 ; 95% CI -1.078 to -0.1670, $p=0.0078$) and NIHSS (difference between means -1.586 ± 0.5126 ; 95% CI -2.600 to -0.5727, $p=0.0024$) were significantly lower, while BI (difference between means 10.19 ± 3.950 ; 95% CI 2.375 to 18.00, $p=0.011$) was significantly higher in the sovateltide compared to the control group. The number of patients with complete recovery (defined

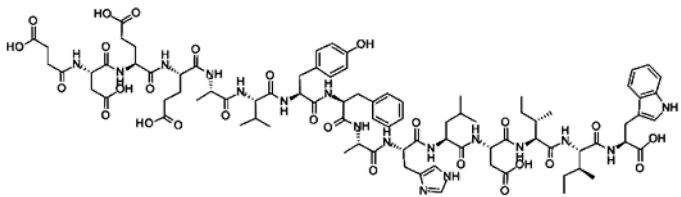
as NIHSS score of 0, mRS of 0, and BI of 100) was numerically greater in the sovateltide group than in the saline group. Sovateltide treatment improved Quality-of-Life as measured by EQ-5D ($p=0.0055$) and SSQoL (0.0673) compared to the control group on day 90. Sovateltide increased the frequency of favorable neurological outcomes at 3 months.

It was concluded that sovateltide is an effective agent for treating acute cerebral ischemic stroke, having a highly favorable ordinal shift in the mRS across the range. In addition, it is safe, well-tolerated, and produced a statistically significant and clinically meaningful improvement in neurological outcomes in acute cerebral ischemic stroke patients 90 days post-treatment.

Our clinical phase II and III results indicate that sovateltide is a first-in-class neuronal progenitor cell therapeutic that promotes quick recovery and significantly improves neurological outcomes in cerebral ischemic stroke patients. With this convincing evidence, we plan to conduct a multicentric, randomized, double-blind, parallel, placebo-controlled phase III clinical study in the United States, Canada, United Kingdom, and Europe (the demographics and standard of treatment being similar in these countries) to further assess the safety and efficacy of sovateltide in patients with acute cerebral ischemic stroke.

2.2 Name and Description of Investigational Product(s)

Active Pharmaceutical Ingredient

Synonym	PMZ-1620; IRL-1620; sovateltide
International nonproprietary name	Sovateltide
IUPAC/Chemical name	(2S,5S,8S,11S,14S,17S,20S,23S,26S,29S,32S,35S,38S,41S)-17-((1H-imidazol-5-yl)methyl)-2-((1H-indol-3-yl)methyl)-23-benzyl-5,8-di-sec-butyl-35,38-bis(2-carboxyethyl)-11,41-bis(carboxymethyl)-26-(4-hydroxybenzyl)-14-isobutyl-29-isopropyl-20,32-dimethyl-4,7,10,13,16,19,22,25,28,31,34,37,40,43-tetradeca-oxo-3,6,9,12,15,18,21,24,27,30,33,36,39,42-tetradecaazahexatetracontane-1,46-dioic acid
Structural Formula	
Amino acid sequence	Suc-Asp-Glu-Glu-Ala-Val-Tyr-Phe-Ala-His-Leu-Asp-Ile-Ile-Trp
Physical appearance	White/off-white lyophilized powder

Counter ion	Trifluoroacetate
Solubility	Soluble to 0.70 mg/ml in water 1.0 mg/ml in 5 % NH ₄ OH in water
Moisture content	NMT 5 %
λ_{max}	215 nm
Elemental Analysis	C, 56.72; H, 6.48; N, 13.08; O, 23.72
Molecular formula	C ₈₆ H ₁₁₇ N ₁₇ O ₂₇
Molecular weight	1820.9 ± 1
Chemical purity	>98.5 % (HPLC)
CAS#	142569-99-1
Manufactured by	Piramal Pharma Limited C-43, MIDC, TTC Industrial Area, Turbhe, Off Thane Belapur Road, Navi Mumbai District: Thane - 400 703, India.
Storage	Store at -20 °C
Biochemical/physiological actions	Highly selective ET _B receptor agonist (K _i values are 0.016 and 1900 nM at ET _B and ET _A receptors, respectively).
Intended usage of the product	For the treatment of cerebral ischemia.

Investigational Product:

The active ingredient sovateltide (IRL-1620) was formulated as a sterile Lyophilized Injection for intravenous use.

Investigational Product Name : IRL-1620 For Injection 30µg (sovateltide)

Composition	: Each vial of IRL-1620 For Injection contains: Sovateltide (IRL-1620)..... 30 µg Excipients..... q.s.
Product Description	: White/off-white lyophilized cake in 5 mL amber USP Type I glass vial.
Active Ingredient	: Sovateltide (IRL-1620)
Product Strength	: Each 5 mL vial contains 30 µg of sovateltide (IRL-1620)
Dosage Form	: Sterile intravenous injection (Lyophilized)
Therapeutic Indication	: For the treatment of Cerebral ischemia
Route of Administration	: Intravenous
Method of Administration	: To be administered as an intravenous bolus dose over 1 to 2 minutes
Dose	: Three doses of sovateltide (each dose of 0.3 µg/kg body weight) are to be administered as an IV bolus over one to two minutes, at an interval of 3 hours ± 1 hour on day 1. The dose will be repeated on day 3 and day 6 post-randomization. The day 6 dosing has to be performed within a time window of ± 1 day.
Direction for Use	: Dissolve the contents of the vial in 5 mL of Normal Saline (0.9% Sodium Chloride Injection). When reconstituted as directed, each vial contains a sterile solution equivalent to 6.0 µg/mL of sovateltide (30 µg/5 mL).

- Stability of Reconstituted Product** : Reconstitute the product just before use. The reconstituted Sovateltide for injection 30µg is stable for 48 hours when stored at 2 to 8 °C
- Storage Conditions** : Store between 2-8 °C. Protect from light and moisture.
- Shelf-life** : Stable for 2 years at recommended storage conditions
- Undesirable Effects** : Phase I study demonstrated that sovateltide is safe and was well tolerated when administered in multiple ascending doses in healthy subjects. A dose of 0.6 µg/kg (Maximum Tolerated Dose) did not produce any adverse events in healthy subjects. The adverse events reported at 0.9 µg/kg dose were uneasiness, sweating, abdominal discomfort, and vomiting. These events were mild, caused transient discomfort to the subject, and were resolved without any sequelae.
- Packaging Information** : Sterile 5 mL, USP Type I tubular amber glass vial.
- Manufactured By** : Pharmazz India Private Limited, At Steril Gene Life Sciences Pvt. Ltd., No. 45, Mangalam Main Road, Mangalam Village, Villianur Commune, Puducherry-605110 INDIA.

2.3 A Summary of Findings from Non-clinical Studies that Potentially Have Clinical Significance

A comprehensive battery of nonclinical studies has been performed with sovateltide. Please refer to the Investigator's Brochure for a complete summary of nonclinical data for sovateltide.

Nonclinical Pharmacology

Specific Pharmacological Actions (Cerebral Ischemia)

We studied the effect of Sovateltide (IRL-1620, PMZ-1620) on levels of brain ET receptors in middle cerebral artery occluded (MCAO) rats. ET_B receptors are present in large numbers in the CNS and play a key role in its development. We have demonstrated that ET_B receptors in the brain are over-expressed at the time of birth, and their expression decreases with the maturity of the brain [42, 43]. We pharmacologically activated a regenerative response in the ischemic brain by stimulating ET_B receptors. Stimulation of ET_B receptors by Sovateltide has been shown to provide a neuroprotective effect in MCAO rats [34-36]. ET_A receptors are increased in the infarcted hemisphere at 24 hours post-ischemia and return to normal levels by one week. Conversely, a significant increase in ET_B receptor expression occurs after 1 week only in the infarcted hemisphere of rats treated with Sovateltide [35]. We determined the effect of Sovateltide on neurological deficit following focal cerebral ischemia in rats. Twenty-four hours after MCAO, there was a significantly ($P < 0.001$) higher neurological deficit and poor motor function than in sham-operated rats, indicative of neurological impairment following induction of cerebral ischemia. Animals treated with Sovateltide showed significant improvement in all neurological and motor function tests compared with vehicle-treated. Treatment with Sovateltide effectively reduced oxidative stress as measured by decreased levels of malondialdehyde (MDA) and increased levels of reduced glutathione (GSH) and superoxide dismutase (SOD) as compared with the vehicle-treated group. Increased MDA levels and decreased levels of antioxidants GSH and SOD are all hallmarks of oxygen-free radical generation occurring as part of the ischemic cascade [44]. In a longer-term study, cerebral ischemia resulted in a distinct loss of motor coordination. Animals treated with Sovateltide showed minimal deficit on day one following occlusion and improved over 7 days [34, 35]. We determined the effect of Sovateltide on the binding characteristics of ET_B receptors in the brain following focal cerebral ischemia in rats. Binding characteristics (K_d and B_{max}) were not altered 24 hours post-MCAO. However, a significant decrease in K_d values of ET_B receptor binding in both left and right hemispheres was observed 7 days post-MCAO. The decrease in K_d in the right (ischemic) hemisphere was significantly ($P < 0.001$) greater compared to the left (non-ischemic) hemisphere. B_{max} was increased in both the left and right hemispheres, with the right hemisphere showing a significantly ($P < 0.001$) greater increase than the left hemisphere. It was concluded that an increase in the density and affinity of ET_B receptors on the 7th day of cerebral ischemia could be an attempt to provide neuroprotection of the ischemic brain.

We determined the effect of Sovateltide on infarct volume in (MCAO) rats. MCAO for 7 days resulted in an infarct volume of $177.06 \pm 13.21 \text{ mm}^3$ in vehicle-treated rats. Administration of Sovateltide significantly reduced infarct volume ($54.06 \pm 14.12 \text{ mm}^3$; $P < 0.05$) as compared with the vehicle. We determined the effect of Sovateltide on the 7-day survival of rats following focal cerebral ischemia. It was found that MCAO rats in the vehicle-treated group presented with 38% mortality by the 7th day. On the other hand, MCAO rats treated with Sovateltide showed no mortality throughout the 7 days [35].

Our preliminary results and supporting literature have prompted us to investigate the mechanism involved in the neuroprotective and neurorestorative effects of stimulating ET_B receptors in rats with cerebral ischemia. Angiogenesis and neurogenesis are the driving forces for neurovascular remodeling, which is essential post-stroke to restore normal brain function [45]. Activation of ET_B receptors with Sovateltide is known to cause cerebral vasodilatation and increased blood flow through the release of nitric oxide (NO) [46-49]. Recent studies have demonstrated that endothelial NO stimulation and increased cerebral blood flow via pharmacological means enhance angiogenesis and improve recovery following a stroke in rats [50, 51]. Along these lines, the ET_B receptor has been shown to enhance the formation of new blood vessels through eNOS [52]. An increase in the number of blood vessels in the hypoxic brain provides the tissue with a better supply of oxygen and nutrients, thus enhancing the survival of the neuronal tissue. We studied the effect of Sovateltide on angiogenesis and neurogenesis following cerebral ischemia in rats. We found that Sovateltide, administered on the day of infarct, is neuroprotective and enhances angiogenic and neurogenic remodeling following cerebral ischemia [36]. Previous studies using an ET_B receptor-deficient model have indicated that this receptor promotes neuronal survival and decreases apoptosis in the hippocampus, dentate gyrus, and olfactory epithelium [53-55]. Thus, Sovateltide administered intravenously was found to be highly effective in preventing damage following stroke and aids in the neurovascular remodeling of the ischemic brain by angiogenesis and neurogenesis [36].

Type 2 Diabetes Mellitus (T2DM) is associated with cardiovascular diseases, such as cerebral ischemic stroke. We conducted a study to determine if Sovateltide is effective in diabetic rats with cerebral ischemic stroke. It was found that survival rates for the Sovateltide group were significantly higher, and the Sovateltide group performed significantly better on neurological evaluations in both non-diabetic and diabetic rats than the vehicle group. In addition, in both non-diabetic and diabetic rats, a significant reduction in infarct volume was seen in the Sovateltide group. Sovateltide (5 µg/kg, intravenous), administered at 4, 6, and 8 hr on days 0, 3, and 6 post-MCAO, is an effective treatment for cerebral ischemia in T2DM, significantly reducing infarct volume, improving neurological and motor function, and enhancing angiogenesis.

We also studied the effect of Sovateltide in a pediatric model of cerebral ischemic stroke [56]. We reported that Sovateltide significantly reduced the neurological and motor deficit and infarct volume while increasing cerebral blood flow and thus promoting recovery from stroke. These findings are consistent with those in adult rats [36]. We observed that cerebral blood flow (CBF)

significantly improved in Sovateltide-treated animals, increased phospho-Akt, and decreased in Bad level 7 h post-occlusion. Anti-apoptotic Bcl-2 expression was decreased, while pro-apoptotic Bax expression was increased in vehicle-treated MCAO rats; these changes were attenuated by Sovateltide. Mitochondrial membrane-bound Bax intensity significantly decreased in Sovateltide-treated MCAO rats. Sovateltide treatment reduced the number of TUNEL-positive cells compared to the vehicle at 24 h and day 7 post-MCAO. The results demonstrated that, Sovateltide is neuroprotective and attenuates neural damage following cerebral ischemia in rats by increasing CBF and reducing apoptosis. In another study we reported, stimulation of brain ET_B receptors with Sovateltide promotes NeuroD1-mediated neurogenesis and may be responsible for improved neurological functions [57].

Studies Supporting Present Indication Neurodegenerative Diseases

We also studied the effect of Sovateltide on other neurodegenerative disorders like Alzheimer's disease (AD), which is a progressive brain disorder leading to impairment of learning and memory. We determine the effect of Sovateltide; in beta-amyloid β (A β) induced oxidative stress and cognitive impairment in normal and diabetic rats: Amyloid β (A β) induced oxidative stress has been implicated in the initiation and progression of AD. Sovateltide treatment of beta-amyloid β (A β) treated rats improved learning and memory on water maze tasks and reduced oxidative stress parameters. This finding indicates that ET_B receptor stimulation prevents cognitive impairment and may be useful in neurodegenerative diseases [58]. We explored PI3K-Akt, MAPK, and Notch signaling pathways in the mechanism of action of Sovateltide to enhance functional recovery in AD. PI3K, pAkt, Bcl2, and Synapsin expression significantly increased in Sovateltide treated rats. Bax and NOTCH 1 expression was significantly decreased. Sovateltide increases cell viability *in vitro* and *ex vivo* by possibly reversing mitochondrial dysfunction. Sovateltide initiates a signaling cascade resulting in the recruitment of PI3K and subsequent activation of Akt via phosphorylation.

Second messengers are involved in the convergence of several pathways in response to activation by Sovateltide, resulting in an overall neuroprotective mechanism and neurogenesis. A recent discovery of niches of endogenous multipotent stem cells within the adult spinal cord has shed light on stem cell therapies for CNS damage and neurodegenerative diseases. We demonstrated the presence of neuronal stem cells in rat normal spinal cord tissue by showing the presence of neural stem cell activator proteins, such as SYN 1, Mushashi, Nestin, and SOX2. We also confirmed the presence of neural progenitor cells and neuroblasts through NeuroD1 and nestin staining. Sovateltide-treated cells showed more neurons and cell nuclei than untreated, indicating the possibility of neuronal multiplication with Sovateltide treatment. We studied the effect of Sovateltide in a rat spinal cord injury (SCI) model to study the effect of Sovateltide on functional recovery of SCI. The selective stimulation of ET_B receptors with the intravenous treatment of Sovateltide significantly improves hindlimb locomotor function following SCI in rats. The substantial recovery of hindlimb function, improving motor function, and reduction in time to improvement indicate that selective stimulation of ET_B receptors with Sovateltide may be a novel therapeutic option for treating SCI.

General Pharmacological Actions

The effects of sovateltide on vital systems such as cardiovascular, respiratory, and central nervous systems were studied through various non-clinical studies in rats. A summary of these studies is given below:

Cardiovascular Effects

The studies were conducted in urethane anesthetized rats to determine the cardiovascular effects of acute repeated (IV) administration of sovateltide. It was found that sovateltide did not significantly affect heart rate but produced a transient fall in BP. Repeated administration of a low dose (1.6 µg/kg, i.v.) of sovateltide produced a fall in mean arterial pressure (MAP) but no tachyphylaxis. However, a fall in MAP was observed when repeated administration of high-dose sovateltide (5.0 µg/kg, i.v) was given [46].

Cardiovascular function assessment of sovateltide was also assessed in telemetry-instrumented conscious beagle dogs following single intravenous administration (**Study No.:** VLL/0721/G/S002). Four groups (G1, G2, G3, and G4) comprising of 1 male beagle dog per group were administered a single intravenous dose (slow bolus) of either the vehicle (phosphate buffer saline, pH 7.4) or Sovateltide in phosphate buffer saline (PBS) at dose levels of 0.5, 1 and 2 µg/kg bw respectively. All animals survived throughout the study period. No abnormal clinical signs were observed during daily observations in any of the animals treated with the vehicle. Animals treated at 0.5 µg/kg revealed transient dullness, nausea, and dyspnoea on respective treatment days. Animals treated at 1 µg/kg revealed transient dullness, emesis, in-coordination, anxiety, and dyspnoea on respective treatment days. Animals treated at 2 µg/kg revealed transient dullness, emesis, in-coordination, anxiety, dyspnoea, and nausea on respective treatment days. The signs in all tested dose levels varied in their frequency of persistence from 5 minutes up to a maximum of 2 hours post-dose. All animals treated with the test item (0.5, 1, and 2 µg/kg bw) as well as the vehicle, displayed normal PR, RR, QRS, and QT/QTcF intervals. No significant changes attributable to the treatment were observed in the heart rate of animals treated at the dose levels of 0.5, 1 and 2 µg/kg bw, compared to those treated with vehicle alone. All animals exhibited normal systolic and diastolic blood pressure at all recorded time points. Intravenous administration of Sovateltide up to 2 µg/kg bw to conscious, freely moving Beagle dogs did not induce any potential effects in any of the cardiovascular parameters tested. Based on findings, the No Observed Effect Level (NOEL) of Sovateltide for the tested cardiovascular parameters, following single intravenous administration in male beagle dogs, is 2 µg/kg bw, under the tested conditions.

Central Nervous System Effects

The effect of sovateltide on neurological functions following permanent MCAO in rats was studied. The rats were assessed for neurological deficits and motor function. Animals treated with sovateltide showed significant improvement in all neurological and motor function tests compared to the vehicle-treated MCAO group [34].

Safety evaluation of the central nervous system after single-dose administration of sovateltide by intravenous route was assessed in male Wistar rats (**Study No.:** G21135). Thirty-two male rats were assigned into four groups of eight animals per group, viz., Vehicle control (G1), Low dose (G2), Mid Dose (G3), and High dose (G4). The test item formulations were prepared in phosphate-buffered saline (1X PBS; pH 7.4). The vehicle alone was administered as a single dose to the vehicle control (G1) group animals. Test item formulations were administered as a single dose at dose levels of 5.0 (G2), 10.0 (G3), and 15.0 (G4) µg/kg to respective groups at a dose volume of 2.0 mL/kg body weight. After administration of the test item, no mortality/morbidity was observed in vehicle control and test item treatment groups during the observation period. Test item-related adverse effects were not observed on functional observation battery test/neurological examination in test item treatment group animals at post-dose observations on day 1 post-dose at 5 min (±2 min), 30 minutes (±10 min), 1 hour (±10 min), 2 hour (±10 min) and 4 hour (±10 min)]. However, the statistically significant decrease observed in physiological measurement i.e., rectal temperature (°C) in mid (G3) and high (G4) dose at 5 min and at 30 min in high dose (G4) were observed when compared with the vehicle control (G1) group. The decrease in rectal temperature was found to be marginal (-2.1%), non-dose dependent, and not consistent throughout the observation period, and values were found to be within the normal range. Hence, these changes were considered incidental. A decrease in mobility/locomotor activity was observed in test item-treated groups and vehicle control group animals. However, the other related measured/observed parameters (locomotor and rearing activity readings of the actinometer) were found to be comparable to the vehicle control group. During the air righting reflex, a slightly uncoordinated landing (i.e., two-legged or unsteady landing) of one animal from the high-dose group (G4) was observed. This observation was considered as incidental as it was inconsistent, and the other FOB parameters examined were found to be comparable to the vehicle control group. Considering these findings, the single dose administration of sovateltide up to 15.0 µg/kg by intravenous route in male Wistar rats was found to have no adverse pharmacodynamic effects on the central nervous system function evaluated.

Respiratory Effects

A study was conducted to assess the effects of sovateltide at (IV) doses of 5, 10, and 20 µg/kg on respiration rates and tidal volumes in rats. Male Sprague Dawley rats (8 rats per group) were assigned to each treatment and were administered either a single (IV) dose of 5, 10, or 20 µg/kg sovateltide, vehicle, or 20 mg/kg morphine (reference substance). Although the tidal volume was decreased, sovateltide did not have any effect on respiration rate in rats [59].

Safety evaluation of the respiratory system after a single dose administration of sovateltide by intravenous route was assessed in male Wistar rats (**Study No:** G21107). Thirty-two male rats were distributed into four groups of eight animals per group, viz., vehicle control (G1, 0.0 µg/kg b.w.), low dose (G2, 5.0 µg/kg b.w.), mid-dose (G3, 10.0 µg/kg b.w.) and high dose (G4, 15.0 µg/kg b.w.). Phosphate buffered saline (1X PBS, pH 7.45) was used as a vehicle for formulation preparation. Test item formulations/vehicle alone were administered as a single dose at a dose volume of 2 mL/kg. After administration of the test item, there was no mortality/morbidity

observed in any animals till the monitoring period. After administration (~5 min), isolated incidences of tachypnea were observed in animals treated with a high dose. Respiratory parameters were monitored in unrestrained rats using whole body plethysmograph chambers prior to dosing (baseline) for 1 hour, 0 to 1 hour, 1 to 2 hours, 2 to 3 hours, and 3 to 4 hours post-dose in all the animals. Statistically significant increase in respiratory rate, peak expiratory flow, minute volume, enhanced pause, and mid-tidal expiratory flow up to 1 hour post-dose was observed at 15 µg/kg bw compared to the control group. A statistically significant decrease compared to the vehicle control group was observed in inspiratory time at 15.0 µg/kg bw at only 1 to 2 hours post-dose, expiratory time at 10.0 µg/kg bw at 0 to 1 hour post-dose and at 15.0 µg/kg bw up to 2 hours post-dose, relaxation time at 10.0 µg/kg bw up to 1 hour and at 15.0 µg/kg bw up to 2 hours post-dose, end-inspiratory pause at 5.0 and 15 µg/kg bw treated groups up to 1 hour post-dose, and rejection index at 15.0 µg/kg bw treated group at 0 to 1 hour post-dose. Peak inspiratory flow, tidal volume, and end-expiratory pause were found comparable to the vehicle control group after test item administration at 5.0, 10.0, and 15.0 µg/kg bw up to 4 hours post-dose. Considering the above, it is concluded that a dose of 10.0 µg/kg bw of sovateltide has no potential undesirable pharmacological effects on the respiratory system under the experimental conditions employed.

Pharmacokinetics

Toxicokinetic evaluation of PMZ-1620 in rats

The Toxicokinetics (TK) profile of sovateltide was determined in three satellite groups of 12 male and 12 female rats, each that received a slow (IV) bolus dose of sovateltide once weekly for four weeks at doses of 5, 10, and 15 µg/kg. C_{max} values were 41.2, 137, and 225 ng/mL in males and 43.0, 126, and 218 ng/mL in females following nominal sovateltide doses of 5.0, 10.0, and 15.0 µg/kg, respectively. The terminal disposition $T_{1/2}$ was similar across doses and gender and ranged from 4.7 to 7.9 minutes. No apparent gender differences in TK of sovateltide were seen. It was concluded that there is dose linearity in rats, using both C_{max} and AUC analysis. Blood clearance was rapid as shown by CL and $T_{1/2}$ [60].

Single and repeated dose 4 weeks toxicokinetics of sovateltide was determined in Wistar rats. Four dose groups with 3 male and 3 female rats in G1 (0.0 µg/kg) and 9 male and 9 female rats in G2 (3.0 µg/kg), G3 (6.0 µg/kg), and G4 (12.0 µg/kg) were used. C_{max} values on day 1 were 0.00, 37.30, 88.60, and 209.00 ng/mL in males and 0.00, 40.00, 77.10, and 174.00 ng/mL in females following single intravenous sovateltide doses of 0, 3, 6, and 12 µg/kg respectively. C_{max} values on day 26 were 0.00, 47.50, 95.10, and 173.00 ng/mL in males and 0.00, 33.70, 80.90, 183.00 ng/mL in females following Repeated intravenous sovateltide doses of 0, 3, 6, and 12 µg/kg t.i.d x 3/week for 4 weeks respectively. $T_{1/2}$ was similar across doses and gender and ranged from 0.06 to 0.09 hours (3.6 to 5.4 min). AUC_{last} was similar in both genders at respective dose levels.

Pharmacokinetic Analysis in Beagle Dogs

Pharmacokinetic analysis was conducted for 20 male and 20 female beagle dogs divided into four dose level groups receiving 3 doses of the vehicle only or sovateltide (0.5 µg/kg, 1 µg/kg, or 2

µg/kg) by slow intravenous bolus 3 to 4 hours apart. This regimen was repeated three times per week for 4 weeks. Sovateltide plasma concentration declined rapidly in a mono-exponential manner, with plasma concentration generally falling below the LLOQ by 30 minutes after injection. Highest mean sovateltide C_{max} across the 6 doses on the two study days was 3.19 ng/mL in females and 3.54 ng/mL in males following 0.5 µg/kg, 6.68 ng/mL in females and 6.28 ng/mL in males following 1 µg/kg, and 15.03 ng/mL in females and 13.05 ng/mL in males following 2 µg/kg. The $t_{1/2-\lambda_z}$ was extremely rapid, ranging from 3.3 to 5.9 min in female dogs and 3.1 to 5.5 min in male dogs. Mean C_{max} , AUC_{0-15} , and $AUC_{0-\infty}$ following 0.5 µg/kg or 1 µg/kg were similar between female and male dogs across study days and dose per study day. Mean sovateltide peak and total systemic exposure across study days and dose per study day were slightly greater following 2 µg/kg in the female compared to male dogs. C_{max} , AUC_{0-15} , and $AUC_{0-\infty}$ increased nearly proportional to the increase in dose from 0.5 µg/kg to 1 µg/kg and 1 µg/kg to 2 µg/kg in male and female dogs and from 0.5 µg/kg to 2 µg/kg in male dogs. CL was comparable between male and female dogs following 0.5 µg/kg and 1 µg/kg, CL was 18% lower in females compared to male dogs following 2 µg/kg, CL was comparable in male dogs across the three dose groups, and CL was lower in female dogs at 2 µg/kg versus 0.5 µg/kg.

Drug Metabolism Studies

Drug metabolism studies were conducted to test Sovateltide in solution properties, in vitro absorption, and in vitro metabolism assays. Sovateltide showed 83-85% protein bound in the protein binding assay. In transporter inhibition assays, Sovateltide showed 57.9% and 56.8% inhibition on OATP1B1 and OATP1B3 but did not show significant inhibition on the other drug transporters tested and is not a substrate of P-gp or BCRP. Additionally, sovateltide has low permeability across cell membranes. Sovateltide also revealed low hepatic intrinsic clearance and no evidence of CYP inhibition or CYP induction of CYP1A2, CYP2B6, or CYP3A4.

Toxicology

Toxicity studies on Sovateltide (IRL-1620, PMZ-1620) were conducted. Female Swiss Webster mice were administered with a single (IV) dose of Sovateltide at the dose levels of 30, 60, 120, and 240 µg/kg showed no mortality. Single and multiple (t.i.d) dose administration of Sovateltide within 24 hours by intravenous route in female Wistar rats revealed maximum tolerated dose (MTD) is 35 µg/kg body weight, and on the administration of three doses per day at approximately three-hour intervals is 30 µg/kg body weight (t.i.d) or 90 µg/kg body weight/day. Repeated intravenous administration of Sovateltide in mice at the dose levels of 1.0, 2.5, and 6.3 µg/kg for 30 days showed no mortality at any dose levels [61]. The single daily intravenous administration of Sovateltide to Wistar rats for 14 days at dose levels of 0, 1, 5, and 15 µg/kg body weight did not affect survival parameters. No adverse effects were observed at 15 µg/kg/day. Intravenous administration of sovateltide thrice a day at approximately three-hour intervals for three days a week in Wistar rats for 28 days at dose levels 0, 3, 6, and 12 µg/kg body weight revealed treatment-related histopathological findings in testes and epididymides of high (12.0 µg/kg body weight) and mid-dose (6.0 µg/kg body weight) groups of male rats only. No adverse effects were observed at

3.0 µg/kg body weight in male rats and at 12.0 µg/kg body weight in female rats. Daily intravenous administration of Sovateltide in Beagle dogs up to 5.0 µg/kg for 14 days showed no adverse effects. Intravenous Sovateltide was administered to Beagle dogs consecutively three times a day every three hours three times a week for 28 days at dose levels 0, 0.5, 1.0, and 2.0 µg/kg body weight. Soft stool was seen in at least 1 animal in all Sovateltide groups, and diarrhea was observed in 1 male in the 2.0 µg/kg x 3 Sovateltide group. Vomiting, soft stool, and diarrhea did not affect Sovateltide-treated animals because these signs were observed only on dosing days, and body weights and food consumption did not change. On Day 29, there were significant increases in mean alanine transferase (ALT) levels in the group administered 2.0 µg/kg x 3 Sovateltide when compared to the control group. Increased ALT levels in the high-dose group corresponded with histopathology findings in the liver samples (single-cell hepatocellular necrosis); hence these observations were correlated to the test article. The increased ALT levels recovered fully by Days 42 and 43. The lowest observed adverse effect level (LOAEL) of Sovateltide was established at the dose level of 2.0 µg/kg x 3. The no observed adverse effect level (NOAEL) of Sovateltide was established at the dose level of 1.0 µg/kg x 3. Intravenous administration of Sovateltide up to 2.0 µg/kg body weight to conscious, freely moving Beagle dogs did not induce any potential effects in any cardiovascular parameters assessed. No sensitization or allergic potential for Sovateltide was observed in Guinea pigs. Sovateltide did not affect the male fertility of Sprague Dawley rats when administered by an intravenous route up to 15 µg/kg body weight. Sovateltide was administered to male and female Sprague Dawley rats at 2.5, 5.0, and 10 (in female rats only) µg/kg body weight before mating, through mating and up to implantation in one set of females (Set A) and up to lactation day 21 in another set of females (Set B) resulted in no treatment-related clinical signs or effects on fertility in male and female rats and early embryonic development to implantation in female rats, while male rats administered 10 µg/kg body weight resulted in a functional effect (e.g., libido, epididymal sperm maturation) on the male reproductive organs. Sovateltide, when administered intravenously to Sprague Dawley female rats and New Zealand White female rabbits, exhibited no maternal or fetal toxicity and is not teratogenic up to 15.0 µg/kg body weight. A Bacterial Reverse Mutation Test was performed to evaluate the mutagenic potential of Sovateltide using *salmonella typhimurium* Strains. Sovateltide is non-mutagenic as it does not induce (point) gene mutations at histidine locus by base pair changes or frame-shift in the presence and absence of metabolic activation system in all the five tester strains of *Salmonella typhimurium* TA1537, TA1535, TA98, TA100, and TA102 up to the concentration of 100 µg/plate. Mammalian Erythrocyte Micronucleus Test was conducted to assess the clastogenic potential of Sovateltide using a mammalian erythrocyte micronucleus test in rats. Sovateltide did not cause micronucleus induction in the polychromatic erythrocyte cells of bone marrow in male and female Wistar rats up to the dose level of 30 µg/kg body weight. There was no statistically significant increase in the percentage of aberrant cells in all tested concentrations of sovateltide-treated CHO-K1 cultures compared to the vehicle; therefore, sovateltide was found to be non-clastogenic and did not exhibit the potential to induce structural aberrations up to the highest tested concentration of 250 µg/mL, in either the presence or absence of metabolic activation system, for short-term exposure of 4 hours and in the absence of metabolic activation system for long-term exposure of 24 hours. In the clot

lysis assay, t-PA exhibited thrombolytic activity in a concentration and time-dependent manner. Sovateltide alone did not have any considerable clot lysis activity at any of the tested concentrations, and it did not influence the clot lysis potential of t-PA when treated in combination.

2.4 Clinical Data to Date

We conducted an open-label phase I study to determine the safety, tolerability, and pharmacodynamics of multiple ascending doses of sovateltide in healthy male volunteers (CTRI/2016/11/007509). The study was designed with 3 cohorts, each cohort having 3 subjects and each subject receiving 3 doses of either 0.3, 0.6, or 0.9 µg/kg administered at an interval of 4 hours as an intravenous bolus over 1 minute. Therefore, in the first cohort, each subject received a total dose of 0.9 µg/kg; in the second cohort, each subject received a total dose of 1.8 µg/kg; and in the third cohort, each subject was to receive 2.7 µg/kg. However, a total of 4 non-serious adverse events (uneasiness, sweating, abdominal discomfort, and vomiting) were reported by the first subject of cohort 3 after the first dose of 0.9 µg/kg of sovateltide. The adverse events were resolved in about 10 minutes without any intervention. Four hours later, a second dose of 0.9 µg/kg was given to the same subject, and again uneasiness, sweating, and abdominal discomfort were reported, but there was no vomiting; all the events resolved within 10 minutes without any intervention. No further dosing was carried out, and the study was concluded. The events observed at both doses were mild, transient, and resolved without any sequelae. No subjects experienced serious adverse effects in any cohort. Sovateltide did not significantly affect the vital signs, ECGs, or laboratory parameters of the healthy male volunteers. Sovateltide was well tolerated and found safe when administered in multiple ascending doses in healthy subjects. The Minimum Intolerable Dose (MID) was established as 0.9 µg/kg, and the Maximum Tolerated Dose (MTD) was 0.6 µg/kg. The therapeutic dose of sovateltide in patients with cerebral ischemia is 0.3 µg/kg, which is lower than the established MTD. The safety of sovateltide has also been established through several clinical studies conducted in India and USA. We are now developing sovateltide to treat cerebral ischemic stroke, spinal cord injury, and Alzheimer's disease. We have completed phase II, and III clinical studies in patients of cerebral ischemic stroke (CTRI/2017/11/010654, CTRI/2019/09/021373) and two-phase II clinical studies in patients of spinal cord injury (CTRI/2018/12/016667) and Alzheimer's disease (CTRI/2017/12/016394) are ongoing. In addition, it was also being developed as an adjunct to chemotherapy (NCT00613691, NCT01773785, and NCT01741155) [62-64] and as a diagnostic agent for the detection of the solid tumor, having completed a Phase III clinical trial (CTRI/2009/091/001006 and CTRI/2013/07/003854). In clinical studies conducted in India, this investigational product was administered intravenously (at 0.8 µg/kg dose) to 45 patients in Phase I/II study (CTRI/2009/091/001006) and 220 patients in phase III study (CTRI/2013/07/003854). Also, in phase I clinical study conducted in the USA (NCT00613691), 36 patients were administered an intravenous dose of sovateltide (Dose range: 0.5 to 15.1 µg/m²). sovateltide was well tolerated and found safe in these studies.

We have conducted a prospective, multi-centric, randomized, double-blind, parallel, saline-controlled phase II clinical study (CTRI/2017/11/010654, NCT04046484) in forty patients with

cerebral ischemic stroke, of which 36 completed 90-day follow-up. All patients received standard treatment and were randomly assigned to either control (saline; n=18; 11 male and 7 female) or sovateltide group (n=18; 15 male and 3 female). Clinical outcome parameters such as the National Institute of Health Stroke Scale (NIHSS), Modified Rankin Scale (mRS), Barthel Index (BI), EuroQol Index, and Stroke-Specific Quality-of-life (SS-QoL) Index for cerebral stroke were determined. Standard treatment and care were provided to all the patients. All the patients received saline or sovateltide within 24 hours of stroke onset. The number of patients that received saline within 20 hours of the onset of stroke was 14 out of 18, whereas those that received sovateltide were 10 out of 18. NIHSS on day 6, when compared to day 1, showed a significance value of $P=0.0495$ in saline and $P<0.0001$ in sovateltide cohorts. Similarly, mRS on day 6 compared to that observed on day 1 showed a significance value of $P=0.0859$ in the saline cohort, while in the sovateltide cohort, it was $P<0.0001$. BI on day 6, when compared to day 1, was similar ($P=0.3948$) in the saline cohort, whereas it significantly ($P<0.0001$) improved in the sovateltide cohort. Sovateltide produced a significantly quicker recovery in patients with acute cerebral ischemic stroke. In saline 12.50% while in sovateltide cohort 87.50% (odds ratio 10.18; 95% CI 10.18 to 120.50; $P=0.0201$), patients showed an improvement of ≥ 6 in NIHSS from baseline (enrollment time) at day 6 of treatment, while at 90 day 43.48% in saline and 56.52% patients in sovateltide cohort (odds ratio 2.275; 95% CI 0.5785 to 8.310; $P=0.2714$) showed an improvement of ≥ 6 in NIHSS. An improvement of ≥ 2 in mRS was observed in 39.13% and 60.87% of patients in saline and sovateltide cohorts, respectively, at 30 days of treatment. Similar improvement in mRS was observed on day 90 of treatment (odds ratio 5.250; 95% CI 1.00 to 27.63; $P=0.0519$). An improvement of ≥ 40 in BI in the saline cohort was 30.00%, while it was 70.00% for patients in the sovateltide cohort at 30 days of treatment (odds ratio 5.833; 95% CI 1.217 to 21.26; $P=0.0172$). At 90 days of treatment, 36.00% and 64.00% of patients in control and sovateltide cohorts showed an improvement of ≥ 40 in BI (odds ratio 12.44; 95% CI 1.502 to 147.10; $P=0.0112$). A change of 80 or more in the EuroQol scale was 0% of patients in saline and 100% of patients in the sovateltide cohort at 60 days (odds ratio ∞ ; 95% CI 0.9391 to ∞ ; $P=0.0389$) and at 90 days (odds ratio ∞ ; 95% CI 1.00 to ∞ ; $P=0.0332$). Similarly, a change of 150 or more in the SS-QoL scale was 22.22% of patients in saline and 77.78% of patients in the sovateltide cohort at 60 days (odds ratio 4.773; 95% CI 0.9278 to 25.06; $P=0.0665$) and at 90 days (odds ratio 5.250; 95% CI 1.00 to 27.63; $P=0.0519$). The number of patients with 100% recovery achieving NIHSS score of 0 ($P=0.04791$), mRS of 0 ($P=0.1193$), and BI of 100 ($P=0.02795$) was more in the sovateltide cohort compared to saline. Sovateltide treatment did not affect hemodynamic, biochemical, or hematological parameters. No incidence of drug-related adverse event was reported. There was no mortality and no incidence of recurrent cerebral ischemic stroke in any enrolled patient. Sovateltide was safe and was well tolerated in patients with cerebral ischemic stroke. Sovateltide has the potential to be a first-in-class neuronal progenitor cell therapy that promotes quicker recovery and significantly improves neurological outcomes in cerebral ischemic stroke patients.

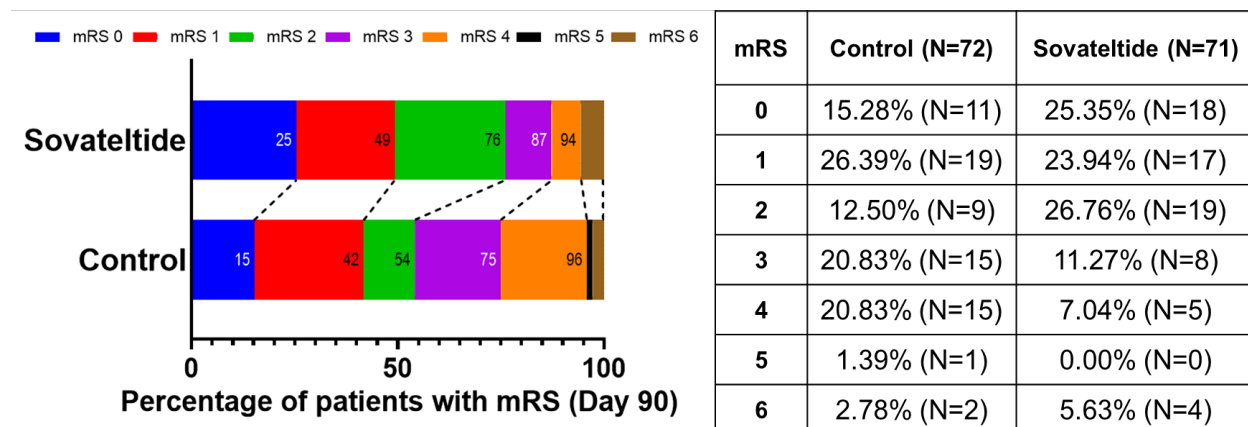
A prospective, multicentric, randomized phase III study was conducted in acute cerebral ischemic stroke patients aged 18 years through 78 years. Clinical Trials Registry (CTRI/2019/09/021373;

NCT04047563). Patients with radiologic confirmation of ischemic stroke could be enrolled if presenting up to 24 hours after the onset of symptoms and with a modified Rankin Score (mRS) of 3 to 4 and NIHSS score of greater than 5. Patients with intracranial hemorrhage and those receiving endovascular therapy were excluded. Patients were randomized in a 1:1 ratio, 80 patients to the sovateltide group and 78 patients to the control (saline) group. Every patient received the standard of care (SOC) and was followed for 90 days. The study drug (normal saline or sovateltide (0.3 µg/kg) was administered in three doses administered as an intravenous bolus over 1 minute at an interval of 3 hours ± 1 hour on day 1, day 3, and day 6 (total sovateltide dose of 0.9 µg/kg/day). The primary objectives were to determine the neurological outcome based on the National Institute of Health Stroke Scale (NIHSS), modified Rankin Scale (mRS), and Barthel Index (BI) scales from day 1 through day 90. In addition, quality-of-Life was measured by EuroQol EQ-5D (EQ-5D) and stroke-specific quality-of-life (SS-QoL) at 60 days and 90 days of treatment.

A total of 158 patients with acute cerebral ischemic stroke were enrolled in this study, of which 137 completed 90-day follow-ups. Patients received saline (n=78; 48 male and 30 female) or Sovateltide (n=79; 53 male and 26 female) within 24 hours of the onset of stroke. Patients received the investigational drug at 17.31 ± 0.74 and 17.48 ± 0.65 hours (mean ± SEM, $p=0.8609$) of stroke onset in control and Sovateltide cohorts, respectively. The baseline characteristics and SOC in both cohorts were similar. Alberta Stroke Program Early Computed Tomographic Score (ASPECTS) mean value was similar in the control (7.44) and Sovateltide (7.61) groups indicating that the extent of infarction was similar in both the groups. Sovateltide treatment resulted in significantly improved neurological outcomes compared to the control cohort. The mRS score is the most widely used primary outcome measure in trials for acute stroke interventions. The median mRS score at 90 days was significantly ($p=0.008$) lower in the Sovateltide group (1.00, 95% CI 1.179 to 1.776) than in the control group (2.00, 95% CI 1.753 to 2.447). The distribution of scores on the mRS at 90 days in the intention-to-treat population shows an ordinal shift across the range that highly favors Sovateltide therapy. An improvement of ≥ 2 points on the mRS was observed in 52.86% and 76.12% of patients in the saline and Sovateltide groups, respectively ($p=0.0045$). A mean improvement in the NIHSS was 0.9610 (95% CI -0.1244 to 2.046, $p=0.1217$) and 1.267 (95% CI 0.3176 to 2.216, $p=0.0017$) in the control and Sovateltide groups, respectively on day 2 from the day of randomization. Similarly, on day 3, an improvement in NIHSS from the day of randomization was 1.981 (95% CI 0.8843 to 3.077) in the control group and 3.050 (95% CI 2.086 to 4.014) in the Sovateltide group. At day 90, an improvement in the NIHSS of ≥ 6 points was seen in 64.29% and 82.09% of the saline and Sovateltide groups, respectively ($p=0.0190$). At day 90, BI score improvement of ≥ 40 points was seen in 76.12% of patients in the Sovateltide group vs. 61.43% in the saline group ($p=0.0639$). At 90 days mRS (difference between means -0.6224 ± 0.2303 ; 95% CI -1.078 to -0.1670, $p=0.0078$) and NIHSS (difference between means -1.586 ± 0.5126 ; 95% CI -2.600 to -0.5727, $p=0.0024$) were significantly lower, while BI (difference between means 10.19 ± 3.950 ; 95% CI 2.375 to 18.00, $p=0.011$) was significantly higher in the Sovateltide compared to the control group. The number of patients with complete recovery (defined as NIHSS score of 0, mRS of 0, and BI of 100) was numerically greater in the Sovateltide

group than in the saline group. Sovateltide treatment improved Quality-of-Life as measured by EQ-5D ($p=0.0055$) and SS-QoL (0.0673) compared to the control group on day 90. Sovateltide increased the frequency of favorable neurological outcomes at 3 months.

It was concluded that Sovateltide is an effective agent for treating acute cerebral ischemic stroke, having a highly favorable ordinal shift in the mRS across the range (figure and table below). In addition, it is safe, well-tolerated, and produced a statistically significant and clinically meaningful improvement in neurological outcomes in acute cerebral ischemic stroke patients 90 days post-treatment.



Distribution of scores on the Modified Rankin Scale at 90 days in the Intention-to-Treat population. The modified Rankin Scale (mRS) score is the most widely used primary outcome measure in trials for acute stroke interventions. A modified Rankin scale score of 0 indicates no disability, 1 no clinically significant disability, 2 slight disability, 3 moderate disability but able to walk unassisted, 4 moderately severe disability, 5 severe disability, and 6 death.

The convincing results of preclinical efficacy studies of sovateltide in ischemic stroke and its safety affirmation from previous human exposure encouraged us to investigate its safety and efficacy in patients of ischemic stroke. Clinical phase II and III results indicate that sovateltide is a novel, first-in-class, highly effective, and safe drug candidate for treating cerebral ischemic stroke. The present phase III clinical study aims to evaluate the safety and efficacy of sovateltide therapy along with the standard of care in patients with acute cerebral ischemic stroke. In the previous phase I study involving 15 subjects, the maximum tolerated dose (MTD) of sovateltide determined was 1.0 $\mu\text{g/kg}$ (IV), and in the phase II study, a dose of 0.8 $\mu\text{g/kg}$ was administered intravenously in 30 patients towards its efficacy evaluation. Therefore, for the present indication of cerebral ischemic stroke, we have proposed a dose of 0.3 $\mu\text{g/kg}$ of sovateltide, which is the same dose administered in the phase III study and is lower than the MTD (1.0 $\mu\text{g/kg}$) and the dose used in phase II study (0.8 $\mu\text{g/kg}$). This dose of 0.3 $\mu\text{g/kg}$ of sovateltide will be administered intravenously, in 3 doses, at an interval every 3 hours on day 1, day 3, and day 6. The day 6 dosing has to be performed within a time window of ± 1 day.

The details of the previous clinical studies on sovateltide are summarized below:

Table: Details of clinical studies on Sovateltide

Drug	Study	Indication	Sample size	Current status (Country)	Route of administration	Dose	NCT/CTRI Number
PMZ-1620 (Sovateltide)	An Open Label, Phase I Study to Determine the Safety, Tolerability and Pharmacodynamics of Multiple Ascending Doses of PMZ-1620 in Healthy Male Volunteers.	Ischemic Stroke	07	Completed (India)	IV	MTD established: 0.6 µg/kg	CTRI/2016/11/007509
PMZ-1620 (Sovateltide)	A Prospective, Multicentric, Randomized, Double Blind, Parallel, Saline Controlled Phase II Clinical Study to Compare the Safety and Efficacy of PMZ-1620 Therapy along with Standard Supportive Care in Subjects of Acute Ischemic Stroke.	Ischemic Stroke	40	Completed (India)	IV	0.3 µg/kg	CTRI/2017/11/010654 NCT04046484
PMZ-1620 (Sovateltide)	A Prospective, Multicentric, Randomized, Double Blind, Parallel, Phase III Clinical Study to Assess Efficacy of PMZ-1620 along with Standard Treatment in Patients of Acute Ischemic Stroke.	Ischemic Stroke	158	Completed (India)	IV	0.3 µg/kg	CTRI/2019/09/021373 NCT04047563
PMZ-1620 (Sovateltide)	A Prospective, Multicentric, Randomized, Double Blind, Placebo Controlled Phase II Clinical Study to Compare the Safety and Efficacy of PMZ-1620 Therapy along with Standard Supportive Care in Subjects of mild to moderate Alzheimer's disease.	Alzheimer's Disease	80	Ongoing (India)	IV	0.3 µg/kg	CTRI/2017/12/016394
PMZ-1620 (Sovateltide)	A Prospective, Multicentric, Randomized, Double Blind, Saline Controlled Phase II Clinical Study to Compare the Safety and Efficacy of PMZ-1620 Therapy along with Standard Supportive Care in Patients of Acute Spinal Cord Injury.	Spinal Cord Injury	40	Ongoing (India)	IV	0.3 µg/kg	CTRI/2018/12/016667
SPI-1620 (Sovateltide)	A Phase I, Open-label, Ascending Dose Study of the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of the ET _B Agonist, SPI-1620, in Patients with Recurrent or Progressive Carcinoma	Carcinoma	36	Completed (USA)	IV	MTD established: 11 µg/m ²	NCT00613691
VRP-1620 (Sovateltide)	An open labeled, single dose, multiple escalation phases I/II study to evaluate the safety, tolerability and enhancement of tumor detection radiologically using VRP-1620 in subjects with solid tumors	Diagnosis of solid tumors	Phase I: 15 Phase II: 30	Completed (India)	IV	MTD established in Phase I: 1.0µg/kg. Dose of 0.8µg/kg used for Phase II	CTRI/2009/091/001006
VRP-1620 (Sovateltide)	Phase III: A double blind, multi-centric, randomized controlled clinical trial to evaluate the efficacy and safety of VRP-1620 in increasing the sensitivity of sonomammography and lateral chest X-ray in patients of breast cancer.	Diagnosis of solid tumors	220	Completed (India)	IV	0.8 µg/kg	CTRI/2013/07/003854

2.5 Rationale for the Current Study

Extensive research is being conducted in search of neuroprotective agents for possible use in the acute phase of stroke and agents that can be used for neurorepair in the later stages of stroke. Several trials have been conducted and are in progress using different pharmacological agents, but none of the studies involve the stimulation of ETB receptors to treat cerebral ischemic stroke. Sovateltide (IRL-1620, PMZ-1620) has been effective in animal models of cerebral ischemic stroke. Its safety and tolerability have been demonstrated in a human phase I study with 7 subjects. Clinical phase II and III results indicate that sovateltide is a novel, first-in-class, highly effective drug candidate for treating cerebral ischemic stroke. A strong safety profile and significant efficacy in improving the National Institutes of Health Stroke Scale (NIHSS), Modified Rankin scale (mRS), and Barthel index (BI) obtained in phase II and III studies in patients with cerebral ischemic stroke in India are convincing and encouraged us to investigate its safety and efficacy in cerebral ischemic stroke patients globally. Therefore, we plan to conduct a phase III clinical study to evaluate the safety and efficacy of sovateltide therapy and the standard of care in patients with acute ischemic stroke.

2.5.1 Potential Benefits and Risks of the Study Drug

Sovateltide is a selective endothelin-B receptor agonist currently under investigation for the treatment of acute ischemic stroke. Preclinical studies and early-phase clinical trials have demonstrated that sovateltide offers neuroprotective and regenerative benefits by promoting neurogenesis, reducing neuronal injury, and enhancing microcirculation in ischemic brain tissue. Clinical data from Phase II and III trials conducted in India suggest that administering sovateltide within 24 hours of stroke onset is associated with improved functional recovery.

As an investigational product, the complete safety profile of sovateltide is not yet fully known. In earlier Phase II and III clinical studies, no sovateltide-related adverse effects were reported. In the Phase I study, sovateltide was well-tolerated at doses up to 0.6 µg/kg. At a higher dose (0.9 µg/kg), some mild, transient side effects, such as uneasiness, sweating, stomach discomfort, and vomiting, were reported and resolved without lasting effects. It is important to note that, as with any drug, there is also a potential risk of allergic reaction. Furthermore, participants receiving a placebo or those for whom sovateltide is ineffective may not experience clinical improvement.

Studies conducted to date indicate that sovateltide is well tolerated, suggesting that the risks are minimal compared to the potential benefits for the treatment of acute ischemic stroke. Sovateltide may offer significant advantages for patients suffering from acute ischemic stroke who require immediate medical attention.

2.6 Compliance Statement for Study Conduct in Accordance with Protocol, GCP, and Applicable Regulatory Requirements

The study protocol, amendments to the protocol (if applicable), patient's information, informed consent form (ICF), and consent form updates (if applicable) will be submitted to the institutional review board (IRB)/institutional ethics committee (IEC), which is constituted according to local

law to obtain approval before initiation of the study and as applicable thereafter. The study will only be initiated after receipt of approval from the IRB/IEC. The investigator will report promptly to the IRB/IEC new information that may adversely affect the safety of the patients or the conduct of the study. The investigator will follow the protocol in conformity with the International Council for Harmonisation-Good Clinical Practice (ICH-GCP) and applicable regulatory requirements. Before admission into the study, the written ICF must be signed and personally dated by the patient/LAR and the investigator or designee who conducts the informed consent discussion. If the original consent process involved anyone other than the participant, and if required by local standards, consent will be sought for the remaining procedures from the participant once they have been deemed to have regained capacity. In obtaining and documenting informed consent, the investigator must comply with the applicable regulatory requirement(s) and must adhere to GCP. The investigator must inform the patient of all pertinent aspects of the study, including the written information approved/favorably assessed by the IRB.

2.7 Description of the Study Population

A total of 514 patients (257 in each treatment group), male or female, age 18 to 80 years, with clinically and/or radiologically confirmed acute cerebral ischemic stroke and NIHSS score ≥ 8 and < 20 as well as NIHSS Level of consciousness (1A) score < 2 are to be evaluated in this study.

3 STUDY OBJECTIVES AND PURPOSE

3.1 Primary Objective

The primary objective of the study is:

- To determine whether sovateltide causes a good functional outcome in patients with acute cerebral ischemic stroke assessed by modified Rankin Scale (mRS) score of 0-2 at day 90 post-randomization.

3.2 Secondary Objectives

The secondary objectives of the study are to determine whether sovateltide causes:

- A good functional outcome in patients with acute cerebral ischemic stroke assessed by the National Institute of Health Stroke Scale (NIHSS) score of < 6 at day 90 post-randomization.
- A good functional outcome in patients with acute cerebral ischemic stroke assessed by the Barthel Index (BI) score of ≥ 90 at day 90 post-randomization.
- An excellent functional outcome in patients with acute cerebral ischemic stroke assessed by a modified Rankin Scale score of 0-1 at day 90 post-randomization.
- Change in Quality-of-life (QoL) as assessed by EuroQol EQ-5D and by Stroke-Specific Quality of Life (SS-QOL) at days 30, 60, and 90 post-randomization.
- Incidence of recurrent cerebral ischemic stroke within 90 days post-randomization
- Incidence of mortality within 90 days post-randomization.

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- Incidence of symptomatic intracerebral hemorrhage (ICH) within 24 ($\pm$ 6) hours of randomization.
 - Incidence of radiographic intracerebral hemorrhage (ICH) within 24 ($\pm$ 6) hours of randomization.
 - Alteration in cognition at day 30 and day 90 measured by Montreal Cognitive Assessment (MoCA) Test.
 - Any adverse events (AE) or serious adverse events (SAEs) associated with sovateltide.

3.3 Exploratory Objectives

The exploratory objectives of the study are to determine:

- To determine whether sovateltide causes a good functional outcome in patients with acute cerebral ischemic stroke assessed by mRS score of 0-2 at day 30 post-randomization.
- A good functional outcome in patients with acute cerebral ischemic stroke assessed by the NIHSS score of <6 at day 30 post-randomization.
- A good functional outcome in patients with acute cerebral ischemic stroke assessed by the BI score of ≥ 90 at day 30 post-randomization.
- Alberta Stroke Program Early CT (ASPECT) Score for stroke severity at baseline and identify sites of ischemic lesions.
- The subtype of the index acute cerebral ischemic stroke according to TOAST classification by day 30 post randomization.

4 STUDY DESIGN

4.1 Efficacy Endpoints

4.1.1 Primary Endpoint

The primary efficacy endpoint of the study is:

- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale score of 0-2 on day 90 post-randomization

4.1.2 Secondary Endpoints

The secondary efficacy endpoints of the study are:

- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with NIHSS score of <6 on day 90 post-randomization.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with BI score of ≥ 90 on day 90 post-randomization.
- The proportion of acute cerebral ischemic stroke patients having an excellent functional outcome with a modified Rankin Scale score of 0-1 on day 90 post-randomization.
- Change in QoL as assessed by EuroQol EQ-5D and by SS-QOL from baseline to days 30, 60, and 90 post-randomization.

-
- The proportion of patients with recurrent ischemic stroke within 90 days post-randomization.
 - Number of deaths within day 90 post-randomization.
 - The proportion of patients with symptomatic ICH within 24 ($\pm$ 6) hours of randomization.
 - The proportion of patients with radiographic ICH within 24 ($\pm$ 6) hours of randomization.
 - Change in MoCA score at days 30 and 90 post-randomization.
 - The proportion of patients with adverse events (AEs) and serious adverse events (SAEs).

4.1.3 Exploratory Endpoints

The exploratory endpoints of the study are to:

- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a mRS score of 0-2 on day 30 post-randomization
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with NIHSS score of <6 on day 30 post-randomization.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with BI score of ≥ 90 on day 30 post-randomization.
- Compare baseline Alberta Stroke Program Early CT (ASPECT) Score for stroke severity and identify sites of ischemic lesions.
- Compare the subtype of the index acute cerebral ischemic stroke by TOAST classification by day 30 post-randomization.

4.2 Description of Type and Design of Study

This is a multicentric, randomized, double-blind, placebo-controlled phase III clinical study to assess the safety and efficacy of sovateltide along with the standard of care in patients with acute cerebral ischemic stroke. A total of 514 patients (257 in each treatment arm) with acute cerebral ischemic stroke will be enrolled in this study.

The enrolment period of the study will be approximately 15 months, and the total duration of the study will be approximately 18 months. For an individual patient, the duration of the study will be 3 months (90 days), including 4 study visits: visit 1/Day 1 (screening/baseline/treatment visit), visit 2/Day 30 $\pm$ 5, visit 3/Day 60 $\pm$ 5 (telephonic visit), and visit 4/Day 90 $\pm$ 5 (End of Study (EOS)) (Table 1). Approximately 514 patients will be randomized 1:1 to one of the 2 treatment groups after meeting the eligibility criteria:

- Group 1 (Active Group): sovateltide + standard of care
- Group 2 (Control Group): matching placebo for sovateltide + standard of care

Central randomization will be stratified by early vs. late enrollment (<12 hours vs. ≥ 12 hours) as well as usage or non-usage of IV thrombolytic agents at enrollment. Although the usage of IV thrombolytics is recommended within 4.5 hours after symptom onset (varying by regional approvals), operationally, there will be some patients who receive IV thrombolytics on time but may enroll late. Therefore, the stratification of IV thrombolytics will be done among the patients

that will be enrolled either early (< 12 hours) or late (≥ 12 hours). To allow adequate opportunity for enrollment into each of the strata, a minimum of 200 patients per stratum in early vs. late enrollment and a minimum of 70 patients will be included in each sub-stratum of IV thrombolytic usage and non-usage in the early (< 12 hours) but not in the late (≥ 12 hours) enrollment group.

Sovateltide or placebo will be administered as an IV bolus dose over 1 to 2 minutes within the window of 24 hours after the onset of the stroke. In the sovateltide group, 3 doses of sovateltide at a dose of $0.3 \mu\text{g/kg}$ will be administered as an IV bolus over 1 to 2 minutes every 3 hours ± 1 hour on days 1, 3, and 6 (total dose/day: $0.9 \mu\text{g/kg}$). In the control group, 3 doses of the placebo will be administered as an IV bolus over 1 to 2 minutes every 3 hours ± 1 hour on days 1, 3, and 6 post-randomization. Patients will be provided the standard of care in both groups. Every effort will be made to have drug administration at the same time on days 1, 3, and 6. The day 6 dosing has to be performed within a time window of ± 1 day. Those patients discharged prior to day 6 will be called back to the healthcare facility for the remaining doses.

The standard care to be provided to the patients shall be the one used in the particular hospital setup for acute cerebral ischemic stroke.

Each patient will be monitored closely throughout his/her hospitalization for the qualifying stroke and will be followed for 3 months from randomization. Each patient will be assessed for efficacy and safety parameters over 3 months from randomization at a clinic visit. Each patient will be contacted by telephone for a brief (< 30 minutes) clinical and interviewer-administered QoL assessments at visit 3 (telephonic)/Day 60 from randomization.

Table 1: Study flow chart

Study Day	1	2	3	6£	30†	60† 	90†	Unscheduled visit
Study Visit	1¥				2	3	4 (End of Study)	
Informed Consent	X*							
Inclusion/Exclusion	x							
Demographic/Medical History	x							
Columbia-Suicide Severity Rating Scale (C-SSRS)	x	x	x	x	x	x	x	x
Concomitant Illnesses	x							
CT/MRI** ASPECT Score	x							
Subtype of index acute cerebral ischemic stroke (TOAST)					x			
Vital signs (respiratory rate, heart rate, systolic and diastolic blood pressure)	x	x	x	x	x		x	x
Physical Examination	x	x	x	x	x		x	x
Blood Tests ^Ω	x				x		x	x
Immunogenicity Assessment	x [£]		x	x	x		x	x
Electrocardiogram	x		x	x	x		x	x
Urine Pregnancy Test	x							
Investigational Drug Administration	x		x	x				
Adverse Event	x	x	x	x	x	x	x	x
Concomitant Therapy	x	x	x	x	x	x	x	x
Mortality and Time-to-Death	x	x	x	x	x	x	x	x
NIH Stroke Scale	x				x		x	x
Modified Rankin scale	x [†]				x		x	x
Barthel Index	x				x		x	x
Interviewer-administered EuroQol, interviewer-administered SS-QOL	x				x	x	x	
Recurrent Ischemic Stroke		x	x	x	x	x	x	x
Cognition (MoCA)				x	x		x	

†The Visit 2, 3, and 4 has to be performed within a time window of ± 5 days.

£ The day 6 dosing has to be performed within a time window of ±1 day.

¥The screening/baseline assessment and the first dose administration will be performed within a window of 24 hours after the onset of the stroke. Laboratory and imaging assessments conducted within 24 hours prior to consenting can be considered as baseline values. Visit 1 will be from day 1 to day 6 ±1 of the Study (even for those patients who get discharged prior to day 6 and re-admit for remaining doses).

*If the original consent process involved anyone other than the participant (and if required), the PI/delegates will make ongoing efforts until: (1) Regained capacity consent is obtained from the participant; (2) death, or (3) completion of the Day 90 assessments**Repeat CT/MRI will be done at the investigator's discretion if the patient deteriorates neurologically or intracranial hemorrhage is suspected for any reason.

^Ω Blood tests to be conducted include:

Day 1: Complete blood count (CBC)– hemoglobin (Hb), red blood cells (RBC), white blood cells (WBC), including differential count, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), and platelet count; Blood glucose test; Lipid profile (LDL, HDL, VLDL), blood cholesterol; Serum electrolytes – Na+, K+, Ca++; Liver function test – serum bilirubin, aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), albumin; Renal function test – serum creatinine and blood urea nitrogen; Partial thromboplastin time (PTT); Prothrombin time (PT)/ International normalized ratio (INR).

Day 30: CBC – Hb, RBC, WBC, including differential count, MCV, MCH, MCHC, and platelet count; Blood glucose test; Lipid profile (LDL, HDL, VLDL), blood cholesterol; Serum electrolytes – Na⁺, K⁺, Ca⁺⁺; Liver function test – serum bilirubin, AST, ALT, ALP, albumin; Renal function test – serum creatinine and blood urea nitrogen; PTT; PT/INR.

Day 90: CBC – Hb, RBC, WBC, including differential count, MCV, MCH, MCHC, and platelet count; Blood glucose test; Lipid profile (LDL, HDL, VLDL), blood cholesterol; Serum electrolytes – Na⁺, K⁺, Ca⁺⁺; Liver function test – serum bilirubin, AST, ALT, ALP, albumin; Renal function test – serum creatinine and blood urea nitrogen; PTT; PT/INR.

[€] To establish the baseline, the sample for immunogenicity assessment on Day 1 will be collected after informed consent is obtained and before the study drug is administered.

[‡] includes a pre-morbid mRS score (immediately prior to stroke).

The date of admission to the hospital and the date of discharge from the hospital will be recorded whenever applicable.

4.2.1 Central Imaging

This trial is using only imaging that would be required as per the patient's standard of care treatment. This will include baseline and follow up brain and neurovascular imaging for the treatment of the index stroke. Key images taken will be rendered anonymous and sent to the central core imaging lab. The core imaging lab will review the imaging against the inclusion and exclusion criteria. Central adjudication will be used to determine and confirm baseline ASPECT scoring.

Baseline imaging(s) used to enroll patients from another affiliated hospital are allowed and are expected to be transferred to the central imaging lab if appropriate.

4.3 Measures Taken to Avoid Bias

All patients and investigators will be masked to the treatment assigned until completion of the trial to minimize any patient- or investigator-induced bias in the outcome measures.

4.3.1 Randomization

An enrollment ratio of 1:1 is considered for the present phase III study. The sample size for this study is calculated based on the results of our recently completed phase III clinical study in patients of cerebral ischemic stroke in India (CTRI/2019/09/021373; NCT04047563).

A total of approximately 514 (257 per arm) patients with acute cerebral ischemic stroke will be enrolled and randomized in a ratio of 1:1 to either:

- Group 1 (Active Group): sovateltide + standard of care
- Group 2 (Placebo Group): matching placebo for sovateltide + standard of care

The derivation of the sample size is described in section 9.2. Central randomization will be stratified by early vs. late enrollment (<12 hours vs. ≥12 hours) as well as usage or non-usage of IV thrombolytic agents at enrollment. Although the usage of IV thrombolytics is recommended within 4.5 hours after symptom onset (varying by regional approvals), operationally, there will be some patients who receive IV thrombolytics on time but may enroll late. Therefore, the stratification of IV thrombolytics will be done among the patients that will be enrolled either early (< 12 hours) or late (≥ 12 hours). To allow adequate opportunity for enrollment into each of the strata, a minimum of 200 patients per stratum in early vs. late enrollment and a minimum of 70 patients will be included in each sub-stratum of IV thrombolytic usage and non-usage in the early (< 12 hours) but not in the late (≥ 12 hours) enrollment group. . The randomization list will be prepared and maintained blinded by an independent team who does not have interaction with the clinical trial team. An interactive web response system (IWRS) method containing randomization codes will be used to randomize/unblind the eligible patient to the treatment groups.

4.3.2 Blinding

In this double-blind study, the patient and all relevant personnel involved with the conduct and interpretation of the study (including the investigator, investigational site personnel, and the sponsor or designee's staff) will remain blinded to the identity of the investigational product (IP) and the randomization codes. In addition, the final randomization list will be kept strictly confidential, filed securely by the independent team, and accessible only to authorized persons as per the sponsor's standard operating procedures until the completion of the study.

4.4 Study Medication, Dose, Dosage Regimen, Dosage Form, Packaging, and Labeling

4.4.1 Dosage Regimen

The investigational product (IP) sovateltide will be provided by Pharmazz India Private Limited. It is available as IRL-1620 for Injection containing 30 µg of sovateltide in a 5.0 mL vial, to be reconstituted in 5 mL normal saline (0.9% Sodium Chloride Injection). The study drug will be

administered within 24 hours of the stroke onset. Three doses of 0.3 µg/kg will be administered as an IV bolus over one to two minutes every 3 hours on day 1. The dose will be repeated on days 3 and 6 post-randomization. The day 6 dosing has to be performed within a time window of ±1 day. Those patients discharged prior to day 6 will be called back to the healthcare facility for remaining doses.

The reference product is a matching placebo for sovateltide. It is a sterile lyophilized injection consisting of excipients without sovateltide in a 5.0 mL vial, to be reconstituted in 5 mL of normal saline (0.9% Sodium Chloride Injection). Three doses will be administered as an IV bolus over one to two minutes every 3 hours on day 1. The dose will be repeated on days 3 and 6 post-randomization. The day 6 dosing has to be performed within a time window of ±1 day. Those patients discharged prior to day 6 will be called back to the healthcare facility for remaining doses.

4.4.2 Dispensing (Labeling) and Compliance

The IP will be dispensed by the sites blinded pharmacist/delegate as per the randomization arm indicated by the Randomization and Trial Supply Management (RTSM) system. Access to the dispensed IP must be provided only to the investigator's designee/pharmacist.

The RTSM will assign a 6 digit carton ID to dedicate to the randomized patient. Each patient will receive 1 or 2 cartons (each with 9 vials) based on their indicated weight at randomization. Once a carton is assigned to a patient, all vials associated with that carton are assigned to the patient. Record which vial(s) is administered to the patient at each of the 9 dosing intervals. The order of vial administration from a carton is agnostic. In the event a patient is designated 2 cartons, it is recommended to use all vials from one carton before opening the second carton.

Refer to the RTSM User Guide for additional information regarding how to request an unscheduled dispensation in the system.

Vials will be labeled with the same 6 digit carton ID with an additional hyphenated digit (1 thru 9).

The label on IP and reference product will have the following information:

- Investigational Product
- Vial Number
- Batch Number
- Manufacture Date
- Expiry Date
- Storage Temperature
- The statement “For Clinical Study Use Only”, or similar wording based on the regulatory authority.

4.5 Description of the Sequence and Duration of Events during Study

4.5.1 Visit 1 (Day 1, Screening/Baseline/Randomization/Treatment Visit)

The screening/baseline assessment and the first dose administration (study treatment) have to be performed within a window of 24 hours after the onset of the stroke. The following procedures and assessments will be performed at this visit:

- Consent obtained as per Section 12.3: Patient Information and Consent.
- Inclusion and exclusion criteria will be checked
- Recording of medical history
- Collection of demographic information
- Recording of concomitant illness
- Recording of concomitant medications
- Computed tomography (CT)/Magnetic Resonance Imaging (MRI) (standard of care imaging will be accepted)
- ASPECT Score
- Vital signs (respiratory rate, heart rate, systolic and diastolic BP, body temperature) measurement
- Physical examination
- Recording of date of admission (DOA) to the hospital
- Randomization
- IP administration within 24 hours from the time of stroke: 3 doses 3±1 hour apart (total dose/day: 0.9 µg/kg body weight)
- Blood tests
 - Complete blood count – hemoglobin (Hb), red blood cells (RBC), white blood cells (WBC), including differential count, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), and platelet count
 - Blood glucose test
 - Lipid profile (LDL, HDL, VLDL), blood cholesterol
 - Serum electrolytes – Na⁺, K⁺, Ca⁺⁺
 - Liver function test – serum bilirubin, aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), albumin
 - Renal function test – serum creatinine and blood urea nitrogen
 - Partial thromboplastin time
 - Prothrombin time and International normalized ratio
- Blood sample for immunogenicity assessment– To establish the baseline, the sample for immunogenicity assessment on Day 1 will be collected after informed consent is obtained and before the study drug is administered.
- Electrocardiogram (ECG)
- Urine pregnancy testing (for females), if applicable

- NIHSS scoring
- mRS scoring including pre-morbid mRS
- BI scoring
- QoL assessment using interviewer-administered EuroQol and SS-QOL
- Recording of AE/SAE, if any
- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality at baseline using the Baseline version of C-SSRS.
- Mortality and time-to-death

Participants will be identified using usual standard of care screening methods at the stroke hospital. All participants will undergo an acute clinical assessment, blood laboratory assessment, and baseline imaging. A screening log will be maintained for potential participants that are approached for consent into this trial after initial patient triage through standard stroke processes.

A screen failure will be considered as a patient that has consented to the trial, but did not receive a single dose of IP.

If a subject was initially determined to be eligible for the trial, consented, and dosed, but subsequently it was revealed that an inclusion/exclusion was not met, the patient will still receive their full dose (9 doses as applicable) of IP and complete protocol required follow ups.

4.5.2 Visit 1 (Day 2, Day 3 & Day 6 Assessments)

4.5.2.1 Day 2

- Vital signs (respiratory rate, heart rate, Systolic and diastolic BP, and body temperature) measurement
- Physical examination
- Recording of AE/SAE, if any
- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality for post-dose assessments using the Since Last Visit version of C-SSRS.
- Recording of concomitant medications
- Mortality and time-to-death post-randomization
- Clinical evaluation for signs or symptoms of recurrent ischemic stroke

4.5.2.2 Day 3

- Vital signs (respiratory rate, heart rate, systolic and diastolic BP, and body temperature) measurement
- Physical examination
- IP administration (3 doses 3 hourly) (total dose/day: 0.9 µg/kg body weight)
- Blood sample for immunogenicity assessment
- Electrocardiogram (ECG)
- Recording of AE/SAE, if any

- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality for post-dose assessments using the Since Last Visit version of C-SSRS.
- Recording of concomitant medications
- Mortality and time-to-death post-randomization
- Clinical evaluation for signs or symptoms of recurrent ischemic stroke

4.5.2.3 Day 6 ± 1 day

- Vital signs (respiratory rate, heart rate, systolic and diastolic BP, and body temperature) measurement
- Physical examination
- IP administration (3 doses 3 hourly) (total dose/day: 0.9 µg/kg body weight)
- Blood sample for immunogenicity assessment
- Electrocardiogram (ECG)
- Recording of AE/SAE, if any
- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality for post-dose assessments using the Since Last Visit version of C-SSRS.
- Recording of concomitant medications
- Mortality and time-to-death post-randomization
- Clinical evaluation for signs or symptoms of recurrent ischemic stroke
- MoCA

4.5.3 Visit 2 (Day 30 ± 5)

The following procedures and assessments will be performed at this visit:

- Vital signs (respiratory rate, heart rate, systolic and diastolic BP, and body temperature) measurement
- Physical examination
- Blood tests
 - Complete blood count – Hb, RBC, WBC, including differential count, MCV, MCH, MCHC, and platelet count
 - Blood glucose test
 - Lipid profile (LDL, HDL, VLDL), blood cholesterol
 - Serum electrolytes – Na⁺, K⁺, Ca⁺⁺
 - Liver function test – serum bilirubin, AST, ALT, ALP, albumin,
 - Renal function test – serum creatinine and blood urea nitrogen
 - Partial thromboplastin time
 - Prothrombin time and International normalized ratio
- ECG
- Recording of AE/SAE, if any
- Subtype of the index acute cerebral ischemic stroke by TOAST classification
- Blood sample for immunogenicity assessment

-
- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality for post-dose assessments using the Since Last Visit version of C-SSRS.
 - Recording of concomitant medications
 - Mortality and time-to-death
 - NIHSS scoring
 - mRS scoring
 - BI scoring
 - Clinical evaluation for signs or symptoms of recurrent ischemic stroke
 - QoL assessment using interviewer-administered EuroQol and SS-QOL
 - MoCA

The Day 30 visit is intended to be conducted in person. If it is not possible to have the visit conducted in person, where permitted, the visit can be conducted via telemedicine. This will permit the timely collection of efficacy endpoints and safety data.

4.5.4 Visit 3 (Day 60 ± 5)

The site personnel will contact patients telephonically 60 days post-randomization for brief clinical and QoL assessments. The following are the assessments:

- Recording of AE/SAE, if any
- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality for post-dose assessments using the Since Last Visit version of C-SSRS.
- Recording of concomitant medications
- Mortality and time-to-death
- QoL assessment using interviewer-administered EuroQol and SS-QOL
- Clinical evaluation for signs or symptoms of recurrent ischemic stroke

4.5.5 Visit 4 (Day 90 ± 5, End of Study Visit)

The following procedures and assessments will be performed at this visit:

- Vital signs (respiratory rate, heart rate, systolic and diastolic BP and body temperature) measurement
- Physical examination
- Blood tests
 - Complete blood count – Hb, RBC, WBC, including differential count, MCV, MCH, MCHC, and platelet count
 - Blood glucose test
 - Lipid profile (LDL, HDL, VLDL), blood cholesterol
 - Serum electrolytes – Na⁺, K⁺, Ca⁺⁺
 - Liver function test – serum bilirubin, AST, ALT, ALP, albumin,
 - Renal function test – serum creatinine and blood urea nitrogen
 - Partial thromboplastin time
 - Prothrombin time and International normalized ratio

- ECG
- Recording of AE/SAE, if any
- Blood sample for immunogenicity assessment
- The C-SSRS, a measure of suicidal ideation and behavior, will be used to document suicidality for post-dose assessments using the Since Last Visit version of C-SSRS.
- Recording of concomitant medications
- Recording of date of discharge (DOD) from the hospital. The total number of days stayed in the hospital is calculated by DOD-DOA
- Mortality and time-to-death
- NIHSS scoring
- mRS scoring
- BI scoring
- QoL assessment using interviewer-administered EuroQol and SS-QOL
- Clinical evaluation for signs or symptoms of recurrent ischemic stroke
- MoCA

4.5.6 Unscheduled Visits

An unscheduled visit is defined as any visit that occurs outside the schedule of visits. Testing and procedures may be performed at the discretion of the investigator. Data collected during the unscheduled visit will be recorded on the CRF document with the ‘Unscheduled’ visit ID box marked but will not be included in the analysis. If the unscheduled visit results in premature termination/withdrawal, then all procedures enlisted under “Visit 4” will be performed.

4.5.7 End of Study (EOS)

The end of the study is defined as the Last Patient Last Visit (LPLV), i.e., when the last patient completes Visit 4 (Day 90 $\pm$ 5, End of Study Visit), passes away, or withdraws from the study.

4.6 Accountability Procedures for the Investigational Product

An IP accountability log will be maintained by the study staff. This record will document the receipt, dispensing, and return or destruction of the IP. Accountability will be performed by reviewing the accountability records and by checking the randomization number entered in the electronic case report form (eCRF). At the EOS, all used IP will be destroyed on site after monitor verification. And unused IP will either be destroyed on site after monitoring if appropriate SOPs are in place, or returned to the Sponsor’s depot for destruction.

4.7 Maintenance of Study Treatment Randomization Codes and Procedures for Breaking Codes

A central randomization schedule will be generated by an independent team. Patients will be randomized 1:1 to one of the two treatment groups. Randomization will be performed with a computer program-generated randomization schedule assigning the patients in the sequence of their appearance to treatment groups.

4.7.1 Emergency Unblinding of an Individual Patient

If it is medically imperative to know which study treatment the patient is receiving, emergency unblinding will be done through RTSM. The investigator or his/her designee will unblind the code when medically needed without identifying the patient's treatment. The person who unblinds the randomization code must record the date, time, and reason for emergency unblinding in the patient's medical record on the Note-to-File. In such cases, the IP will be permanently discontinued, and the patient will be followed until completion of the trial.

4.8 Identification of Any Data to be Directly Recorded on Case Report Forms

Not applicable since eCRF will be used in the study.

5 SELECTION AND WITHDRAWAL OF PATIENTS

5.1 Inclusion Criteria

A patient will be considered eligible for inclusion in the study if they fulfill the following criteria:

1. Adult males or females aged 18 – 80 years old
2. Consent obtained per national laws and regulations, and in accordance with the applicable ethics committee requirements prior to study procedures
3. Stroke is ischemic in origin that is diagnosed clinically and/or radiologically (CT scan or diagnostic MRI) prior to enrolment. No hemorrhage as proved by cerebral CT/MRI scan.
4. Cerebral ischemic stroke patients presenting within 24 hours after onset of symptoms with NIHSS score ≥ 8 and < 20 , NIHSS Level of consciousness (1A) score < 2 at the time of screening. This includes cerebral ischemic stroke patients who completely recovered from earlier episodes before having a new or fresh stroke having pre-stroke historical measure of mRS score 0-2.
5. The patient is < 24 hours from the time of stroke onset when the first dose of sovateltide is administered. Time of onset is when symptoms began; for stroke that occurred during sleep, time of onset is when the patient was last seen or was self-reported to be normal
6. Reasonable expectation of availability to receive the full sovateltide course of therapy and to be available for subsequent follow-up visits

5.2 Exclusion Criteria

A patient will not be considered eligible for inclusion in this study if they meet any of the following criteria:

1. Patients receiving endovascular therapy or is a candidate for any surgical intervention for the treatment of stroke, which may include but not limited to endovascular techniques
2. Patients classified as comatose are defined as a patient who requires repeated stimulation to attend or is obtunded and requires strong or painful stimulation to make movements (NIHSS Level of Consciousness (1A) score ≥ 2)

3. Evidence of intracranial hemorrhage (intracerebral hematoma, intraventricular hemorrhage, SAH, epidural hemorrhage, acute or chronic subdural hematoma (SDH) on the baseline CT or MRI scan
4. Known pregnancy and lactating women
5. Known medical history of neurological (other than current acute ischemic stroke) or psychiatric condition that, in the investigator's opinion, would confound the neurological and functional evaluations, lead to further deterioration of neurological status, or interfere with participation in this study
6. Concurrent participation in any other therapeutic clinical trial
7. Evidence of any other major life-threatening or serious medical condition that would prevent completion of the study protocol impair the assessment of outcome, or in which sovateltide therapy would be contraindicated or might cause harm to the patient

5.3 Patient Withdrawal Criteria

Patients will have the right to withdraw from the study at any time and for any reason without prejudice to her/his future medical care by the physician or at the institution. Withdrawal of consent means that the patient does not wish to or is unable to continue further study participation. The investigator will discuss with the patient the most appropriate way to withdraw to ensure the patient's health. Data up to the point of withdrawal of consent will be kept for trial purposes unless specified otherwise by the patient.

The investigator can withdraw patients from the study in the following situations: in the event of intercurrent illness, AEs, treatment failure, or protocol violations. However, patients will be followed up to day 90 for safety assessment.

Patients who exhibit signs of hypersensitivity, such as an anaphylactic reaction, will have their treatment terminated if any doses are still pending. Nevertheless, these patients will still be followed for their safety and efficacy outcomes per protocol.

If any inclusion/exclusion criteria were missed upon enrollment that would preclude the patient's inclusion into the trial, at the discretion of the PI, treatment will be stopped immediately. Follow up visits will still occur to collect all relevant safety and efficacy data as appropriate.

For a patient prematurely discontinuing from the study, besides scheduled study assessments for that visit, the investigator will perform all (EOS) assessments at that visit itself. However, patients will be followed up to day 90 for safety assessment.

The reasons for discontinuation of study drug treatment might include:

- Withdrawal of consent
- As per the investigator or sponsor's discretion
- Significant protocol deviation
- Patient non-compliance
- AE or SAE
- Study termination by the sponsor

- Any treatment unblinding by the investigator will lead to permanent treatment discontinuation

Patients with discontinuation of their treatment will still be followed for the duration of the trial to collect safety and efficacy data unless the patient withdraws consent.

5.3.1 Withdrawn Patient Data Collection

The principal investigator and/or other investigator involved in the study will document the reason for the patient's withdrawal as follows:

- Lost to follow-up: patients who leave the study unnotified, do not attend the study visit, or cannot be contacted by phone. Intensive efforts should be made to locate and recall them if possible and to determine their health status at a minimum
- Adverse event: an AE form must be completed. All patients, irrespective of the time of dropout, will be considered for safety analysis
- Adverse laboratory event: this reason should be stated when a laboratory value is interpreted as a clinically significant abnormal value not explained by a laboratory error or being not a known or abnormal value commonly observed in this type of population
- Protocol violation (e.g., incorrect inclusion process)
- Consent withdrawn
- Other: if no reasons mentioned above are applicable, then the reason will be specified

6 TREATMENT OF PATIENTS

6.1 Investigational Product including Name(s), Dose(s), Dosing Schedule(s), Route(s), Treatment Period(s), Follow-up Period(s)

Product Name : IRL-1620 for injection (sovateltide)

Product strength : Each 5.0 mL vial contains 30 µg of sovateltide

Composition and information on ingredients : Each 5.0 mL vial contains,

Sovateltide 30 µg

Excipients q.s.

Manufacturing company : Pharmazz India Private Limited, At Steril Gene Life Sciences Pvt. Ltd., No. 45, Mangalam Main Road, Mangalam Village, Villianur Commune, Puducherry-605110 INDIA.

Therapeutic category : For the treatment of cerebral ischemia

Dose/Dosing Schedule : 0.3 µg/kg body weight at an interval every 3 hours on days 1, 3, and 6. The study drug will be administered as an IV bolus dose over 1 to 2 minutes within 24 hours of the stroke onset (total dose/day: 0.9 µg/kg body weight)

Intended route of administration : Intravenous

Shelf Life : Stable for 2 years at recommended storage conditions

Storage condition : Store between 2-8 °C and protect from light and moisture

6.2 Medication(s)/Treatment(s) Permitted and Not Permitted Before and/or During the Study

6.2.1 Medication(s)/Treatment(s) Permitted

- Standard treatment for the management of cerebral ischemic stroke.

6.2.2 Medications/Treatment(s) Not Permitted

- Patients will not be prohibited from taking any medications during the study.

6.2.3 Procedure(s) for Monitoring Patient Compliance

A dispensing log of test preparations will be maintained at the study site to ensure accountability of all medications supplied. The dispensing log will include the amount of drug received, lot and/or batch number, date drug received, patient number and initials, date drug dispensed, amount used, the amount wasted (if applicable), and the amount remaining in stock. In addition, the study drug administration and accountability logs (individual and master) will be regularly maintained and filed. The investigator is responsible for the accountability of the study drug, reconciliation, and recording maintenance throughout the course of the study in accordance with all applicable regulatory requirements.

7 STUDY ASSESSMENTS

7.1 Eligibility Assessments

- Adult male or female age 18- 80 years of age.
- Confirmation of cerebral ischemic stroke by CT/MRI
- NIHSS score ≥ 8 and < 20 , as well as NIHSS Level of consciousness (1A) score < 2 .
- ECG
- Blood tests
 - Complete blood count – Hb, RBC, WBC, including differential count, MCV, MCH, MCHC, and platelet count
 - Blood glucose test
 - Lipid profile (LDL, HDL, VLDL), blood cholesterol
 - Serum electrolytes – Na⁺, K⁺, Ca⁺⁺
 - Liver function test – serum bilirubin, AST, ALT, ALP, albumin
 - Renal function test – serum creatinine and blood urea nitrogen
 - Partial thromboplastin time
 - Prothrombin time and International normalized ratio
- Blood sample for immunogenicity assessment
- Urine pregnancy testing (for females)
- The patient is < 24 hours from the time of stroke onset when the first dose of sovateltide is administered
- Reasonable expectation of their availability to receive the full sovateltide course of therapy and to be available for subsequent follow-up visits

7.2 Efficacy Assessments

7.2.1 National Institutes of Health Stroke Scale (Appendix 1)

NIHSS is the most widely used scale to assess stroke severity and subsequent stroke outcome. It is a 42-point scale that measures neurological deficits and contains 15 items, including the level of consciousness, language function neglect, visual fields, eye movements, facial symmetry, motor strength, sensation, and coordination. The examination can be performed quickly.

The NIHSS scoring is as follows:

- Score 25: Very severe neurological impairment
- Score 15-24: Severe impairment
- Score 5-14: Moderately severe impairment
- Score <5: Mild impairment

NIHSS has been used extensively in clinical studies to measure stroke outcomes and has been validated and standardized to reduce inter-observer error. It is assessed on day 1, 30, and 90 post-randomization. In the case of dosing day 1 the NIHSS score will be measured prior to administering the first study dose.

7.2.2 Modified Rankin Scale (Appendix 2)

The mRS measures independence rather than the performance of specific tasks. In this way, mental as well as physical adaptations to neurological deficits are incorporated. The scale consists of 7 grades, from 0 to 6, with 0 corresponding to no symptoms and 6 corresponding to death.

- 0: No symptoms at all
- 1: No significant disability despite symptoms; able to carry out all usual duties and activities
- 2: Slight disability; unable to carry out all previous activities, but able to look after own affairs without assistance
- 3: Moderate disability; requiring some help but able to walk without assistance
- 4: Moderately severe disability; unable to walk without assistance and unable to attend to own bodily needs without assistance
- 5: Severe disability; bedridden, incontinent, and requiring constant nursing care and attention
- 6: Dead

The mRS has been shown to be reliable and valid for use in stroke. A score of ≤ 2 on the mRS is considered a favorable outcome with minimal or no disability. The mRS scoring is assessed at day 1, 30, and 90 post-randomization.

7.2.3 Barthel Index (Appendix 3)

The BI is considered a reliable disability scale for stroke patients. It is a 10-item scale on which disability is assessed by various aspects of self-care (feeding, grooming, bathing, dressing, bowel and bladder care, and toilet use) and a group related to mobility (ambulation, transfers, and stair climbing). Patients who score 100 (maximum score) on BI are fully independent in physical functioning. They are continent, can feed and dress themselves, get out of bed, walk more than 1 block, and perform activities of daily living. The lowest score is 0, representing a totally dependent bedridden state. The Barthel index is assessed at day 1, 30, and 90 post-randomization.

7.2.4 Quality-of-life as assessed by

Quality of life will be assessed by interviewer-administered EuroQol and SS-QoL.

7.2.4.1 EuroQol-5D-5L (Appendix 4)

EQ-5D-5L is a generic measure of health status developed by an international research group, the EuroQol Group. The EQ-5D-5L defines health in terms of mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension is divided into 5 levels: no problem, slight problem, moderate problem, severe problem, and extreme problem. The scale is numbered from 0 to 100; 100 means the best health you imagine, and 0 is the worst health you imagine. The Interviewer-administered EQ-5D-5L will be used, where a rater who has completed the study specific rater training program will ask the assessment questions and, based on the answers from patients, mark the appropriate responses on the source document. The rater will determine the patient's health score by asking, "I would like to ask you to say how good or bad your health is TODAY"? The rater will then mark a Visual Analog Scale numbered from 0 to 100, where 100 means the best health patient can imagine and 0 means the worst health patient can imagine.

7.2.4.2 Stroke-Specific Quality-of-life (Appendix 5)

The SS-QOL is a questionnaire consisting of 49 items in the 12 domains of energy, family roles, language, mobility, mood, personality, self-care, social roles, thinking, upper extremity function, vision, and work/productivity. The domains are scored separately, and a total score is provided, with a minimum value of 1 (meaning the worst outcome) and a maximum value of 5 (meaning the best outcome). For the trial, a rater who has completed the study specific rater training program will ask the SS-QOL questions of the patient in interview format and record the patient's responses on the source document.

7.2.5 Montreal Cognitive Assessment (MoCA) score (Appendix 6)

The Montreal Cognitive Assessment (MoCA) was designed as a rapid screening instrument for mild cognitive dysfunction. It assesses different cognitive domains: attention and concentration, executive functions, memory, language, visuo-constructional skills, conceptual thinking, calculations, and orientation. The time to administer the MoCA is approximately 10 minutes. The total possible score is 30 points; a score of 26 or above is considered normal. MoCA scores will be assessed on days 6, 30, and 90.

7.2.6 The Alberta Stroke Programme Early Computed Tomography Score (ASPECTS)

The Alberta Stroke Programme Early Computed Tomography Score (ASPECTS) is widely used to determine the extent of early ischemic changes in brain imaging for acute stroke and is recognized by the American Heart Association guidelines for managing acute stroke [65]. The ASPECT score is a 10-point quantitative score wherein 1 point is deducted for each defined middle cerebral artery (MCA) area, showing any evidence of early ischemic change. The ten defined MCA regions include M1 – Anterior MCA cortex, M2 – MCA cortex lateral to the insular ribbon, M3 – posterior MCA cortex, M4 – anterior MCA territory, M5 – lateral MCA territory, M6 – posterior MCA territory, C – Caudate, L – Lentiform Nucleus, I – Insula, and IC – Internal Capsule [66]. A score of 0 would indicate infarction of all 10 regions.

7.2.7 TOAST Classification of Subtypes of Ischemic Stroke

The Toast classification [67] includes five categories:

1. Large-artery atherosclerosis
2. Cardioembolism
3. Small-vessel occlusion
4. Stroke of other determined etiology
5. Stroke of undetermined etiology

8 ASSESSMENT OF SAFETY

8.1 Adverse Events

8.1.1 Definition of Adverse Event

An AE is defined in the ICH-GCP as “any untoward medical occurrence in a patient administered a medicinal product and which does not necessarily have to have a causal relationship with this treatment.”

An AE can, therefore, be any unfavorable and unintended sign (including an abnormal laboratory finding, for example), symptom, or disease temporally associated with the use of an IP, whether or not considered related to the IP. (ICH E6:1.2).

Any abnormal laboratory findings identified as clinically significant by the investigator or delegate which occur after the administration of an investigational product (IP) will be identified as a treatment emergent adverse event.

Worsening of a pre-existing medical condition (e.g., diabetes, migraine headaches, and gout) should be considered as an AE if there is either an increase in severity, frequency, or duration of the condition or an association with significantly worse outcomes.

Pre-existing conditions that have not worsened are not considered AEs.

Any AE that is a symptom of an immune response, in the opinion of the principal investigator, will require immunogenicity assessment.

8.1.1.1 Adverse Events of Special Interest (AESI)

AESIs are required to be reported by the investigator to the sponsor immediately (i.e., no more than 24 hours after learning of the event). An AE such as "intracranial hemorrhage" will be considered as an AESI.

8.1.2 Serious Adverse Event

An SAE is any untoward medical occurrence that, at any dose:

- results in death
- is life-threatening*
- requires in-patient hospitalization or prolongation of existing hospitalization
- results in persistent or significant disability/incapacity

- is a congenital anomaly/birth defect
- “Important medical events” that may not result in death, be life-threatening, or require hospitalization may be considered serious when, based upon appropriate medical judgment, they may jeopardize the patient and may require medical or surgical intervention to prevent one of the outcomes listed in this definition

* NOTE: The term “life-threatening” in the definition of “serious” refers to an event in which the patient was at risk of death at the time of the event; it does not refer to an event that hypothetically might have caused death if it were more severe.

Following are the few, but not limited to, complications that may arise during the natural course of the disease:

- Paresis/muscle weakness
- Dysarthria
- Pain/numbness in any part of the body
- Memory loss
- Visual field defects
- Vertigo
- Impaired levels of consciousness.

Such events will be considered treatment emergent adverse effects if they occur after IP administration. The investigator will then determine causality of the complication and report this information.

8.1.2.1 Serious Unexpected Serious Adverse Reactions (SUSAR)

AEs that meet all of the following criteria will be classified as suspected unexpected serious adverse reactions (SUSARs) and will be reported to the appropriate regulatory authorities in accordance with applicable regulatory requirements for expedited reporting:

- Serious
- Unexpected. Based on the clinical data to date, non-serious dyspnea and hypotension are considered expected. All serious adverse reactions (SARs) will be considered unexpected for the purpose of regulatory reporting.
- There is at least a reasonable possibility that there is a causal relationship between the event and the study treatment.

The investigator will assess whether an event is causally related to study treatment. The sponsor (or Syneos Health) will consider the investigator’s assessment and determine whether the event meets the criteria for being reportable as a 7 day or 15 day safety report. SUSARs that are fatal or life-threatening must be reported to the regulatory authorities and the IEC/IRBs (where required) within 7 days after the sponsor (or Syneos Health) has first knowledge of them, with a follow-up report submitted within a further 8 calendar days. Other SUSARs must be reported to the relevant regulatory authorities and the IEC/IRBs within 15 calendar days after the sponsor (or Syneos Health) first has knowledge of them.

The sponsor (or Syneos Health) is responsible for reporting SUSARs and any other events required to be reported in an expedited manner to the regulatory authorities and for informing investigators of reportable events, in compliance with applicable regulatory requirements within specific timeframes. Investigators will notify the relevant IEC/IRBs of reportable events within the applicable timeframes.

8.1.3 Methods and Timing for Assessing, Recording, and Analyzing Safety Parameters

All AEs will be collected on eCRFs from the time the patient signs the ICF until the study exit. Any SAE that is ongoing at the time the patient exits the study must be followed until the event is resolved, reaches a sequelae or there is a satisfactory explanation that meets one of the following outcomes:

- Resolved (With residual effects or without residual effects)
- Death
- Unknown (Despite adequate follow-up)

8.1.4 Documentation and Reporting of Immediately Reportable Adverse Events

All SAEs will be recorded from the time a patient is enrolled in the study till the end of the study.

- Regardless of a causal relationship, the site will report all SAEs to the sponsor/designee. Any such SAE due to any cause whether or not related to the study drug, must be reported within 24 hours of occurrence or when the investigator becomes aware of the event. Notification can be made using the dedicated fax line or email for the Syneos Health Safety and Pharmacovigilance group:
- Syneos Health Safety and Pharmacovigilance fax number: +1 877 464-7787
- Syneos Health Safety and Pharmacovigilance email address: safetyreporting@syneoshealth.com
- If the investigator does not have all information regarding an SAE, he/she will not wait to receive additional information before completing and submitting eSAE form.
- Additional relevant information or clinical follow-up should be sent via fax or e-mail to Syneos Health Safety and Pharmacovigilance as soon as it becomes available.
- All SAEs will be followed up until resolution (i.e., asymptomatic, stabilization, or death)
- Syneos Health is responsible for reporting SUSARs (Suspected Unexpected Serious Adverse Reactions) and any other events required to be reported in an expedited manner to the regulatory authorities and for informing investigators of reportable events, in compliance with applicable regulatory requirements within specific time frames. Investigators will notify the relevant IEC/IRBs of reportable events within the applicable time frames.

8.1.5 Data Safety Monitoring Board

Data Safety Monitoring Board (DSMB) is an independent advisory committee established by the sponsor/sponsor's designee of a clinical trial (CT) to assess the trial's progress and data on safety and critical efficacy endpoints at pre-determined intervals or on the request of the Ethics

Committee or the Regulatory Authorities when there are safety concerns and to recommend to the sponsor whether to continue, modify or stop the trial.

A DSMB will be constituted to assess the conduct of safety/efficacy data emerging from the study. Un-blinded data will be reviewed by the study team upon recommendation of the DSMB or if the review of blinded data is inconclusive to decide on. The independent team will provide unblinded data to DSMB as and when recommended. The independent team will not disclose unblinded data to the study team.

The DSMB will be utilized to determine whether the study should be terminated due to SAEs. The DSMB will receive all SAE reports as described as well as reports for any other AE that the investigator, medical monitor, or study monitor believes is appropriate. The DSMB will also review cases with intracerebral hemorrhage.

Periodic DSMB meetings are planned for after study completion of 100, 200, and 300 trial participants. DSMB may call the meetings whenever as required based on the SAE and AE reports submitted to them .

Safety laboratory data and adverse events will be monitored continuously throughout the study using data management's data review reports.

Below is the outline of the data required for the DSMB meeting.

Table 2: Specification of Data Outputs for DSMB Meetings

Output	Analysis set	DSMB
Baseline		
Patient disposition	ITT	Yes
Protocol deviations	ITT	Yes
Protocol deviations by patient level	ITT	Yes
Analysis population	ITT	Yes
Demographic characteristics	ITT	Yes
Background disease characteristics	ITT	Yes
Safety		
Concomitant medications/significant non-drug therapies	SAF	Yes
Relevant medical history/current medical condition	SAF	Yes
Drug exposure	SAF	Yes
Overview of adverse events	SAF	Yes
All adverse events and Serious adverse events by SOC and PT	SAF	Yes
All adverse events by subgroup age and sex	SAF	Yes
All adverse events by patient level	SAF	Yes
Adverse events of special interest (AESI) by SOC and PT	SAF	Yes

Output	Analysis set	DSMB
Newly occurring liver enzyme abnormalities (categorical analysis)	SAF	Yes
Hematology/liver parameters summary statistics	SAF	Yes
Hematology shift table	SAF	Yes
ECG summary statistics	SAF	Yes
ECG notables	SAF	Yes
Vital signs summary statistics	SAF	Yes
Vital signs notables	SAF	Yes
Abnormal lab/ECG/vital signs listing	SAF	Yes
Efficacy		
Modified Rankin Scale score of 0-2 on day 90 post-randomization	ITT	Yes
Summarization of death events	ITT	Yes

8.1.6 Procedures for Eliciting Reports of and Reporting Adverse Events and Intercurrent Illnesses

8.1.6.1 Eliciting, Documentation, and Reporting of Adverse Events

Information on AE will be derived by questioning the patients in general terms (e.g., "How do you feel?" or "How have you been feeling since the last questioning?" respectively), by patients' spontaneous reports, or by observation.

AEs will be documented on the source documents and eCRFs. Trained monitors will check the entries in the source documents. The AE will be transcribed to the AE eCRF pages. The following information will be given for each AE: Description of the AE, severity, start date, end date, incidence, relationship, seriousness, action taken, and outcome of the study drug.

8.1.7 Rating of Adverse Events

The investigator should describe an AE regarding severity, relatedness, and outcome. Severity of AE's will be categorized according to CTC Toxicity Grade 1 to 5. General definitions are included below:

- 0 = no adverse event or within normal limits
- 1 = mild adverse event
- 2 = moderate adverse event
- 3 = severe and undesirable adverse event
- 4 life-threatening or disabling adverse event
- 5 = death related to adverse event

The duration of the event will be described by the start date and end date.

The causal relationship of the event to the use of the IP is described in terms of related or not related to the IP, wherein related AEs include ‘Certain,’ ‘Probable,’ and ‘Possible,’ and not related includes ‘Unlikely’ and ‘Unassessable.’

Causality term	Assessment criteria
Certain	• Event or laboratory test abnormality with a plausible time relationship to drug intake • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e., an objective and specific medical disorder or a recognized pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable	• Event or laboratory test abnormality, with a reasonable time relationship to drug intake • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible	• Event or laboratory test abnormality, with a reasonable time relationship to drug intake • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear
Unlikely	• Event or laboratory test abnormality, with a time to drug intake that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations
Unassessable	• Report suggesting an adverse reaction • Cannot be judged because the information is insufficient or contradictory • Data cannot be supplemented or verified

The outcome of all AEs will be reported based on the following definitions, independent of whether they are serious or non-serious AEs, their severity, or their relationship to the IP:

- Recovered: Fully recovered, or by medical or surgical treatment, the condition has returned to the level observed at the first study-related activity after the patient signed the informed consent
- Recovering: The condition is improving, and the patient is expected to recover from the event. This term should only be used when the patient has completed the study
- Recovered with sequelae: As a result of the AE, the patient suffered persistent and significant disability/incapacity (e.g., became blind, deaf, paralyzed)
- Not recovered
- Fatal
- Unknown: This term should only be used in cases where the patient is lost to follow-up

Each AE will be followed to satisfactory clinical resolution. An IP-related event (adverse drug reaction) is an AE that is judged by the investigator or the sponsor/sponsor's designee to be either "Certain", "possibly related" or "probably related" to the study IP.

The Columbia-Suicide Severity Rating Scale (C-SSRS), a measure of suicidal ideation and behavior, will be used to document suicidality at baseline (Visit 1/Day 1) and post-dosing on Visit 1/Day 2, 3, 6, Visit 2/Day 30 $\pm$ 5, visit 3/Day 60 $\pm$ 5 (telephonic visit) and visit 4/Day 90 $\pm$ 5 30. The Baseline version of the C-SSRS will be used at baseline, and the Since Last Visit version will be used for post-dosing assessments. (Appendix 7). In completing the Since Last Visit C-SSRS assessments, a score of 4 or 5 with regards to suicidal ideation or any suicidal behavior reported should be brought to the attention and further explored by the investigator and should be handled using the appropriate standard of care.

8.2 Physical Examination

A physical examination of the following body systems will be performed at every study visit:

- Head-eyes-ear-nose-throat
- Cardiovascular system
- Respiratory system
- Nervous system
- Gastrointestinal system
- Musculoskeletal system
- Urogenital system
- Others, if required

8.3 Vital Signs

Respiratory rate, heart rate, systolic and diastolic BP, and body temperature will be measured at every study visit.

8.4 Pregnancy Control and Testing

Pregnancy is an exclusion criterion for enrolment in this study, but a subject could become pregnant during participation. Female participants are considered a WOCBP (Woman of Child Bearing Potential) following menarche and until becoming postmenopausal, unless permanently sterile. A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. A WOCBP must follow ≥ 1 highly effective method of contraception for the complete duration of the study, as outlined below. Additionally, male participants who are considered fertile must follow ≥ 1 highly effective method of contraception use for the duration of the trial as outlined below:

Highly Effective Methods of Contraception:

- Sexual abstinence (not having sex),
- Combination hormonal contraception associated with inhibition of ovulation and used in conjunction with a barrier method (e.g., male condom),
Progestogen-only hormonal contraception associated with inhibition of ovulation and used in conjunction with a barrier method (e.g., male condom),
- Intrauterine device (IUD),
- Intrauterine hormone-releasing system,
- bilateral tubal ligation or occlusion,
- Vasectomy (provided that the male participant or the vasectomized partner has received a medical assessment of surgical success).

In the event the partner of a male participant becomes pregnant during the trial, the outcome of the pregnancy must be followed. Participants not willing to use a highly effective contraception method will be excluded. (Refer to Appendix 8)

All pregnancy cases should be reported if they occurred during the study. Study staff must then maintain contact with the subject to obtain information about the outcome i.e., details about the delivery and the newborn, or pregnancy termination. This information should be provided to the Sponsor/Sponsor designee within one month of delivery.

8.5 Emergency Contact –Available 24 hours a day and 7 days a week

In case of any emergency, the following personnel should be contacted:

Name : Dr. Anil Gulati
 Address : Pharmazz, Inc.
 50 W 75th Street, Suite 105
 Willowbrook, IL 60527, USA
 Phone : +1 (630) 780-6087
 E-mail : Anil.gulati@pharmazz.com
 Mobile : +1 (630) 881-4840

The medical monitor will provide guidance in case of any emergency related to the safety of the patient. In addition, the medical monitor will also provide guidance on emergency unblinding. The sponsor should be contacted for an opinion before the code is unblinded.

9 STATISTICS

9.1 Statistical Method

All the statistical tests will be carried out as two-sided on a 5 % level of significance unless otherwise stated. All efficacy endpoints intended for inclusion in labeling will have an overall Type I error rate controlled at a two-sided 0.05 level. The descriptive statistics for continuous variables will be presented with the number (n) of non-missing observations, mean, standard deviation (SD), median, minimum, and maximum or range. For categorical data, the descriptive statistics will be presented with the number of exposed patients and the number (n) with the percentage of observations in the various categories of the endpoint, where the percentage will be based on the exposed patients. Descriptive analyses will also include graphical presentations of data where appropriate. Individual data listings will be provided.

A detailed statistical analysis plan (SAP) will be finalized, providing detailed methods for the analyses outlined below before the database is locked and prior to the analysis of the study being carried out. Any deviations from the analyses described below will be included in the SAP.

9.1.1 Study Population

The Randomized (RAN) population is defined as all patients randomized in the study.

The intent-to-treat (ITT) population is defined as all randomized patients who receive the study medication.

The per-protocol (PP) population is defined as those patients in the ITT population who completed Day 90 of treatment and for whom there were no serious protocol deviations.

The safety (SAF) population is defined as all randomized patients who receive at least one application of study medication.

9.1.2 Demographic and Other Baseline Characteristics

Demographic variables and patient characteristics will be summarized descriptively by treatment assignment and overall. Demographic variables will include age, weight, height, and BMI categories, and sex. Variables that are measured on a continuous scale, such as the age of the patient at the time of enrollment, the number of non-missing observations (n), mean, median, SD, minimum, and maximum, will be tabulated by treatment assignment, and overall for the RAN set. Variables that are measured on a categorical scale will be summarized using frequencies and

percentages by treatment assignment and overall for the ITT set. The Unpaired t-test will be used to compare the continuous variables between the groups at baseline.

9.1.3 Decoding of Randomization

The study blind may be broken in case of an emergency when the information on the treatment assigned is essential for the medical management of the patient. The investigator must notify the sponsor within 24 hours after determining that it is necessary to unblind the treatment assignment. The investigator must also indicate in source documents and in the eCRF that the blind was broken and provide the date, time, and reason for breaking the blind. Any AE or SAE that requires breaking the blind must be recorded and reported as specified in this protocol.

9.1.4 Safety Analysis

Adverse Effects

All safety parameters will be analyzed based on the SAF set.

The number and percentage of patients with treatment-emergent adverse events will be presented. The treatment-emergent adverse events (events started on or after the first dose of study medication or events present prior to the first dose of study medication but increased in severity based on preferred term) will be summarized by MedDRA primary System Organ Class (SOC) and Preferred Term (PT).

Adverse events will be summarized by presenting the number and percentage of patients having at least one AE, having an AE in each primary system organ class, and having each individual AE (preferred term). Summaries will also be presented for AEs by severity (CTC Toxicity grade) and for study treatment-related AEs. If a patient reported more than one adverse event with the same preferred term, the adverse event with the greatest severity will be presented. If a patient reported more than one adverse event within the same primary system organ class, the patient will be counted only once with the greatest severity at the system organ class level, where applicable.

The MedDRA version used for reporting the adverse events will be described in a footnote.

The most common adverse events reported ($\geq z$ % in any group for each preferred term in the table by SOC and PT) will be presented in the clinical study report by descending frequency according to their incidence in the overall group, starting from the most common event. Here the threshold value z is set to 5 (%), but it may be updated following a review of the dry run outputs.

Separate summaries will be provided for death, serious adverse events, other significant adverse events leading to treatment discontinuation, and adverse events leading to dose adjustment.

For the legal requirements of ClinicalTrials.gov, two required tables on treatment-emergent adverse events which are not serious adverse events with an incidence greater than 5% and on

treatment-emergent serious adverse events and SAE suspected to be related to study treatment will be provided by system organ class and preferred term on the safety set.

If, for the same patient, several consecutive AEs (irrespective of study treatment causality, seriousness, and severity) occurred with the same SOC and PT:

- A single occurrence will be counted if there is ≤ 1 -day gap between the end date of the preceding AE and the start date of the consecutive AE
- More than one occurrence will be counted if there is > 1 -day gap between the end date of the preceding AE and the start date of the consecutive AE

For occurrence, the presence of at least one SAE/SAE suspected to be related to study treatment/non-SAE has to be checked in a block, e.g., among AE's in a ≤ 1 -day gap block, if at least one SAE is occurring, then one occurrence is calculated for that SAE.

The number of deaths resulting from SAEs suspected to be related to study treatment and the number of deaths resulting from SAEs irrespective of causality will be provided.

The adverse events will be summarized for the following subgroup.

- Sex (male vs. female)
- Age group (< 65 , ≥ 65 yrs)
- IV thrombolytics vs. no IV thrombolytics
- Race

The adverse events of special interest (AESI) are defined in the Case retrieval Sheet (CRS), which is updated for each MedDRA dictionary. The latest version of MedDRA at the time of the final database lock will be used.

Table 3: List of Safety Topics of Interest

Safety Topic Of Interest	SOC	SOC code	MedDRA Term	Preferred term code
ICH(Intracranial haemorrhage)	Nervous system disorders	10029205	Hemorrhage intracranial	10018985

The number and percentage of patients with these special AEs will be summarized. In addition, listings of related adverse events will be provided.

Adverse events of special interest are also analyzed as flagged by the investigator on the eCRF.

Newly occurring liver enzyme abnormalities and QTcF (QT interval corrected for heart rate according to Fredericia) abnormalities will be summarized. QTcF will be summarized in post-text tables and will be presented in the listing.

Laboratory Tests

Descriptive statistics will be generated for all clinical laboratory tests performed (actual values and changes from baseline, where applicable) for three groups of laboratory tests (Hematology, Biochemistry, and Liver function test) by laboratory test and treatment group. Change from baseline will only be summarized for patients with both baseline and post-baseline values and will be calculated as follows:

$$\text{change from baseline} = \text{post-baseline value} - \text{baseline value}$$

The following laboratory parameters will be analyzed for the hematology test group: hemoglobin (Hb), red blood cells (RBC), white blood cells (WBC), including differential count, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), and platelet count.

The following laboratory parameters will be analyzed for the Liver function test group: Alkaline phosphatase, ALT, AST, Total Bilirubin, PTT, PT/INR, Albumin, BUN, and serum creatinine.

The following laboratory parameters will be summarized for the Biochemistry test group at the baseline visit: Blood glucose test, Lipid profile (LDL, HDL, VLDL), blood cholesterol, Sodium, Potassium, and Calcium. No change from baseline will be presented as these assessments are done only at the baseline visit.

Shift tables using the low/normal/high classification will be used to compare the baseline to each post-baseline value of hematology and liver function test parameters.

Furthermore, the number and percentage of patients with newly occurring or worsening abnormalities in liver function tests will be summarized by treatment and at any time post-baseline considering all post-baseline data from scheduled, unscheduled, and premature discontinuation visits. Abnormalities in liver function tests will be based on normal ranges.

Vital Signs

The following quantitative variables will be summarized: weight (kg), body temperature (°C), heart rate (beats/min), respiratory rate (breaths/min), supine systolic blood pressure (mmHg), and supine diastolic blood pressure (mmHg).

Analysis of vital sign measurement using descriptive summary statistics for the change from baseline for each post-baseline visit will be performed by the vital signs and treatment group. Change from baseline will only be summarized for patients with both baseline and post-baseline values and will be calculated as follows:

$$\text{Change from baseline} = \text{post-baseline value} - \text{baseline value}$$

The number and percentage of patients with newly occurring notable vital signs will be presented at any time post-baseline. Criteria for notable vital sign abnormalities are provided.

Heart rate will be summarized categorically by computing the number and percentage of patients at each time point and the maximum post-baseline value for the following categories:

Heart Rate:

- Increase from baseline >25% and to a value >100 bpm
- Decrease from baseline >25% and to a value <50 bpm

ECG

The following quantitative variables will be summarized for averaged triplicate ECG: RR interval, PR interval, QRS interval, QT interval, QTcB, and QTcF. In addition, summary statistics for change from baseline will be presented for ECG variables by timepoint and treatment group.

QT intervals will be summarized categorically by computing the number and percentage of patients at each time point and the maximum post-baseline value for the following categories:

- QT, QTcF
 - New ≥ 450 and < 480 ms
 - New ≥ 480 and < 500 ms
 - New ≥ 500 ms
 - Increase from baseline of ≥ 30 ms to < 60 ms
 - Increase from baseline of ≥ 60 ms
- HR
 - Increase from baseline >25% and to a value >100 bpm
 - Decrease from baseline >25% and to a value <50 bpm
- PR
 - Increase from baseline >25% and to a value >200 ms

- The new value of >200 ms
- QRS
 - Increase from baseline >25% and to a value >120 ms
 - New >120 ms

A listing of all newly occurring or worsening abnormalities and a by-patient listing of all quantitative ECG parameters will be provided.

C-SSRS - The C-SSRS assessment will be summarized by visit.

Immunogenicity Potential - The immunogenicity assessments will be summarized by visit.

Clinical laboratory parameters, vital signs, and physical examination will be measured descriptively. Summary statistics for absolute results, change from baseline, and shift tables will be generated. The Unpaired t-test will be used to compare the discrete variables between the groups at baseline and at follow-ups. Descriptive analyses will also include graphical presentations of data where appropriate. Individual data listings will be provided. The statistical test will be used as per the nature of the data and carried out as two-sided on a 5 % level of significance.

9.1.5 Efficacy Analysis

Primary Estimand

The estimand for primary analysis is detailed in the SAP and is in line with ICH E9(R1) addendum. An estimand defines the target of estimation for a particular trial objective (i.e., ‘what is to be estimated’) and is described by the following attributes: the target population, the variable, the specification on how to account for intercurrent events, and the summary measure. The SAP will have the details on the handling of intercurrent events along with other aspects of the estimand like target population, the variable of interest, and summary measure.

The associated estimand for the primary objective is to measure the effect of treatment with sovateltide as assessed by the proportion of acute cerebral ischemic stroke patients having a modified Rankin Scale score of 0-2 on day 90 post-randomization.

The following four attributes more comprehensively describe the primary estimand:

1. The target population is adult patients aged 18 – 80 years of age with Cerebral ischemic stroke presenting within 24 hours after the onset of symptoms with NIHSS score of ≥ 8 and < 20 as well as NIHSS Level of Consciousness (1A) score < 2 at the time of screening. This includes cerebral ischemic stroke patients who completely recovered from earlier episodes before having a new or fresh stroke with pre-stroke historical measure of mRS score of 0-2.

2. The **primary variable** is the proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale (mRS) score of 0-2 on day 90 post-randomization.
3. **Intercurrent events** of interest in this study are:
 - a. lost to follow-up
 - b. subjects who do not receive the third series of doses
4. The **summary measure** is the odds ratio of the mRS score of 0-2 on day 90 post-randomization between sovateltide and placebo.

The primary comparison of interest is the proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale (mRS) score of 0-2 on day 90 post-randomization. In addition, treatment comparisons between sovateltide + standard of care and placebo + standard of care will be made using logistic regression with baseline stratification and treatment group as factors and baseline value as a covariate in the model. The primary endpoint will be assessed in the ITT population. In line with the treatment effect of interest, the primary observation period will include all the available data for patients, including those who missed the dose on day 6 and those who discontinued it. Since the treatment effect of interest is on treatment, participants who were randomized but not treated will be excluded from this analysis.

The primary observation period will be defined as the time from the date of the randomization to the last study day. The last study day will be EOS/90 days, and in case of discontinuation/lost to follow-up, the last study day will be the date of discontinuation/lost to follow-up.

A logistic regression model with stratification and treatment group as factors and baseline value as a covariate will be used for analysis. The estimates of the odds ratio for the mRS score of 0-2 on day 90 post-randomization between treatment groups and their 95% confidence intervals will be provided.

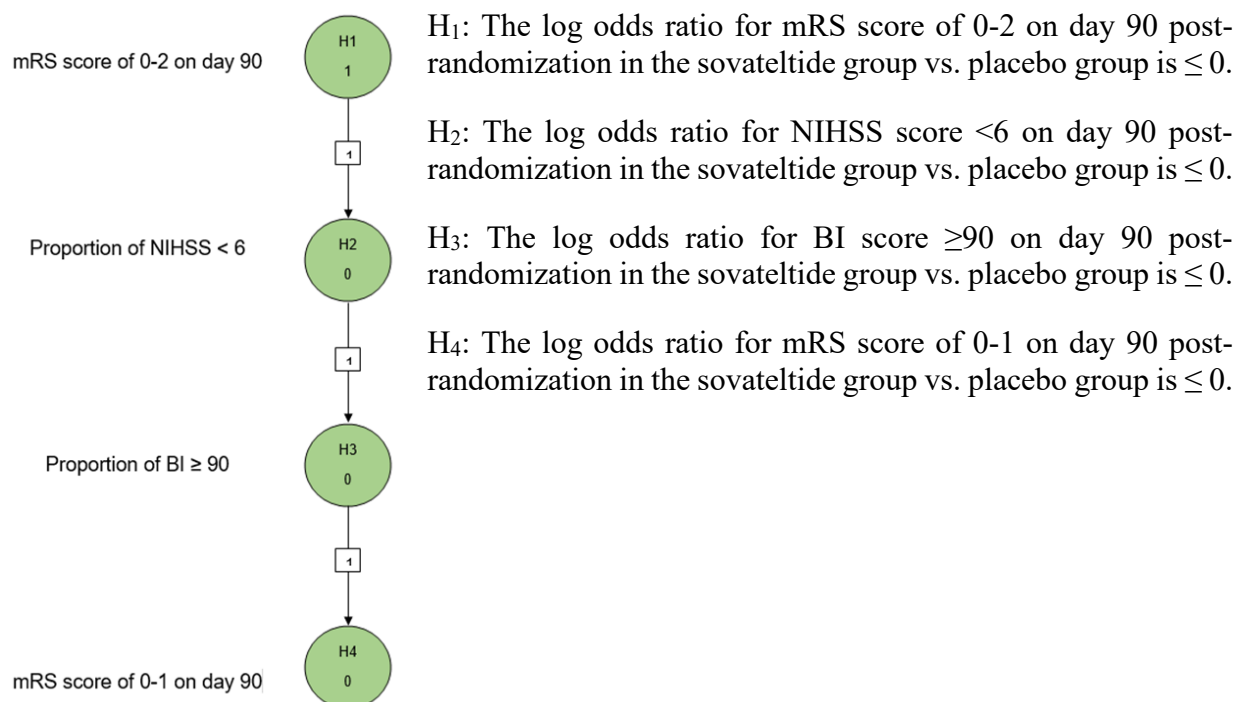
The following hypothesis will be tested:

The primary statistical null hypothesis is:

- H_1 : The log odds ratio for mRS score of 0-2 on day 90 post-randomization in the sovateltide group vs. placebo group is ≤ 0
- H_{1A} : The log odds ratio for mRS score of 0-2 on day 90 post-randomization in the sovateltide group vs. placebo group is > 0

This hypothesis will be tested using a one-sided test, with a z-score corresponding to the log of the odds ratio for the mRS score of 0-2 on day 90 post-randomization between the two groups with 0.025 α -level.

The following hypotheses will be tested sequentially and are included in the hierarchical testing strategy and type-I-error will be set such that a family-wise type-I-error of $\alpha = 2.5\%$ (one-sided) is kept. The graphical approach of (Bretz et al. 2009) for sequentially rejective testing procedures is used to illustrate the testing strategy:



The secondary efficacy variables involved in the above testing strategy will be analyzed analogously to the primary endpoints. i.e., logistic regression model with stratification and treatment group as factors and baseline value as a covariate. The estimates of the odds ratio on day 90 post-randomization between treatment groups and their 95% confidence intervals will be provided. Missing data will be imputed as specified in section 2.5.3 of the statistical analysis plan. The Multiple Imputation (MI) method with 50 iterations creating 100 copies will be used to impute the missing assessments of the respective endpoint at day 90 if the day 90 assessment is missing.

The testing sequence will continue to H2 at familywise Type I error rate at a two-sided 0.05 level only if H1 has been rejected at 0.025 (one-sided). Similarly, the testing sequence will continue to H3 at familywise Type I error rate at a two-sided 0.05 level only if H2 testing has been rejected and the testing sequence will continue to H4 at familywise Type I error rate at a two-sided 0.05 level only if H3 testing has been rejected.

All efficacy endpoints intended for inclusion in labeling will have an overall Type I error rate controlled at a two-sided 0.05 level of significance.

Logistic regression

The same analyses as for the primary endpoint will be repeated for:

- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with National Institute of Health Stroke Scale (NIHSS) score <6 on Day 90.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with Barthel Index (BI) score ≥ 90 on Day 90.
- The proportion of acute cerebral ischemic stroke patients having an excellent functional outcome with a modified Rankin Scale score of 0-1 on day 90 post-randomization.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale score of 0-2 on day 30 post-randomization.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with National Institute of Health Stroke Scale (NIHSS) score <6 on Day 30.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with Barthel Index (BI) score ≥ 90 on Day 30.

Mixed model repeated measures analyses (MMRM)

Mixed model repeated measures analyses (MMRM) will be used for analyzing the below parameters with treatment, stratification factor, visit, treatment*visit interaction as factors, and baseline score as a covariate.

- Change in QoL as assessed by EuroQol EQ-5D-5L from baseline to day 30 and 90 post-randomization
- Change in QoL as assessed by SS-QOL from baseline to day 30 and 90 post-randomization
- Change in MoCA score from day 6 to days 30 and 90 post-randomization

Cochran-Mantel-Haenszel test

The Cochran–Mantel–Haenszel test (CMH) for stratified categorical data of proportion will be used for analyzing the below parameters for comparison between treatment groups if the number allows.

- The proportion of patients with recurrent ischemic stroke within 90 days post-randomization

- Number of deaths within day 90 post-randomization.
- The proportion of patients with symptomatic ICH within 24 ($\pm$ 6) hours of randomization.
- The proportion of patients with radiographic ICH within 24 ($\pm$ 6) hours of randomization.
- The proportion of patients with adverse events (AEs) and serious adverse events (SAEs).

Other Exploratory Analyses

The exploratory objectives are to determine:

- A good functional outcome in patients with acute cerebral ischemic stroke assessed by mRS score of 0-2 at day 30 post-randomization.
- A good functional outcome in patients with acute cerebral ischemic stroke assessed by the NIHSS) score of <6 at day 30 post-randomization.
- A good functional outcome in patients with acute cerebral ischemic stroke assessed by the Barthel Index (BI) score of ≥ 90 at day 30 post-randomization.
- Alberta Stroke Program Early CT (ASPECT) Score for stroke severity at baseline and identify sites of ischemic lesions
- The subtype of the index acute cerebral ischemic stroke according to the TOAST classification by day 30 post randomization

The endpoints for achieving these objectives are:

- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with a modified Rankin Scale score of 0-2 on day 30 post-randomization.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with National Institute of Health Stroke Scale (NIHSS) score <6 on Day 30.
- The proportion of acute cerebral ischemic stroke patients having a good functional outcome with Barthel Index (BI) score ≥ 90 on Day 30.
- Compare baseline Alberta Stroke Program Early CT (ASPECT) Score for stroke severity and identify sites of ischemic lesions
- Compare the subtype of the index acute cerebral ischemic stroke by TOAST classification by day 30 post randomization

The ASPECT score is a 10-point score. The ASPECT scores can only be compared at baseline using a simple t-test.

The subtypes of TOAST classification, which are binary in nature, will also be compared only at day 30 post randomization. These subtypes will be compared using the chi-square test.

9.2 Sample Size

The sample size for this study is calculated based on the results of our recently completed phase III clinical study in patients of cerebral ischemic stroke in India (CTRI/2019/09/021373; NCT04047563). An enrollment ratio of 1:1 is considered for the present phase III study.

The primary endpoint of the study is the **modified Rankin Scale score** at day 90 post-randomization. The key secondary endpoints are **(NIHSS) score of <6** and **Barthel Index (BI) score of ≥90** at day 90 post-randomization. Treatment with placebo is expected to obtain 55.71% of patients having an mRS score ≤ 2 at day 90 post-randomization, and treatment with sovateltide is hypothesized to obtain 80.60% of patients to have an mRS score ≤ 2 at day 90 post-randomization. In the referenced phase III study, an odds ratio of 3.302 was obtained; however, to account for variation in different countries, a conservative approach was used and set the odds ratio to 2.45. Therefore, a sample size of 274 is required to achieve 90% power for the primary analysis to show significance in log-scale relative risk, i.e., $\log(\text{OR}) > 0$ in the proportion of patients with favorable outcomes between the Sovateltide and placebo arms. The above sample size calculation assumes above mentioned proportions for placebo and sovateltide with a Type I error probability of 0.025 (one-sided for assessing the patients who completed the 90-day follow-up assessments).

For the National Institute of Health Stroke Scale (NIHSS) score of <6 at day 90 post-randomization a sample size of 166 is required to achieve 90% power for the analysis to show significance in log-scale relative risk i.e., $\log(\text{OR}) > 0$ in the proportion of patients with favorable outcomes between the Sovateltide and placebo arms. The above sample size calculation assumes a placebo proportion of 0.7, Sovateltide proportion of 0.9104 and Type I error probability of 0.025 (one-sided for assessing the patients who completed the 90-day follow-up assessments).

For the Barthel Index (BI) score of ≥ 90 at day 90 post-randomization a sample size of 436 is required in order to achieve 90% power for the analysis to show significance in log-scale relative risk i.e., $\log(\text{OR}) > 0$ in the proportion of patients with favorable outcomes between the Sovateltide and placebo arms. The above sample size calculation assumes a placebo proportion of 0.4714, Sovateltide proportion of 0.6269 and Type I error probability of 0.025 (one-sided for assessing the patients who completed the 90-day follow-up assessments). The ITT principle will be applied to the analysis and, therefore, to safeguard against dilution of the treatment effect associated with an approximate 15% non-adherence rate (due to lost to follow-up, consent withdrawal, treatment crossovers, and stroke mimics), the above sample needs to be inflated by a factor of 1/0.85 or 1.1765. Therefore, a total sample size of 512.94 rounding it to 514, is required for this study.

9.3 Level of Significance

The data will be summarized descriptively, including the sample number, mean, SD, minimum, median, maximum, and 95 % CI. Applicable comparative tests will have a 0.05 significance level.

9.4 Study Termination Criteria

The sponsor reserves the right to discontinue the study at any time upon EC approval. Reasons for the termination will be provided to the patients and EC. The investigator reserves the right to discontinue the study at any time for the reasons of the patient's safety and welfare. In addition, the IEC may terminate the study if there are major violations of ethics or due to any serious adverse event (s). An alternative therapy should be provided to patients on discontinuation from the study. The sponsor will notify the investigator and ethics committees in writing, including the reason(s).

9.5 Missing Data

The Multiple Imputation (MI) method with 50 iterations creating 100 copies will be used to impute the missing assessments of mRS at day 90 on the primary endpoint if day 90 assessment is missing. No missing data imputation will be done for the endpoints while performing secondary objectives-related analysis.

9.6 Procedures for Reporting Deviations from Original Statistical Plan

Any post-hoc, or unplanned analyses not identified in SAP will be clearly identified in the respective clinical study report (CSR).

10 DIRECT ACCESS TO SOURCE DATA/DOCUMENTS

10.1 Monitoring

In accordance with applicable regulations and ICH-GCP guidelines, the sponsor or its designee's monitors will contact the site prior to the start of the study to review with the site staff the protocol, study requirements, and their responsibilities to satisfy regulatory, ethical, and sponsor's requirements. When reviewing the data collection procedures, the discussion will also include identification, agreement, and documentation of data items for which the eCRF will serve as the source document.

The sponsor and or its designee will monitor the study for protocol compliance verifying the following, but not limited to:

- Safety and rights of patients are being protected
- Study is conducted in accordance with the currently approved protocol and any other study agreements, ICH-GCP guidelines, and all applicable regulatory requirements

- Data reported is authentic, accurate, and complete

The investigator agrees to allow the monitor direct access to all relevant documents for the purpose of verification of available data.

10.2 Documentation at the Study Site

All data relating to the study will be documented in the eCRF. This eCRF is developed to record the data requested by the protocol. The investigator will ensure the accuracy, completeness, legibility, and timeliness of the data recorded in the eCRF. At the beginning of the study, a site master file will be established at the investigational site. The investigator will maintain the study documents as specified in the ICH-GCP and as required by the applicable regulatory requirements. The investigator will take measures to prevent the accidental or premature destruction of these documents. Prior to the start of the study, a signature and delegation list will be completed showing the signatures and hand-written initials of all who are authorized to enter data or make corrections in the eCRF. The investigator will permit study-related monitoring, audits, and regulatory inspection, providing direct access to source data/documents.

10.3 Patient's Data and Data Protection

To protect the patient's identity, patient initials and a patient number will be assigned by the investigator to each study patient and used in lieu of the patient's name when the investigator reports AEs and/or other study-related data. Personal information will be treated as confidential but may need to be reviewed by authorized representatives of the sponsor (monitor and auditor, respectively) and regulatory authorities. The patient's consent for direct access to his original medical records for data verification purposes has to be obtained prior to a patient's participation in the study.

The investigator must maintain a list of names and identifying information (e.g., initials, date of birth, patient identification code, and date of study entry) of all patients enrolled in the study. The investigator will keep the patient identification code list in the site master file.

10.4 Data Management and Analysis

Data management based on GCP refers to the activities defined for achieving routines for entering patient information into a database in an efficient and error-free manner.

Data management routines include procedures for handling the eCRFs, database setup and handling, data entry and verification, data validation, and documentation of activities performed, including information on discrepancies in the procedure.

All of the AEs, including SAEs reported in the study, will be entered into the study database. The initial notification of SAEs will also be entered into the safety database for coding, medical evaluation, and notification to regulatory authorities according to national regulatory requirements. Before database lock, SAE reconciliation will be performed between the safety and study databases.

Data will be entered and verified in a validated database. Data will be validated continuously by running logic checks and manual reviews. For any discrepancies identified in the study database, queries will be generated in the clinical data management system for resolution. When the data have been entered, verified, and validated, the database will be locked for analysis.

10.5 Amendments to the Protocol

The investigator will not implement any deviation from or changes to the protocol without agreement by the sponsor and prior review and documented approval/favorable opinion from the IRB/IEC of an amendment, except where necessary to eliminate immediate hazards to study patient or when the changes involve only logistical or administrative aspects of the study (e.g., change in monitor(s), change in telephone numbers).

All changes or deviations of the trial must be confirmed in writing. Amendment(s) will be signed off in the same way as the protocol and submitted to the applicable authorities and IRB/IEC.

11 QUALITY CONTROL AND QUALITY ASSURANCE

11.1 Audit

To ensure compliance with ICH-GCP guidelines and all applicable regulatory requirements, the sponsor may conduct a quality assurance audit following intimation and appointment. Regulatory agencies may also conduct a regulatory inspection of this study. Such audits/inspections can occur at any time during or after the completion of the study. If an audit or inspection occurs, the investigator and institution agree to allow the auditor/inspector direct access to all relevant documents and to allocate his/her time and the time of his/her staff to the auditor/inspector to discuss findings and any relevant issues.

12 ETHICS

Before initiation of the study, the sponsor will seek permission from the regulatory authorities to conduct the study. All documents required by the appropriate authorities will be submitted. Any notification/submission will be dated and contain sufficient information to identify the protocol.

12.1 Institutional Ethics Committee or Institutional Review Board

The study protocol, amendments to the protocol (if applicable), IP information, patient information, ICF, and consent form updates (if applicable) will be submitted to the IRB/IEC, which is constituted according to local law to obtain approval before initiation of the study. This study will be initiated after the protocol is reviewed and approved by the concerned IRB/IEC. The approval should be kept on file in the site master file with a copy in the trial master file. The investigator will report promptly to the IRB/IEC new information that may adversely affect the safety of the patients or the conduct of the trial.

The present study is registered on clinicaltrials.gov (NCT05691244).

12.2 Ethical Conduct of the Study

The Declaration of Helsinki, the consolidated ICH-GCP, the European Union (EU) Clinical Trials Regulation No. 536/2014, and other national law(s) in the countries where the trial takes place shall constitute the main reference guidelines for ethical and regulatory conduct.

Given that the IP is intended to be administered during the acute phases following a stroke (starting within 24 hours and continuing to 6 days), there will be no provision for the continued administration of the IP to trial participants once the trial is complete. This information is indicated in the ICF provided to participants. Additionally, this information is in line with the ethical guidelines set forth by the World Medical Association Declaration of Helsinki, which is entitled “Ethical Principles for Medical Research Involving Human Subjects (1996).”

At the conclusion of the trial, the results and treatment allocations will be shared with the PIs and disseminated to participants as required locally.

If sufficient evidence exists at the conclusion of the trial, we will seek a marketing authorization from the EMA to provide sovateltide as a possible treatment for future stroke patients.

12.3 Patient Information and Consent

The patient will be screened and included in the study only after giving them adequate and appropriate information and obtaining written informed consent. The patient will be given sufficient time to consider the study’s implications before deciding to participate. The patient will be provided with a copy of the signed informed consent form. The confidentiality of the patient’s records will be maintained.

Should there be any amendments to the protocol, such that would directly affect the patient’s participation in the study, e.g., a change in any procedure, the ICF will be amended to incorporate

this modification, and the patient must agree to sign this amended form indicating that they re-consent to continue their participation in the study.

The investigator is responsible for obtaining the patient's freely given written consent. The consent form must be signed by the patient or LAR, and the investigator-designated research professional obtaining the consent

The consent process will follow a hierarchical approach.

1. The first option for obtaining informed consent will be from the patient.
2. If the patient does not have the capacity to consent, then it will be obtained from the LAR prior to any protocol specified procedures.

All methods of conducting the informed consent process must comply with ICH GCP E6. Approval for all method(s) of consent must be obtained prior to implementation. (Appendix 8).

12.3.1 Regained Capacity Consent

If the original consent process involved anyone other than the participant, and if required by local standards, consent will be sought for the remaining procedures from the participant once they are deemed to have regained capacity. The PI/delegates will make ongoing efforts to obtain consent until (1) regained capacity consent is obtained from the participant once reasonable, (2) death, or (3) completion of the Day 90 assessments.

12.3.2 Amendment to the ICF

In the event that new information is available necessitating an ICF amendment, and if required by local Health Authorities and/or applicable ethics committees, active participants must be re-consented to the most current version of the approved ICF. A copy of each ICF will be provided to the participant.

13 DATA HANDLING AND RECORD KEEPING

The investigator must maintain all documentation relating to this study. Essential documents (as defined in the ICH-GCP) must be retained until at least 2-years after the last approval of a marketing application in an ICH region and until there are no pending or contemplated marketing applications in an ICH region or at least 2-years have elapsed since the formal discontinuation of clinical development of the investigational product. These documents must be retained for a longer period, however, if required by the applicable regulatory requirement(s) or if needed by the sponsor.

In any case, all study records such as but not limited to eCRF, regulatory documents, patient identification code list, patient files, and other source data that support eCRFs, must be retained for at least 15 years after the completion or discontinuation of the study. The investigator must notify the sponsor or its designee of any changes in the archival arrangements, including, but not limited to, archival at an off-site facility and transfer of ownership of the records in the event the investigator leaves the site. These documents must be retained for a longer period, however, if required by the applicable regulatory requirement(s) or if needed by the sponsor.

The sponsor will notify the investigator in writing when the study-related records are no longer needed.

14 FINANCING AND INSURANCE

Pharmazz, Inc. is the sponsor of this study, and all agreements between the investigator and sponsor or its designee must be signed prior to the inclusion of the first patient in the clinical study. The agreement must clearly state the rights and obligations of the parties concerned and include a detailed financial settlement.

Every patient participating in the study will be insured in accordance with the local legal requirements against study-related injuries to health which may occur during the study.

Excluded from this, however, are injuries to health and deterioration of illnesses already present, which would have occurred or continued to exist even if the patient had not taken part in this clinical study. Any injury to health, which might have occurred as a result of participation in the clinical study, must be reported by the investigator.

The sponsor is self-insured, which covers product liability and patient injuries.

15 PUBLICATION POLICY

A CSR, according to the ICH E3 Guidelines for Structure and Content of CSRs, will be prepared by Pharmazz Inc./Designee upon full completion of the study. This CSR, as well as any discovery and improvement related to IP, shall be the property of the sponsor.

Neither the complete nor any part of the results of the study carried out under this protocol, nor any of the information provided by the sponsor for the purposes of performing the study will be published or passed on to any third party without the consent of the sponsor. Any investigator involved with this study is obligated to provide the sponsor with complete test results and all data derived from the study.

The investigator agrees to keep strictly confidential all unpublished information and results concerning this study. Unpublished information must not be published or disclosed without the sponsor's prior written approval.

16 APPENDICES

Appendix 1: National Institutes of Health Stroke Scale

Appendix 2: Modified Rankin Scale

Appendix 3: Barthel Index

Appendix 4: EuroQoL-5D-5L

Appendix 5: Stroke-Specific Quality of Life

Appendix 6: Montreal Cognitive Assessment (MoCA) Scale

Appendix 7: Columbia-Suicide Severity Rating Scale (C-SSRS)

Appendix 8: Contraception Definitions, Contraceptive Requirements, and Pregnancy Testing

APPENDIX 1: NATIONAL INSTITUTE OF HEALTH (NIH) STROKE SCALE

Instructions	Score
1a. Level of Consciousness* The investigator must choose a response if a full evaluation is prevented by such obstacles as an endotracheal tube, language barrier, orotracheal trauma / bandages. A 3 is scored only if the patient makes no movement (other than reflexive posturing) in response to noxious stimulation.	☐ 0 = Alert; keenly responsive. ☐ 1 = Not alert; but arousable by minor stimulation to obey, answer, or respond. ☐ 2 = Not alert; requires repeated stimulation to attend or is obtunded and requires strong or painful stimulation to make movements (not stereotyped). ☐ 3 = Responds only with reflex motor or autonomic effects or totally unresponsive, flaccid, and areflexic.
1b. LOC Questions The patient is asked the month and his/her age. The answer must be correct - there is no partial credit for being close. Aphasic and stuporous patients who do not comprehend the questions will score 2. Patients unable to speak because of endotracheal intubation, orotracheal trauma, severe dysarthria from any cause, language barrier, or any other problem not secondary to aphasia are given a 1. It is important that only the initial answer be graded and that the examiner not "help" the patient with verbal or non-verbal cues.	☐ 0 = Answers both questions correctly. ☐ 1 = Answers one question correctly. ☐ 2 = Answers neither question correctly.
1c. LOC Commands The patient is asked to open and close the eyes and then to grip and release the non-paretic hand. Substitute another one step command if the hands cannot be used. Credit is given if an unequivocal attempt is made but not completed due to weakness. If the patient does not respond to command, the task should be demonstrated to him or her (pantomime), and the result scored (i.e., follows none, one or two commands). Patients with trauma, amputation, or other physical impediments should be given suitable one-step commands. Only the first attempt is scored.	☐ 0 = Performs both tasks correctly. ☐ 1 = Performs one task correctly. ☐ 2 = Performs neither task correctly.

Instructions	Score
2. Best Gaze Only horizontal eye movements will be tested. Voluntary or reflexive (oculocephalic) eye movements will be scored, but caloric testing is not done. If the patient has a conjugate deviation of the eyes that can be overcome by voluntary or reflexive activity, the score will be 1. If a patient has an isolated peripheral nerve paresis (CN III, IV or VI), score a 1. Gaze is testable in all aphasic patients. Patients with ocular trauma, bandages, pre-existing blindness, or other disorder of visual acuity or fields should be tested with reflexive movements, and a choice made by the investigator. Establishing eye contact and then moving about the patient from side to side will occasionally clarify the presence of a partial gaze palsy.	☐ 0 = Normal. ☐ 1 = Partial gaze palsy; gaze is abnormal in one or both eyes but forced deviation or total gaze paresis is not present. ☐ 2 = Forced deviation, or total gaze paresis not overcome by the oculocephalic maneuver.
3. Visual Visual fields (upper and lower quadrants) are tested by confrontation, using finger counting or visual threat, as appropriate. Patients may be encouraged, but if they look at the side of the moving fingers appropriately, this can be scored as normal. If there is unilateral blindness or enucleation, visual fields in the remaining eye are scored. Score 1 only if a clear-cut asymmetry, including quadrantanopia, is found. If patient is blind from any cause, score 3. Double simultaneous stimulation is performed at this point. If there is extinction, patient receives a 1, and the results are used to respond to item 11.	☐ 0 = No visual loss. ☐ 1 = Partial hemianopia. ☐ 2 = Complete hemianopia. ☐ 3 = Bilateral hemianopia (blind including cortical blindness)

Instructions	Score
4. Facial Palsy Ask – or use pantomime to encourage – the patient to show teeth or raise eyebrows and close eyes. Score symmetry of grimace in response to noxious stimuli in the poorly responsive or non-comprehending patient. If facial trauma/bandages, orotracheal tube, tape or other physical barriers obscure the face, these should be removed to the extent possible.	☐ 0 = Normal symmetrical movements. ☐ 1 = Minor paralysis (flattened nasolabial fold, asymmetry on smiling). ☐ 2 = Partial paralysis (total or near-total paralysis of lower face). ☐ 3 = Complete paralysis of one or both sides (absence of facial movement in the upper and lower face).
5. Motor Arm The limb is placed in the appropriate position: extend the arms (palms down) 90 degrees (if sitting) or 45 degrees (if supine). Drift is scored if the arm falls before 10 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic arm. Only in the case of amputation or joint fusion at the shoulder, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	☐ 0 = No drift ; limb holds 90 (or 45) degrees for full 10 seconds. ☐ 1 = Drift ; limb holds 90 (or 45) degrees, but drifts down before full 10 seconds; does not hit bed or other support. ☐ 2 = Some effort against gravity ; limb cannot get to or maintain (if cued) 90 (or 45) degrees, drifts down to bed, but has some effort against gravity. ☐ 3 = No effort against gravity ; limb falls. ☐ 4 = No movement . ☐ UN = Amputation or joint fusion, explain: _____ Score: 5a. Left Arm _____ 5b. Right Arm _____
6. Motor Leg The limb is placed in the appropriate position: hold the leg at 30 degrees (always tested supine). Drift is scored if the leg falls before 5 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic leg. Only in the case of amputation or joint fusion at the hip, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	☐ 0 = No drift ; leg holds 30-degree position for full 5 seconds. ☐ 1 = Drift ; leg falls by the end of the 5-second period but does not hit bed. ☐ 2 = Some effort against gravity ; leg falls to bed by 5 seconds but has some effort against gravity. ☐ 3 = No effort against gravity ; leg falls to bed immediately. ☐ 4 = No movement . ☐ UN = Amputation or joint fusion, explain: _____ Score: 6a. Left Leg _____ 6b. Right Leg _____

Instructions	Score
7. Limb Ataxia This item is aimed at finding evidence of a unilateral cerebellar lesion. Test with eyes open. In case of visual defect, ensure testing is done in intact visual field. The finger-nose-finger and heel-shin tests are performed on both sides, and ataxia is scored only if present out of proportion to weakness. Ataxia is absent in the patient who cannot understand or is paralyzed. Only in the case of amputation or joint fusion, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice. In case of blindness, test by having the patient touch nose from extended arm position.	☐ 0 = Absent. ☐ 1 = Present in one limb. ☐ 2 = Present in two limbs. ☐ UN = Amputation or joint fusion, explain:
8. Sensory Sensation or grimace to pinprick when tested, or withdrawal from noxious stimulus in the obtunded or aphasic patient. Only sensory loss attributed to stroke is scored as abnormal and the examiner should test as many body areas (arms [not hands], legs, trunk, face) as needed to accurately check for hemisensory loss. A score of 2, "severe or total sensory loss," should only be given when a severe or total loss of sensation can be clearly demonstrated. Stuporous and aphasic patients will, therefore, probably score 1 or 0. The patient with brainstem stroke who has bilateral loss of sensation is scored 2. If the patient does not respond and is quadriplegic, score 2. Patients in a coma (item 1a=3) are automatically given a 2 on this item.	☐ 0 = Normal; no sensory loss. ☐ 1 = Mild-to-moderate sensory loss; patient feels pinprick is less sharp or is dull on the affected side; or there is a loss of superficial pain with pinprick, but patient is aware of being touched. ☐ 2 = Severe to total sensory loss; patient is not aware of being touched in the face, arm, and leg.
9. Best Language A great deal of information about comprehension will be obtained during the preceding sections of the	☐ 0 = No aphasia; normal. ☐ 1 = Mild-to-moderate aphasia; some obvious loss of fluency or facility of comprehension, without

Instructions	Score
examination. Comprehension is judged from responses, as well as to all of the commands in the preceding general neurological exam. If visual loss interferes with the tests, ask the patient to identify objects placed in the hand, repeat, and produce speech. The intubated patient should be asked to write. The patient in a coma (item 1a=3) will automatically score 3 on this item. The examiner must choose a score for the patient with stupor or limited cooperation, but a score of 3 should be used only if the patient is mute and follows no one-step commands.	significant limitation on ideas expressed or form of expression. Reduction of speech and/or comprehension, however, makes conversation about provided materials difficult or impossible. For example, in conversation about provided materials, examiner can identify picture or naming card content from patient's response. ☐ 2 = Severe aphasia; all communication is through fragmentary expression; great need for inference, questioning, and guessing by the listener. Range of information that can be exchanged is limited; listener carries burden of communication. Examiner cannot identify materials provided from patient response. ☐ 3 = Mute, global aphasia; no usable speech or auditory comprehension.
10. Dysarthria If patient is thought to be normal, an adequate sample of speech must be obtained by asking patient to read or repeat words. If the patient has severe aphasia, the clarity of articulation of spontaneous speech can be rated. Only if the patient is intubated or has other physical barriers to producing speech, the examiner should record the score as untestable (UN), and clearly write an explanation for this choice. Do not tell the patient why he or she is being tested.	☐ 0 = Normal. ☐ 1 = Mild-to-moderate dysarthria; patient slurs at least some words and, at worst, can be understood with some difficulty. ☐ 2 = Severe dysarthria; patient's speech is so slurred as to be unintelligible in the absence of or out of proportion to any dysphasia or is mute/anarthric. ☐ UN = Intubated or other physical barrier, explain:
11. Extinction and Inattention (formerly Neglect) Sufficient information to identify neglect may be obtained during the prior testing. If the patient has a severe visual loss preventing visual double simultaneous stimulation, and the cutaneous stimuli are normal, the score is normal. If the patient has aphasia but does appear to attend to both sides, the score is normal. The presence of visual spatial neglect or anosognosia may also be taken as evidence of abnormality. Since the	☐ 0 = No abnormality. ☐ 1 = Visual, tactile, auditory, spatial, or personal inattention or extinction to bilateral simultaneous stimulation in one of the sensory modalities. ☐ 2 = Profound hemi-inattention or extinction to more than one modality; does not recognize own hand or orients to only one side of space

Instructions	Score
abnormality is scored only if present, the item is never untestable.	

Total Score: __ __

* NIHSS score ≥ 8 and < 20 , as well as NIHSS Level of Consciousness (1A) score must be < 2 for patient's inclusion in the study.

APPENDIX 2: MODIFIED RANKIN SCALE (MRS)

Choose the score that most appropriately describes the patient's current state

Description	Score
No symptoms at all	☐ 0
No significant disability despite symptoms; able to carry out all usual duties and activities	☐ 1
Slight disability; unable to carry out all previous activities, but able to look after own affairs without assistance	☐ 2
Moderate disability; requiring some help, but able to walk without assistance	☐ 3
Moderately severe disability; unable to walk without assistance and unable to attend to own bodily needs without assistance	☐ 4
Severe disability; bedridden, incontinent and requiring constant nursing care and attention	☐ 5
Dead	☐ 6

APPENDIX 3: BARTHEL INDEX (BI)

Guidelines:

1. The index should be used as a record of what a patient does, not as a record of what a patient could do.
2. The main aim is to establish degree of independence from any help, physical or verbal, however minor and for whatever reason.
3. The need for supervision renders the patient not independent.
4. A patient's performance should be established using the best available evidence. Asking the patient, friends/relatives and nurses are the usual sources, but direct observation and common sense are also important. However direct testing is not needed.
5. Usually the patient's performance over the preceding 24-48 hours is important, but occasionally longer periods will be relevant.
6. Middle categories imply that the patient supplies over 50 per cent of the effort.
7. Use of aids to be independent is allowed.

Please circle the appropriate score for each scale item.

Item	Score	Scoring Criteria
1. Feeding (if food needs to be cut = help)	10	Independent. The patient can feed themselves a meal from a tray or table when someone puts the food within their reach. They must put on an assistive device if this is needed, cut up the food, use salt and pepper, spread butter, etc. They must accomplish this in a reasonable time.
	5	Some help is necessary (with cutting up food, etc., as listed above).
	0	Unable to perform this activity or N/A
2. Moving from wheelchair to bed and returning (includes sitting up in bed)	15	Independent in all phases of this activity. Patient can safely approach the bed in their wheelchair, lock brakes, lift footrests, move safely to bed, lie down, come to a sitting position on the side of the bed, change the position of the wheelchair, if necessary, to transfer back into it safely, and return to the wheelchair.
	10	Either some minimal help is needed in some step of this activity or the patient needs to be reminded or supervised for safety of one or more parts of this activity.
	5	Patient can come to a sitting position without the help of a second person but needs to be lifted out of bed, or they transfer with a great deal of help.
	0	Unable to perform this activity or N/A

3. Personal toilet (washing face, combing hair, shaving, cleaning teeth)	5	Patient can wash hands and face, comb hair, clean teeth, and shave. They may use any kind of razor but must put in blade or plug in razor without help as well as get it from drawer or cabinet. Female patients must put on own makeup, if used, but need not braid or style hair.
	0	Unable to perform this activity or N/A
4. Getting on and off toilet (handling clothes, wiping, flushing)	10	Patient is able to get on and off toilet, fasten and unfasten clothes, prevent soiling of clothes, and use toilet paper without help. They may use a wall bar or other stable object for support if needed. If it is necessary to use a bed pan instead of a toilet, they must be able to place it on a chair, empty it, and clean it.
	5	Patient needs help because of imbalance or in handling clothes or in using toilet paper.
	0	Unable to perform this activity or N/A
5. Bathing self	5	Patient may use a bath tub, a shower, or take a complete sponge bath. They must be able to do all the steps involved in whichever method is employed without another person being present.
	0	Unable to perform this activity or N/A
6. Walking on a level surface (or if unable to walk, propelling wheelchair) * Score only if unable to walk	15	Patient can walk at least 45 meters without help or supervision. They may wear braces or prostheses and use crutches, canes, or a walkerette but not a rolling walker. They must be able to lock and unlock braces if used, assume the standing position and sit down, get the necessary mechanical aides into position for use, and dispose of them when they sit. (Putting on and taking off braces is scored under dressing.)
	10	Patient needs help or supervision in any of the above but can walk at least 45 meters with a little help.
	5*	Patient cannot ambulate but can propel a wheelchair independently. They must be able to go around corners, turn around, maneuver the chair to a table, bed, toilet, etc. They must be able to push a chair at least 45 meters.
	0*	Unable to perform this activity or N/A
7. Ascending and descending stairs	10	Patient is able to go up and down a flight of stairs safely without help or supervision. They may and should use handrails, canes, or crutches when needed. They must be able to carry canes or crutches as they ascend or descend stairs.
	5	Patient needs help with or supervision of any one of the above items.
	0	Unable to perform this activity or N/A

8. Dressing (includes tying shoes, fastening fasteners)	10	Patient is able to put on and remove and fasten all clothing, and tie shoe laces (unless it is necessary to use adaptations for this). The activity includes putting on and removing and fastening corset as suspenders, loafer shoes, dresses that open down the front may be used when necessary.
	5	Patient needs help in putting on and removing or fastening any clothing. They must do at least half the work themselves. They must accomplish this in a reasonable time. Women need not be scored on use of a brassiere or girdle unless these are prescribed garments.
	0	Unable to perform this activity or N/A
9. Controlling bowels	10	Patient is able to control their bowels and have no accidents. They can use a suppository or take an enema when necessary (as for spinal cord injury patients who have had bowel training).
	5	Patient needs help in using a suppository or taking an enema or has occasional accidents.
	0	Unable to perform this activity or N/A
10. Controlling bladder	10	Patient is able to control their bladder day and night. Spinal cord injury patients who wear an external device and leg bag must put them on independently, clean and empty bag, and stay dry day and night.
	5	Patient has occasional accidents or cannot wait for the bed pan or get to the toilet in time or needs help with an external device.
	0	Unable to perform this activity or N/A

Total Score (0-100): _ _ _

APPENDIX 4: INTERVIEWER-ADMINISTERED EQ-5D-5L (EUROQOL)



Health Questionnaire

English version for the USA

VERSION FOR INTERVIEWER ADMINISTRATION

Note to interviewer: although allowance should be made for the interviewer's particular style of speaking, the wording of the questionnaire instructions should be followed as closely as possible. In the case of the EQ-5D descriptive system on page 2 of the questionnaire, the precise wording must be followed.

If the respondent has difficulty regarding which response to choose, or asks for clarification, the interviewer should repeat the question word for word and ask the respondent to answer in a way that most closely resembles his or her thoughts about his or her health today.

INTRODUCTION

(Note to interviewer: please read the following to the respondent.)

We are trying to find out what you think about your health. I will explain what to do as I go along, but please interrupt me if you do not understand something or if things are not clear to you. There are no right or wrong answers. We are interested only in your personal view.

First, I am going to read out some questions. Each question has a choice of five answers. Please tell me which answer best describes your health TODAY.

Do not choose more than one answer in each group of questions.

(Note to interviewer: first read all five options for each question. Then ask the respondent to choose which one applies to him/herself. Repeat the question and options if necessary. Mark the appropriate box under each heading. You may need to remind the respondent regularly that the timeframe is TODAY.)

EQ-5D DESCRIPTIVE SYSTEM

First, I would like to ask you about mobility. Would you say that:

1. You have no problems walking? ☐
 2. You have slight problems walking? ☐
 3. You have moderate problems walking? ☐
 4. You have severe problems walking? ☐
 5. You are unable to walk? ☐
-

Next, I would like to ask you about self-care. Would you say that:

1. You have no problems washing or dressing yourself? ☐
 2. You have slight problems washing or dressing yourself? ☐
 3. You have moderate problems washing or dressing yourself? ☐
 4. You have severe problems washing or dressing yourself? ☐
 5. You are unable to wash or dress yourself? ☐
-

Next, I would like to ask you about usual activities, such as work, study, housework, family or leisure activities. Would you say that:

1. You have no problems doing your usual activities? ☐
 2. You have slight problems doing your usual activities? ☐
 3. You have moderate problems doing your usual activities? ☐
 4. You have severe problems doing your usual activities? ☐
 5. You are unable to do your usual activities? ☐
-

Next, I would like to ask you about pain or discomfort. Would you say that:

1. You have no pain or discomfort? ☐
 2. You have slight pain or discomfort? ☐
 3. You have moderate pain or discomfort? ☐
 4. You have severe pain or discomfort? ☐
 5. You have extreme pain or discomfort? ☐
-

Finally, I would like to ask you about anxiety or depression. Would you say that:

1. You are not anxious or depressed? ☐
 2. You are slightly anxious or depressed? ☐
 3. You are moderately anxious or depressed? ☐
 4. You are severely anxious or depressed? ☐
 5. You are extremely anxious or depressed? ☐
-

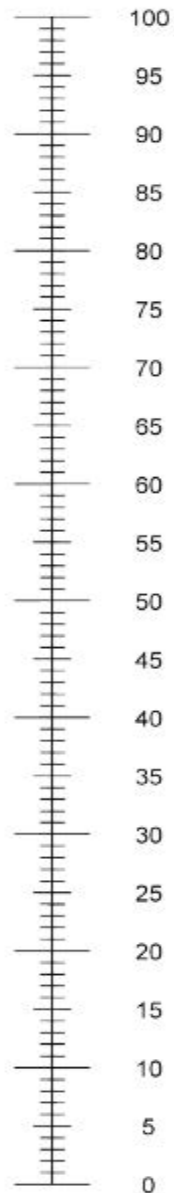
EQ-5D VAS

- Now, I would like to ask you to say how good or bad your health is TODAY.
- I would like you to try to picture in your mind a scale that looks like a thermometer.
(Note to interviewer: if interviewing face-to-face, please show the person the VAS scale.)
- The best health you can imagine is marked 100 (one hundred) at the top of the scale and the worst health you can imagine is marked 0 (zero) at the bottom.
- I would now like you to tell me the point on this scale where you would put your health TODAY.
(Note to interviewer: mark the scale at the point indicating the respondent's 'health today'. Now, please write the number you marked on the scale in the box below.)

THE RESPONDENT'S HEALTH TODAY =

Thank you for taking the time to answer these questions.

The best health
you can imagine



The worst health
you can imagine

APPENDIX 5: INTERVIEWER-ADMINISTERED STROKE-SPECIFIC QUALITY OF LIFE SCALE (SS-QOL)

We would like to know how you're doing with activities or feelings that can sometimes be affected by stroke. Each question will ask about a specific activity or feeling. For each question, think about how that activity or that feeling has been in the past week.

The first group of questions asks about how much trouble you have with a specific activity. Each question deals with problems that some people have after their stroke. Please choose the response that best describes how much trouble you have had with that activity in the past week.

DURING THE PAST WEEK:

	Couldn't do it at all	A lot of trouble	Some trouble	A little trouble	No trouble at all
SC1. Did you have trouble preparing food?	1	2	3	4	5
SC2. Did you have trouble eating, for example, cutting food or swallowing?	1	2	3	4	5
SC4. Did you have trouble getting dressed, for example, putting on socks or shoes, buttoning buttons, or zipping?	1	2	3	4	5
SC5. Did you have trouble taking a bath or shower?	1	2	3	4	5
SC8. Did you have trouble using the toilet?	1	2	3	4	5
V1. Did you have trouble seeing the television well enough to enjoy a show?	1	2	3	4	5
V2. Did you have trouble reaching for things because of poor eyesight?	1	2	3	4	5
V3. Did you have trouble seeing things off to one side?	1	2	3	4	5
L2. Did you have trouble speaking, for example, get stuck, stutter, stammer, or slur your words?	1	2	3	4	5
L3. Did you have trouble speaking clearly enough to use the telephone?	1	2	3	4	5

SS-QOL V2.0--

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DURING THE PAST WEEK:

	Couldn't do it at all	A lot of trouble	Some trouble	A little trouble	No trouble at all
L5. Did other people have trouble understanding what you said?	1	2	3	4	5
L6. Did you have trouble finding the word you wanted to say?	1	2	3	4	5
L7. Did you need to repeat yourself so others could understand you?	1	2	3	4	5
M1. Did you have trouble walking? (If you can't walk, circle 1 and go to question M7)	1	2	3	4	5
M4. Did you lose your balance when bending over or reaching for something?	1	2	3	4	5
M6. Did you have trouble climbing stairs?	1	2	3	4	5
M7. Did you have trouble with needing to stop and rest when walking or using a wheelchair?	1	2	3	4	5
M8. Did you have trouble with standing?	1	2	3	4	5
M9. Did you have trouble getting out of a chair?	1	2	3	4	5
W1. Did you have trouble doing daily work around the house?	1	2	3	4	5
W2. Did you have trouble finishing jobs that you started?	1	2	3	4	5
W3. Did you have trouble doing the work you used to do?	1	2	3	4	5

DURING THE PAST WEEK:

	Couldn't do it at all	A lot of trouble	Some trouble	A little trouble	No trouble at all
UE1. Did you have trouble writing or typing?	1	2	3	4	5
UE2. Did you have trouble putting on socks?	1	2	3	4	5
UE3. Did you have trouble buttoning buttons?	1	2	3	4	5
UE5. Did you have trouble zipping a zipper?	1	2	3	4	5
UE6. Did you have trouble opening a jar?	1	2	3	4	5

The next set of questions asks about how much you agree or disagree with each statement. Each question deals with a problem or feeling that some people have after their stroke. Please choose the response that best says how you felt about each statement **during the past week**.

DURING THE PAST WEEK:

	Strongly agree	Moderately agree	Neither agree nor disagree	Moderately disagree	Strongly disagree
T2. It was hard for me to concentrate.	1	2	3	4	5
T3. I had trouble remembering things.	1	2	3	4	5
T4. I had to write things down to remember them.	1	2	3	4	5
P1. I was irritable.	1	2	3	4	5

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DURING THE PAST WEEK:

	Strongly agree	Moderately agree	Neither agree nor disagree	Moderately disagree	Strongly disagree
P2. I was impatient with others	1	2	3	4	5
P3. My personality has changed.	1	2	3	4	5
MD2. I was discouraged about my future.	1	2	3	4	5
MD3. I wasn't interested in other people or activities.	1	2	3	4	5
FR5. I didn't join in activities just for fun with my family.	1	2	3	4	5
FR7. I felt I was a burden to my family.	1	2	3	4	5
FR8. My physical condition interfered with my family life.	1	2	3	4	5
SR1. I didn't go out as often as I would like.	1	2	3	4	5
SR4. I did my hobbies and recreation for shorter periods of time than I would like.	1	2	3	4	5
SR5. I didn't see as many of my friends as I would like.	1	2	3	4	5
SR6. I had sex less often than I would like.	1	2	3	4	5
SR7. My physical condition interfered with my social life.	1	2	3	4	5

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DURING THE PAST WEEK:

	Strongly agree	Moderately agree	Neither agree nor disagree	Moderately disagree	Strongly disagree
MD6. I felt withdrawn from other people.	1	2	3	4	5
MD7. I had little confidence in myself.	1	2	3	4	5
MD8. I was not interested in food.	1	2	3	4	5
E2. I felt tired most of the time.	1	2	3	4	5
E3. I had to stop and rest often during the day.	1	2	3	4	5
E4. I was too tired to do what I wanted to do.	1	2	3	4	5

Now, we would like to ask how you feel you are doing today in some general areas compared to how you were **before your stroke**. Please choose a response to show whether each area is a lot worse, a little worse, or the same as your **before your stroke**. Please remember to compare how you are doing **today** with how you were **before your stroke happened**.

	a lot worse than before my stroke	somewhat worse than before my stroke	a little worse than before my stroke	the same as before my stroke
1E. My energy level is				
2L. My speech is				
3M. My walking is				
4V. My vision is				
5UE. The use of my arms or hands is				
6T. My thinking is				
7MD. My mood is				
8P. My personality is				
9W. I do my jobs at home or at work				
10SC. I can take care of myself				
11FR. I do things for my family				
12SR. I do things for my friends				
13. Overall, my quality of life is				

SS-QOL V2.0--

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APPENDIX 6: MONTREAL COGNITIVE ASSESSMENT (MOCA) SCALE



Version 8.3 English

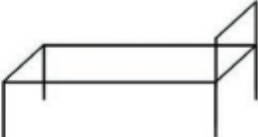
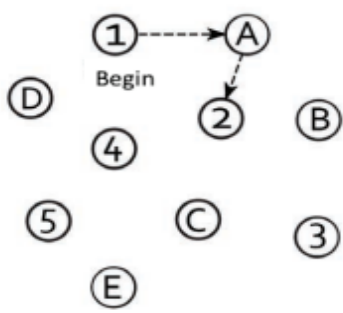
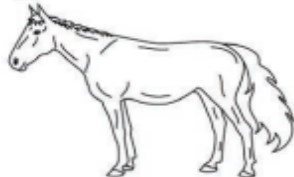
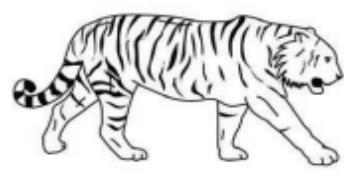
Name:

Education:

Sex:

Date of birth :

DATE :

VISUOSPATIAL / EXECUTIVE								POINTS																													
Copy bed 				Draw CLOCK (Five past ten) (3 points) [] Contour [] Numbers [] Hands				___/5																													
Begin  End []																																					
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<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr style="background-color: #000080; color: white;"> <th style="width: 20%;">MEMORY</th> <th style="width: 40%;">LEG</th> <th style="width: 10%;">COTTON</th> <th style="width: 10%;">SCHOOL</th> <th style="width: 10%;">TOMATO</th> <th style="width: 10%;">WHITE</th> </tr> </thead> <tbody> <tr> <td rowspan="2" style="font-size: small;">Read list of words, subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.</td> <td>1st TRIAL</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>2nd TRIAL</td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>									MEMORY	LEG	COTTON	SCHOOL	TOMATO	WHITE	Read list of words, subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.	1st TRIAL					2nd TRIAL					NO POINTS											
MEMORY	LEG	COTTON	SCHOOL	TOMATO	WHITE																																
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Read list of digits (1 digit/sec.). Subject has to repeat them in the forward order.																																					
Subject has to repeat them in the backward order.	[] 4 2 7																																				
Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors. [] F B A C M N A A J K L B A F A K D E A A A J A M O F A A B									___/1																												
Serial 7 subtraction starting at 60. [] 53 [] 46 [] 39 [] 32 [] 25 4 or 5 correct subtractions: 3 pts, 2 or 3 correct: 2 pts, 1 correct: 1 pt, 0 correct: 0 pt									___/3																												
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Language Fluency. Name maximum number of words in one minute that begin with the letter B. [] _____ (N ≥ 11 words)									___/1																												
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ORIENTATION	Date	Month	Year	Day	Place	City																															
	[]	[]	[]	[]	[]	[]																															

COLUMBIA-SUICIDE SEVERITY RATING SCALE (C-SSRS)

Baseline

Version 1/14/09

***Posner, K.; Brent, D.; Lucas, C.; Gould, M.; Stanley, B.; Brown, G.;
Fisher, P.; Zelazny, J.; Burke, A.; Oquendo, M.; Mann, J.***

Disclaimer:

This scale is intended to be used by individuals who have received training in its administration. The questions contained in the Columbia-Suicide Severity Rating Scale are suggested probes. Ultimately, the determination of the presence of suicidal ideation or behavior depends on the judgment of the individual administering the scale.

*Definitions of behavioral suicidal events in this scale are based on those used in **The Columbia Suicide History Form**, developed by John Mann, MD and Maria Oquendo, MD, Conte Center for the Neuroscience of Mental Disorders (CCNMD), New York State Psychiatric Institute, 1051 Riverside Drive, New York, NY, 10032. (Oquendo M. A., Halberstam B. & Mann J. J., Risk factors for suicidal behavior: utility and limitations of research instruments. In M.B. First [Ed.] Standardized Evaluation in Clinical Practice, pp. 103 -130, 2003.)*

For reprints of the C-SSRS contact Kelly Posner, Ph.D., New York State Psychiatric Institute, 1051 Riverside Drive, New York, New York, 10032; inquiries and training requirements contact posnerk@nyspi.columbia.edu

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SUICIDAL IDEATION	
Ask questions 1 and 2. If both are negative, proceed to "Suicidal Behavior" section. If the answer to question 2 is "yes", ask questions 3, 4 and 5. If the answer to question 1 and/or 2 is "yes", complete "Intensity of Ideation" section below.	Lifetime: Time He/She Felt Most Suicidal
1. Wish to be Dead Subject endorses thoughts about a wish to be dead or not alive anymore, or wish to fall asleep and not wake up. Have you wished you were dead or wished you could go to sleep and not wake up? If yes, describe:	Yes No ☐ ☐
2. Non-Specific Active Suicidal Thoughts General, non-specific thoughts of wanting to end one's life/commit suicide (e.g., "I've thought about killing myself") without thoughts of ways to kill oneself/associated methods, intent, or plan. Have you actually had any thoughts of killing yourself? If yes, describe:	Yes No ☐ ☐
3. Active Suicidal Ideation with Any Methods (Not Plan) without Intent to Act Subject endorses thoughts of suicide and has thought of at least one method during the assessment period. This is different than a specific plan with time, place or method details worked out (e.g., thought of method to kill self but not a specific plan). Includes person who would say, "I thought about taking an overdose but I never made a specific plan as to when, where or how I would actually do it...and I would never go through with it." Have you been thinking about how you might do this? If yes, describe:	Yes No ☐ ☐
4. Active Suicidal Ideation with Some Intent to Act, without Specific Plan Active suicidal thoughts of killing oneself and subject reports having <u>some intent to act on such thoughts</u> , as opposed to "I have the thoughts but I definitely will not do anything about them." Have you had these thoughts and had some intention of acting on them? If yes, describe:	Yes No ☐ ☐
5. Active Suicidal Ideation with Specific Plan and Intent Thoughts of killing oneself with details of plan fully or partially worked out and subject has some intent to carry it out. Have you started to work out or worked out the details of how to kill yourself? Do you intend to carry out this plan? If yes, describe:	Yes No ☐ ☐
INTENSITY OF IDEATION	
The following features should be rated with respect to the most severe type of ideation (i.e., 1-5 from above, with 1 being the least severe and 5 being the most severe). Ask about time he/she was feeling the most suicidal.	Most Severe
Most Severe Ideation: <u> </u> <u> </u> Type # (1-5) Description of Ideation	
Frequency How many times have you had these thoughts? (1) Less than once a week (2) Once a week (3) 2-5 times in week (4) Daily or almost daily (5) Many times each day	_____
Duration When you have the thoughts, how long do they last? (1) Fleeting - few seconds or minutes (4) 4-8 hours/most of day (2) Less than 1 hour/some of the time (5) More than 8 hours/persistent or continuous (3) 1-4 hours/a lot of time	_____

Controllability <i>Could/can you stop thinking about killing yourself or wanting to die if you want to?</i> (1) Easily able to control thoughts (2) Can control thoughts with little difficulty (3) Can control thoughts with some difficulty (4) Can control thoughts with a lot of difficulty (5) Unable to control thoughts (0) Does not attempt to control thoughts	_____
Deterrents <i>Are there things - anyone or anything (e.g., family, religion, pain of death) - that stopped you from wanting to die or acting on thoughts of committing suicide?</i> (1) Deterrents definitely stopped you from attempting suicide (2) Deterrents probably stopped you (3) Uncertain that deterrents stopped you (4) Deterrents most likely did not stop you (5) Deterrents definitely did not stop you (0) Does not apply	_____
Reasons for Ideation <i>What sort of reasons did you have for thinking about wanting to die or killing yourself? Was it to end the pain or stop the way you were feeling (in other words you couldn't go on living with this pain or how you were feeling) or was it to get attention, revenge or a reaction from others? Or both?</i> (1) Completely to get attention, revenge or a reaction from others (2) Mostly to get attention, revenge or a reaction from others (3) Equally to get attention, revenge or a reaction from others stop the pain (you couldn't go on and to end/stop the pain. how you were feeling) (4) Mostly to end or stop the pain (you couldn't go on living with the pain or how you were feeling) (5) Completely to end or living with the pain or (0) Does not apply	_____

SUICIDAL BEHAVIOR (Check all that apply, so long as these are separate events; must ask about all types)	Lifetime
Actual Attempt: A potentially self-injurious act committed with at least some wish to die, <i>as a result of act</i> . Behavior was in part thought of as method to kill oneself. Intent does not have to be 100%. If there is any intent/desire to die associated with the act, then it can be considered an actual suicide attempt. <i>There does not have to be any injury or harm</i> , just the potential for injury or harm. If person pulls trigger while gun is in mouth but gun is broken so no injury results, this is considered an attempt. Inferring Intent: Even if an individual denies intent/wish to die, it may be inferred clinically from the behavior or circumstances. For example, a highly lethal act that is clearly not an accident so no other intent but suicide can be inferred (e.g., gunshot to head, jumping from window of a high floor/story). Also, if someone denies intent to die, but they thought that what they did could be lethal, intent may be inferred. Have you made a suicide attempt? Have you done anything to harm yourself? Have you done anything dangerous where you could have died? What did you do? Did you _____ as a way to end your life? Did you want to die (even a little) when you _____? Were you trying to end your life when you _____? Or did you think it was possible you could have died from _____? Or did you do it purely for other reasons / without ANY intention of killing yourself (like to relieve stress, feel better, get sympathy, or get something else to happen)? (Self-Injurious Behavior without suicidal intent) If yes, describe:	Yes No ☐ ☐ Total # of Attempts _____ Yes No ☐ ☐
Has subject engaged in Non-Suicidal Self-Injurious Behavior?	

Interrupted Attempt: When the person is interrupted (by an outside circumstance) from starting the potentially self-injurious act (<i>if not for that, actual attempt would have occurred</i>). Overdose: Person has pills in hand but is stopped from ingesting. Once they ingest any pills, this becomes an attempt rather than an interrupted attempt. Shooting: Person has gun pointed toward self, gun is taken away by someone else, or is somehow prevented from pulling trigger. Once they pull the trigger, even if the gun fails to fire, it is an attempt. Jumping: Person is poised to jump, is grabbed and taken down from ledge. Hanging: Person has noose around neck but has not yet started to hang - is stopped from doing so. Has there been a time when you started to do something to end your life but someone or something stopped you before you actually did anything? If yes, describe:		Yes No ☐ ☐ Total # of interrupted	
Aborted Attempt: When person begins to take steps toward making a suicide attempt, but stops themselves before they actually have engaged in any self-destructive behavior. Examples are similar to interrupted attempts, except that the individual stops him/herself, instead of being stopped by something else. Has there been a time when you started to do something to try to end your life but you stopped yourself before you actually did anything? If yes, describe:		Yes No ☐ ☐ Total # of aborted	
Preparatory Acts or Behavior: Acts or preparation towards imminently making a suicide attempt. This can include anything beyond a verbalization or thought, such as assembling a specific method (e.g., buying pills, purchasing a gun) or preparing for one's death by suicide (e.g., giving things away, writing a suicide note). Have you taken any steps towards making a suicide attempt or preparing to kill yourself (such as collecting pills, getting a gun, giving valuables away or writing a suicide note)? If yes, describe:		Yes No ☐ ☐	
Suicidal Behavior: Suicidal behavior was present during the assessment period?		Yes No ☐ ☐	
Answer for Actual Attempts Only		Most Recent Attempt Date:	Most Lethal Attempt Date:
Actual Lethality/Medical Damage: 0. No physical damage or very minor physical damage (e.g., surface scratches). 1. Minor physical damage (e.g., lethargic speech; first-degree burns; mild bleeding; sprains). 2. Moderate physical damage; medical attention needed (e.g., conscious but sleepy, somewhat responsive; second-degree burns; bleeding of major vessel). 3. Moderately severe physical damage; <i>medical</i> hospitalization and likely intensive care required (e.g., comatose with reflexes intact; third-degree burns less than 20% of body; extensive blood loss but can recover; major fractures). 4. Severe physical damage; <i>medical</i> hospitalization with intensive care required (e.g., comatose without reflexes; third- degree burns over 20% of body; extensive blood loss with unstable vital signs; major damage to a vital area). 5. Death		<i>Enter Code</i>	<i>Enter Code</i>
Potential Lethality: Only Answer if Actual Lethality=0 Likely lethality of actual attempt if no medical damage (the following examples, while having no actual medical damage, had potential for very serious lethality: put gun in mouth and pulled the trigger but gun fails to fire so no medical damage; laying on train tracks with oncoming train but pulled away before run over). 0 = Behavior not likely to result in injury 1 = Behavior likely to result in injury but not likely to cause death 2 = Behavior likely to result in death despite available medical care		<i>Enter Code</i>	<i>Enter Code</i>

COLUMBIA-SUICIDE SEVERITY RATING SCALE (C-SSRS)

Since Last Visit

Version 1/14/09

Posner, K.; Brent, D.; Lucas, C.; Gould, M.; Stanley, B.; Brown, G.; Fisher, P.; Zelazny, J.; Burke, A.; Oquendo, M.; Mann, J.

Disclaimer:

This scale is intended to be used by individuals who have received training in its administration. The questions contained in the Columbia-Suicide Severity Rating Scale are suggested probes. Ultimately, the determination of the presence of suicidal ideation or behavior depends on the judgment of the individual administering the scale.

*Definitions of behavioral suicidal events in this scale are based on those used in **The Columbia Suicide History Form**, developed by John Mann, MD and Maria Oquendo, MD, Conte Center for the Neuroscience of Mental Disorders (CCNMD), New York State Psychiatric Institute, 1051 Riverside Drive, New York, NY, 10032. (Oquendo M. A., Halberstam B. & Mann J. J., Risk factors for suicidal behavior: utility and limitations of research instruments. In M.B. First [Ed.] Standardized Evaluation in Clinical Practice, pp. 103 -130, 2003.)*

For reprints of the C-SSRS contact Kelly Posner, Ph.D., New York State Psychiatric Institute, 1051 Riverside Drive, New York, New York, 10032; inquiries and training requirements contact posnerk@nyspi.columbia.edu

SUICIDAL IDEATION							
Ask questions 1 and 2. If both are negative, proceed to "Suicidal Behavior" section. If the answer to question 2 is "yes", ask questions 3, 4 and 5. If the answer to question 1 and/or 2 is "yes", complete "Intensity of Ideation" section below.	Since Last Visit						
1. Wish to be Dead Subject endorses thoughts about a wish to be dead or not alive anymore, or wish to fall asleep and not wake up. <i>Have you wished you were dead or wished you could go to sleep and not wake up?</i> If yes, describe:	Yes No						
2. Non-Specific Active Suicidal Thoughts General non-specific thoughts of wanting to end one's life/commit suicide (e.g., "I've thought about killing myself") without thoughts of ways to kill oneself/associated methods, intent, or plan during the assessment period. <i>Have you actually had any thoughts of killing yourself?</i> If yes, describe:	Yes No						
3. Active Suicidal Ideation with Any Methods (Not Plan) without Intent to Act Subject endorses thoughts of suicide and has thought of at least one method during the assessment period. This is different than a specific plan with time, place or method details worked out (e.g., thought of method to kill self but not a specific plan). Includes person who would say, "I thought about taking an overdose but I never made a specific plan as to when, where or how I would actually do it....and I would never go through with it". <i>Have you been thinking about how you might do this?</i> If yes, describe:	Yes No						
4. Active Suicidal Ideation with Some Intent to Act, without Specific Plan Active suicidal thoughts of killing oneself and subject reports having <u>some intent to act on such thoughts</u> , as opposed to "I have the thoughts but I definitely will not do anything about them". <i>Have you had these thoughts and had some intention of acting on them?</i> If yes, describe:	Yes No						
5. Active Suicidal Ideation with Specific Plan and Intent Thoughts of killing oneself with details of plan fully or partially worked out and subject has some intent to carry it out. <i>Have you started to work out or worked out the details of how to kill yourself? Do you intend to carry out this plan?</i> If yes, describe:	Yes No						
INTENSITY OF IDEATION							
The following features should be rated with respect to the most severe type of ideation (i.e., 1-5 from above, with 1 being the least severe and 5 being the most severe).	Most Severe						
Most Severe Ideation: <table border="0" style="width: 100%;"> <tr> <td style="width: 30%;"><u> </u></td> <td style="width: 30%;"><u> </u></td> <td style="width: 40%;"><u> </u></td> </tr> <tr> <td>Type # (1-5)</td> <td></td> <td>Description of Ideation</td> </tr> </table>	<u> </u>	<u> </u>	<u> </u>	Type # (1-5)		Description of Ideation	
<u> </u>	<u> </u>	<u> </u>					
Type # (1-5)		Description of Ideation					
Frequency <i>How many times have you had these thoughts?</i> (1) Less than once a week (2) Once a week (3) 2-5 times in week (4) Daily or almost daily (5) Many times each day	_____						
Duration <i>When you have the thoughts how long do they last?</i> (1) Fleeting - few seconds or minutes (4) 4-8 hours/most of day (2) Less than 1 hour/some of the time (5) More than 8 hours/persistent or continuous (3) 1-4 hours/a lot of time	_____						
Controllability <i>Could/can you stop thinking about killing yourself or wanting to die if you want to?</i> (1) Easily able to control thoughts (4) Can control thoughts with a lot of difficulty (2) Can control thoughts with little difficulty (5) Unable to control thoughts (3) Can control thoughts with some difficulty (0) Does not attempt to control thoughts	_____						

[illegible]

When person begins to take steps toward making a suicide attempt, but stops themselves before they actually have engaged in any self-destructive behavior. Examples are similar to interrupted attempts, except that the individual stops him/herself, instead of being stopped by something else. <i>Has there been a time when you started to do something to try to end your life but you stopped yourself before you actually did anything?</i> If yes, describe:	☐ ☐ Total # of aborted
Preparatory Acts or Behavior: Acts or preparation towards imminently making a suicide attempt. This can include anything beyond a verbalization or thought, such as assembling a specific method (e.g., buying pills, purchasing a gun) or preparing for one's death by suicide (e.g., giving things away, writing a suicide note). <i>Have you taken any steps towards making a suicide attempt or preparing to kill yourself (such as collecting pills, getting a gun, giving valuables away or writing a suicide note)?</i> If yes, describe:	Yes No
Suicidal Behavior: Suicidal behavior was present during the assessment period?	Yes No
Suicide:	Yes No
<i>Answer for Actual Attempts Only</i>	Most Lethal Attempt Date:
Actual Lethality/Medical Damage: 0. No physical damage or very minor physical damage (e.g., surface scratches). 1. Minor physical damage (e.g., lethargic speech; first-degree burns; mild bleeding; sprains). 2. Moderate physical damage; medical attention needed (e.g., conscious but sleepy, somewhat responsive; second-degree burns; bleeding of major vessel). 3. Moderately severe physical damage; <i>medical</i> hospitalization and likely intensive care required (e.g., comatose with reflexes intact; third-degree burns less than 20% of body; extensive blood loss but can recover; major fractures). 4. Severe physical damage; <i>medical</i> hospitalization with intensive care required (e.g., comatose without reflexes; third-degree burns over 20% of body; extensive blood loss with unstable vital signs; major damage to a vital area). 5. Death	<i>Enter Code</i>
Potential Lethality: Only Answer if Actual Lethality=0 Likely lethality of actual attempt if no medical damage (the following examples, while having no actual medical damage, had potential for very serious lethality: put gun in mouth and pulled the trigger but gun fails to fire so no medical damage; laying on train tracks with oncoming train but pulled away before run over). 0 = Behavior not likely to result in injury 1 = Behavior likely to result in injury but not likely to cause death 2 = Behavior likely to result in death despite available medical care	<i>Enter Code</i>

APPENDIX 8: CONTRACEPTION DEFINITIONS, CONTRACEPTIVE REQUIREMENTS, AND PREGNANCY TESTING

Definitions

A woman is considered a WOCBP (i.e., Woman of Child Bearing Potential) following menarche and until becoming postmenopausal, unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy. A WOCBP must follow ≥ 1 highly effective method of contraception as outlined below.

A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. A high follicle-stimulating hormone (FSH) level in the postmenopausal range *may* be used to confirm a postmenopausal state in women not using hormonal contraception or hormone replacement therapy (HRT). However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient.

A man is considered fertile after puberty unless permanently sterile by bilateral orchiectomy. A fertile male must follow ≥ 1 highly effective method of contraception as outlined below.

Highly Effective Methods of Contraception

Highly effective methods of contraception are those that have a failure rate of $< 1\%$ (when implemented consistently and correctly) and include:

- combined (containing estrogen and progestogen) hormonal contraception associated with inhibition of ovulation (administration may be oral, intravaginal, or transdermal) and used in conjunction with a barrier method (e.g., male condom)
- progestogen-only hormonal contraception associated with inhibition of ovulation (administration may be oral, injectable, or implantable) and used in conjunction with a barrier method (e.g., male condom)
- intrauterine device
- intrauterine hormone-releasing system
- bilateral tubal ligation or occlusion up to here
- vasectomy (provided that the male participant or the vasectomized partner has received a medical assessment of surgical success [i.e., azoospermia])
- sexual abstinence (defined as refraining from heterosexual intercourse during the entire period of risk associated with the study treatment[s]); the reliability of sexual abstinence

needs to be evaluated in relation to the duration of the clinical study and the preferred and usual lifestyle of the participant.

Participants not willing to use a highly effective contraception method will be excluded

Pregnancy Testing and Contraception Requirements for Women of Childbearing Potential

WOCBP may only be randomized after a negative sensitive urine or serum pregnancy test.

Female participants will be advised that they must report immediately to the study site for pregnancy testing and appropriate management in the event that they may be pregnant. A pregnancy test must be performed monthly, and at the end of relevant systemic exposure.

WOCBP must use at least 1 highly effective method of contraception from randomization until 5 half-lives after the last dose of study treatment

Requirements for Male Subjects With Pregnant Partners or Nonpregnant Women of Childbearing Potential (WOCBP) Partners

For males with pregnant partners or nonpregnant WOCBP partners, male contraception (condom) is recommended during study treatment and for 5 half-lives after the last dose of study treatment.

Donation or Banking of Sperm or Eggs or Retrieval of Eggs

Male participants must refrain from donating or banking sperm during study treatment and for 5 half-lives after the last dose of study treatment.

Female participants must refrain from donating or banking eggs (ova, oocytes) and retrieving eggs for use during study treatment AND for 5 half-lives after the last dose of study treatment.

Breastfeeding

Female participants are ineligible to be randomized if they are breastfeeding (directly or via pump).

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